**Module 5 Heredity**

**Reproduction**
**Inquiry question: How does reproduction ensure the continuity of a species?**

Explain the mechanisms of reproduction that ensure the continuity of a species, by analysing sexual and asexual methods of reproduction in a variety of organisms, including but not limited to:

- ➔ animals: advantages of external and internal fertilisation

**Fertilisation:** the fusion of gametes (sperm and egg/ovum) in sexual reproduction

1. **External:** combination of gametes in the external environment
   - ★ Frogs: The male frog grabs the female's back and fertilises the eggs as the female frog releases them.
   - ★ Bony fish: Eggs and sperm are released into the water. Sperm fuse with eggs outside of body due to pheromones (chemical signals released by the eggs that attract sperm)
2. **Internal:** combination of gametes within the female body
   - ★ Turtles: Male turtle mounts the female and aligns his cloaca with hers. The male has a penis, which he uses to transfer sperm into the female's cloaca. **(oviparous)**
   - ★ Humans: Sperm is released into the vagina where it fuses with an egg in the ampulla **(viviparous)**

**When fertilised internally, development can be:**

1. **Oviparous:** External within an egg
2. **Viviparous:** Internal within the mother
3. **Ovoviviparous:** Internal within an egg within the mother

| **Fertilisation** | **Advantages** | **Disadvantages** |
|---|---|---|
| **External** | Large number of offspring produced<br><br>Reproduction can continue without pause for gestation<br><br>Offspring are dispersed, meaning less competition<br><br>Little energy to find a mate | Lots of energy & resources used in the production and release of many gametes<br><br>Zygotes/embryos are unprotected<br><br>No parental care=offspring may not survive adulthood<br><br>Dehydration if there is not enough water=sperm die on land |

| | | |
|---|---|---|
| **Internal** | Parental care after birth<br><br>Protection of embryos> more embryos survive to gestational maturity<br><br>Sperm shed into a confined space, increasing the chance of fertilisation<br><br>Less gametes produced, which reduces energy expenditure | Higher risk of sexually transmitted diseases/infections<br><br>Fewer offspring produced<br><br>Breeding has to be paused for gestation<br><br>Requires more energy to conceive/find a mate |

| **Characteristics** | **Internal** | **External** | **Similarities** |
|---|---|---|---|
| **Gametes** | Large number of male gametes and fewer female gametes produced | Large numbers of gametes produced | Male and female gametes required |
| **Union** | Occurs inside the reproductive tract of the female in organisms that live mostly or completely on land | Occurs in open water environments | Sperm fertilise the eggs when they unite |
| **Conception mechanism** | Copulation: the male inserts sperm into the female's reproductive tract | No copulation - simultaneous release of gametes | Sperm will fertilise eggs when in very close proximity to each other; gametes require a watery environment for this to occur |
| **Chance of fertilisation** | High, because male gametes are released into a confined space where there is more chance of successfully uniting with female gametes | Low, because male gametes are released into a large open area where there is less chance of successfully uniting with female gametes | If male and female gametes are in close proximity to each other, fertilisation will usually occur |

- ➔ plants: asexual and sexual reproduction

**Reproduction:** production of offspring

1. Asexual: the production of offspring that are genetically identical to a single-parent
2. Sexual: the production of offspring that are genetically different from the two parents

**Asexual reproduction in plants:**

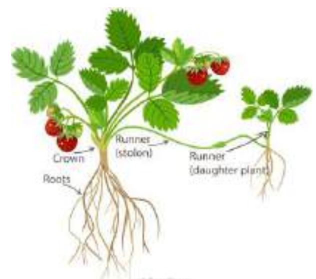
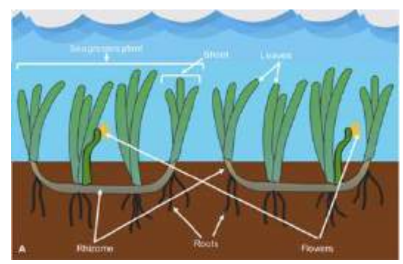
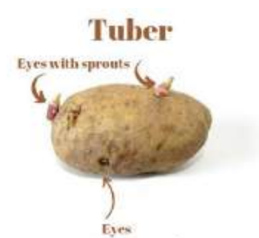
1. Fragmentation: involves new plants growing from small parts of a parent plant that fall to the ground

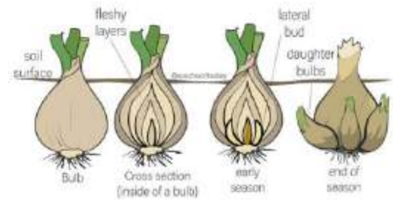
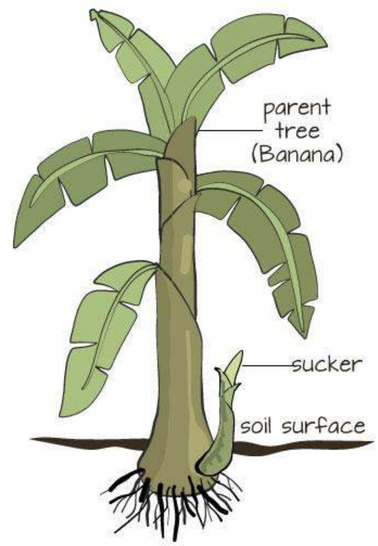
1. Fragments of parent break or split off
2. The broken piece moves to a new location, often carried by wind or water
3. The broken piece develops roots when it lands on suitable environment
4. Roots anchor the piece, and it grows into a new plant

★ Moss

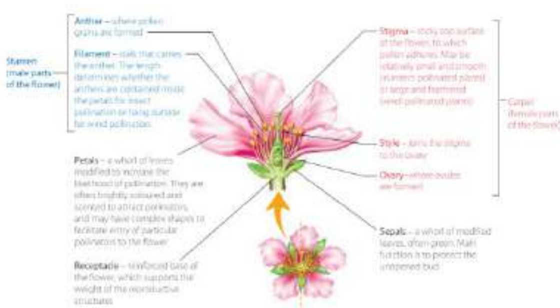
2. Using specialised reproductive structures

➢ **Vegetative propagation (reproduction)**- a new plant grows from a fragment or cutting of the parent plant or specialized reproductive structures

| Type of asexual reproduction | Description | Examples |
|---|---|---|
| Stolons | Stems that grow close to the ground and develop nodes from which new plants form on | <br>Strawberries |
| Rhizomes | Underground plant stems that possess nodes capable of generating shoots that emerge above ground while creating root systems for a new plant | <br>Seagrass |
| Tubers | Thickened part of an underground stem that stores nutrients. It has buds from which new plant shoots can grow | <br>Potato |

| | | |
|---|---|---|
| Bulbs | Underground stem with fleshy leaves that provide nourishment during dormancy |   |
| Suckers | Growth that develops directly from the rootstock of a plant |   Banana |

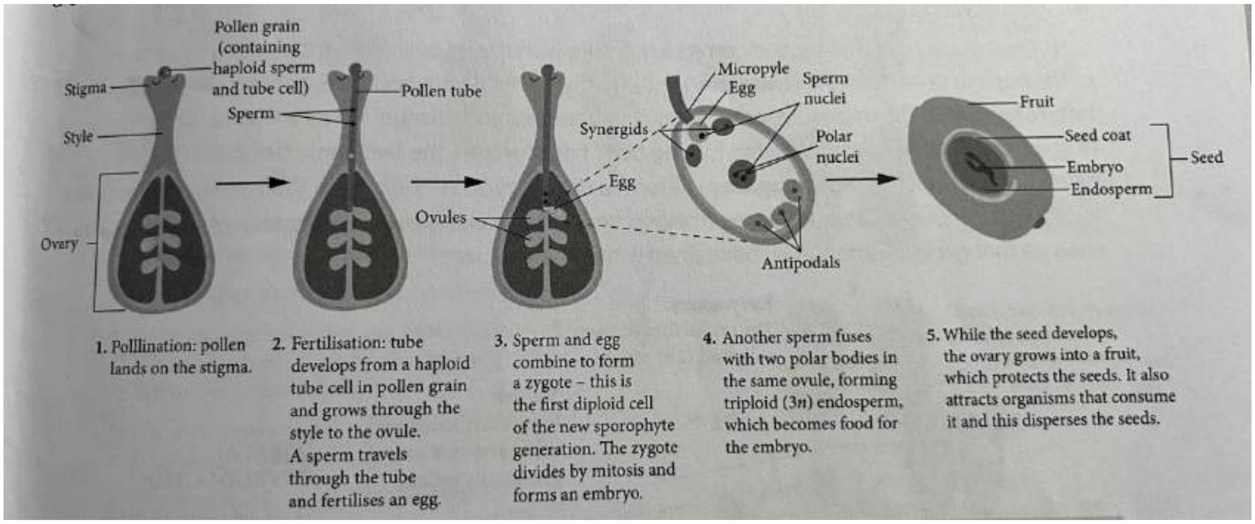
**Sexual reproduction in plants:**




**Pollination:** transfer of pollen from the anther to the stigma of a plant

1. Pollen travels from the anther to the stigma of a plant
2. Once on the stigma, the tube cell germinates and creates a pollen tube that grows down the style, carrying the generative cell that divides into two sperm cells to the ovule
3. One sperm fuses with the egg cell and the second sperm cell joins with the two polar nuclei developing into an endosperm which becomes food for the embryo

4. The ovule becomes a seed, and the ovary grows into a fruit protecting the seed



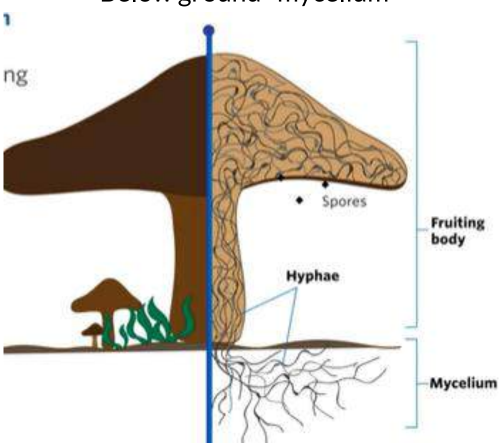

**Methods of pollination**

| Method | Description |
|---|---|
| Self-pollination | • Little variation in offspring<br>• Requires less energy as plants don't need to attract pollinators<br>• Can grow in areas where animals that visit plants are few in numbers |
| Cross-pollination | • increases genetic variation<br>  ○ wind: plants are small and lightweight<br>    ■ wind blows pollen to other plants<br>  ○ water<br>    ■ pollen drifts to other plants on water's surface<br>  ○ animals: plants are colourful and seed bearing<br>    ■ animals attracted by the colours of a flower and feed on the nectar<br>    ■ residual pollen brushes on plant surface<br>    ■ refer to process table for rest (fertilisation on) |

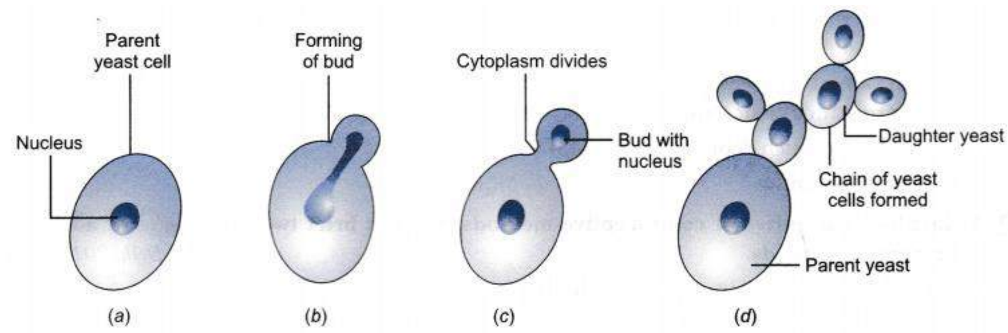
| | |
|---|---|
| | • can have pollen/ovules mature at different times<br>  ○ prevents self pollination ∴ variation & continuity |

➔ fungi: budding, spores

**Fungi:** a diverse group of eukaryotic, unicellular/multicellular, heterotrophic organisms
- Hyphae are the basic structural unit of fungi
- Above ground=fruiting body
- Below ground=mycelium




**Asexual reproduction in fungi:**
**Budding:** a new organism develops from a small part of the parent body
1. When conditions are favourable a small bud develops on the parent body
2. This outgrowth enlarges and the parent cell divides its nucleus where one copy moves into the outgrowth
3. Once reaches a certain size, the bud breaks off and grows into a new cell (complete budding) or it remains attached forming a chain of cells (incomplete budding)
4. Once grown it buds in turn
   ★ Yeast

5. 


**Sexual reproduction in fungi:**

1. When conditions are less favourable fungi use spores to reproduce sexually
2. Spores are released and land on suitable environment to germinate and grow into haploid mycelia
3. Two compatible haploid mycelia meet and their cytoplasm fuses (plasmogamy)
4. Forms a fruiting body
5. Two haploid nuclei fuse forming a diploid zygote (karyogamy)
6. Zygote undergoes meiosis producing new haploid spores
7. Haploid spores are released and the cycle occurs again
   - ➢ Ensures genetic variation

- ★ Mushroom

- ➔ bacteria: binary fission

**Asexual reproduction in bacteria:**
**Binary Fission:** splitting a cell into two

1. Single chromosome and plasmids replicate
2. Cell elongates building more cell wall
3. Replicated chromosomes and plasmids separate moving to a different pole as the cell elongates
4. Cleavage furrow begins to form
5. Two identical daughter cells are produced
   - ★ Escherichia coli (E. coli)

- Can still acquire genetic variation through conjugation
- Conjugation is the most common and is the transfer of DNA from one bacterium to another via a cytoplasmic bridge

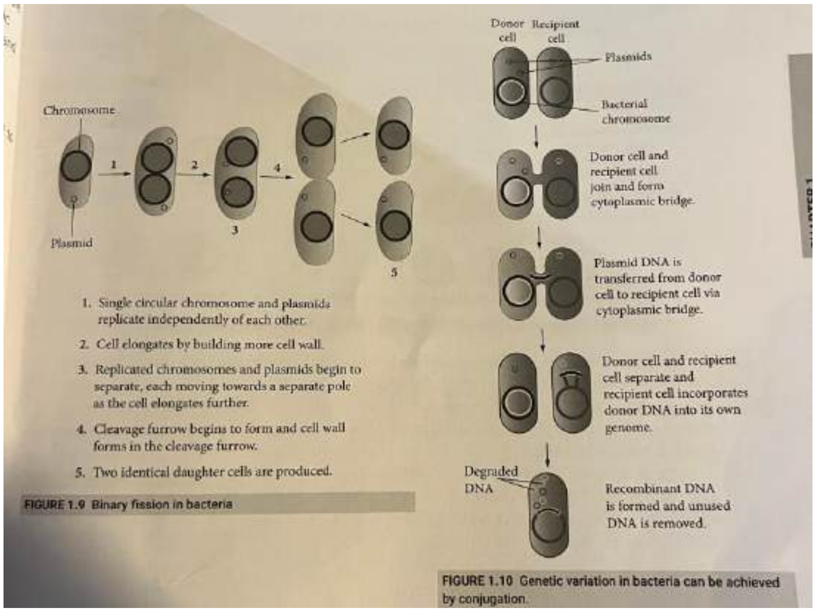


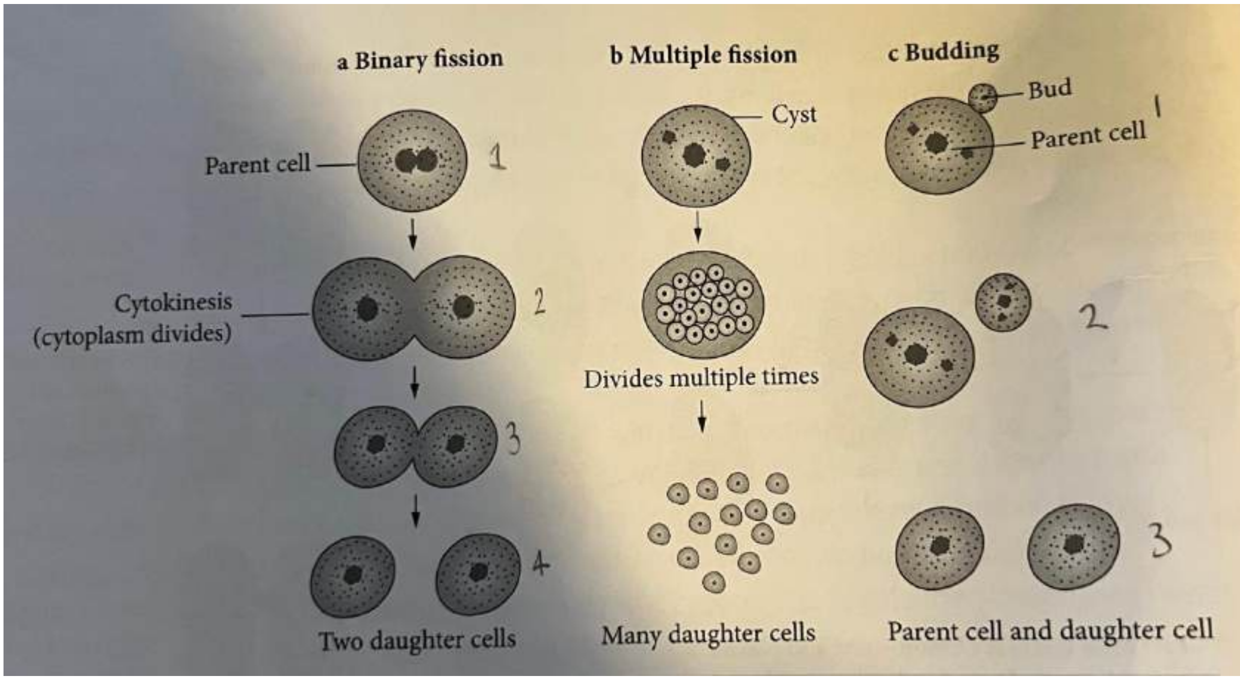
FIGURE 1.9 Binary fission in bacteria

FIGURE 1.10 Genetic variation in bacteria can be achieved by conjugation.

- ➔ protists: binary fission, budding
  - Unicellular eukaryotic organisms that are not classified as animals, plants or fungi
    - ★ Amoeba

**Asexual reproduction in protists:**

- **Binary fission**
  1. Nucleus replicates
  2. Cytokinesis occurs where the cytoplasm divides
  3. Two cells pinch off creating two separate independent organisms
     - ★ Amoeba



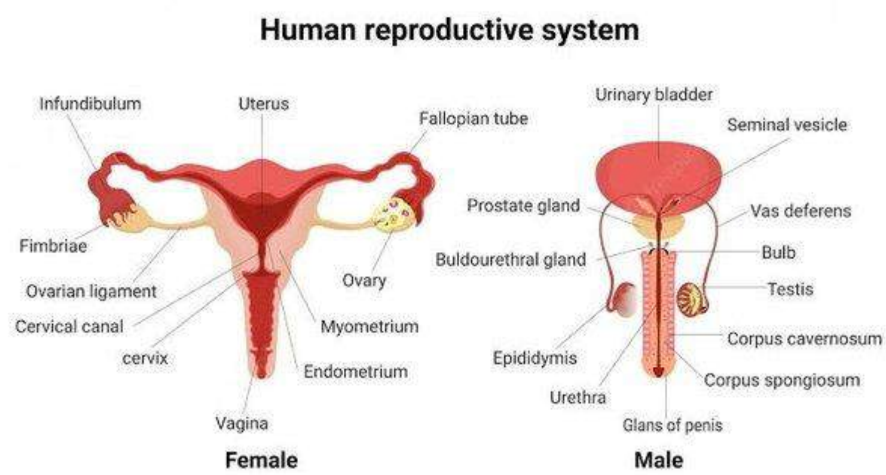

- Budding → same as fungi

★ Amoeba, plasmodium

**Sexual reproduction in protists:**

- Used in unfavourable conditions to introduce genetic variation
    - ★ Syngamy: diploid cell undergoes meiosis to produce four haploid gametes. Gametes from different individuals fuse to produce a new diploid line
    - ★ Conjugation

analyse the features of fertilisation, implantation and hormonal control of pregnancy and birth in mammals

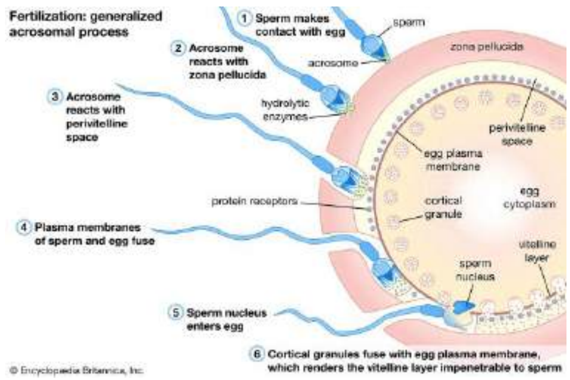



1. Fertilisation
    - Initially male ejaculates into the vagina
        - ➢ Sperm becomes hypermobile in oviduct of uterus (due to progesterone & high pH)
        - ➢ tails beat strongly propelling them **upwards** towards the oocyte (immature egg cell)

- Once inside must penetrate 3 layers
    - ➢ First layer (Corona Radiata): covered in follicle cells which release enzymes to assist in pushing through

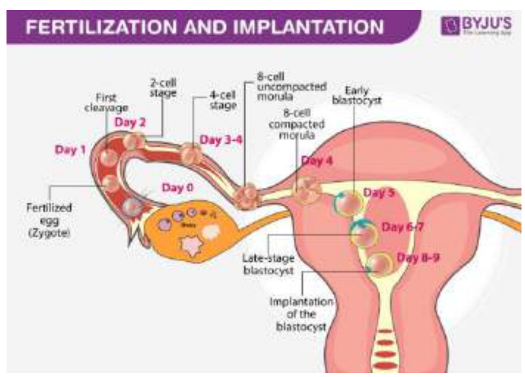
    - ➢ Second layer (Zona Pellucida): the acrosome (protective cap) of the sperm comes into contact with glycoproteins where it fuses with the cell membrane of the sperm head, allowing the tip of the sperm to release enzymes that assist penetration

    - ➢ Third layer (Plasma Membrane): Surface proteins on the egg's plasma membrane allow only one sperm to penetrate this barrier. Once a sperm successfully penetrates, it triggers a response known as the cortical reaction, where cortical granules within the egg release lytic enzymes. These enzymes modify the zona pellucida matrix, specifically destroying glycoproteins, which prevents additional sperm from binding and entering the egg. Simultaneously, the egg undergoes its second meiotic division. After meiosis II is completed, the sperm's nucleus fuses with the egg's nucleus, forming a zygote.




2. Implantation
    - After fertilisation, zygote undergoes several cellular divisions to form a ball of cells (morula)
    - Morula travels down the oviduct until it reaches the uterus
        - ➢ by this stage the morula has differentiated into a hollow fluid-filled ball of cells known as a blastocyst

- Blastocyst embeds into endometrial lining of the uterus
- Once implanted the blastocyst, which contains stem cells, will start to differentiate and an embryo will begin to develop
- Embryonic tissues give rise to the umbilical cord which attaches to the placenta allowing the exchange of blood, nutrients and wastes




3. Hormonal control of pregnancy and birth in mammals

**Hormones:** chemicals that coordinate different functions in your body by carrying messages through your blood to various parts of the body

**Ovarian cycle:**

1. The maturation of a follicle and the egg contained within it
2. The release of the egg during ovulation
3. The subsequent production of the corpus luteum

**The menstrual cycle:** Responsible for the preparation and maintenance of the uterine wall in readiness for implantation of the zygote. If fertilisation does not occur, or if implantation is unsuccessful, the endometrium and egg will be shed during the menstrual period.

<table>
<tr><th>Hormone</th><th>Function</th><th>Location Produced</th></tr>
<tr><td>Follicle Stimulating Hormone</td><td>Stimulates the development of egg within the follicle</td><td rowspan="2">Pituitary Gland (Brain)<ul><li>GnRH causes secretion of FSH & LH</li></ul>
</td></tr>
<tr><td>Luteinising Hormone</td><td>Promotes final maturation of egg within the follicle</td></tr>
<tr><td>Progesterone</td><td>Regrowth of uterine lining to stop contractions & give current egg that's released the opportunity to be fertilised and implant</td><td rowspan="2">Ovaries</td></tr>
<tr><td>Estrogen</td><td>Builds endometrium</td></tr>
</table>

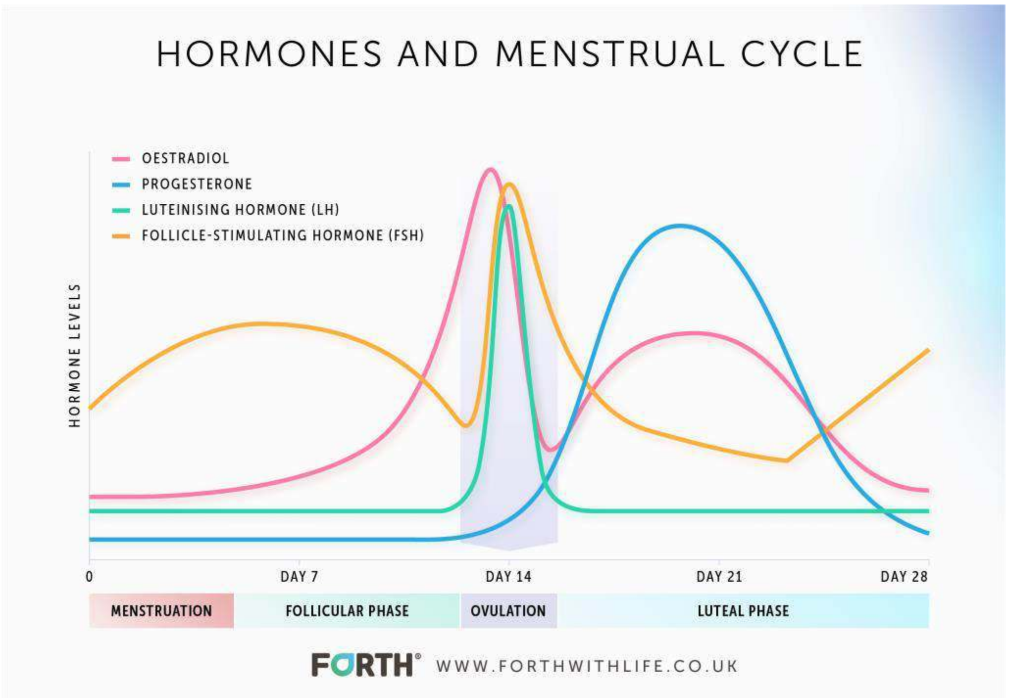
**Key Events in the Menstrual Cycle**

1. **Follicular Phase (Days 1-13)**
   - Gonadotropin-releasing hormones (GnRH) stimulates the pituitary gland to produce follicle stimulating hormone (FSH) to stimulate the growth of ovarian follicles
   - The dominant follicle produces estrogen, which inhibits FSH secretion (negative feedback) to prevent other follicles from growing
   - Estrogen acts on the uterus to stimulate the thickening of the endometrial layer

2. **Ovulation (Day 14)**
   - Midway through the cycle, when estrogen is high, it stimulates the anterior pituitary to secrete hormones (positive feedback)
   - This positive feedback results in a large surge of luteinizing hormone (LH) and a lesser surge of FSH
   - LH causes the dominant follicle to rupture and release an egg (secondary oocyte) – this is called ovulation

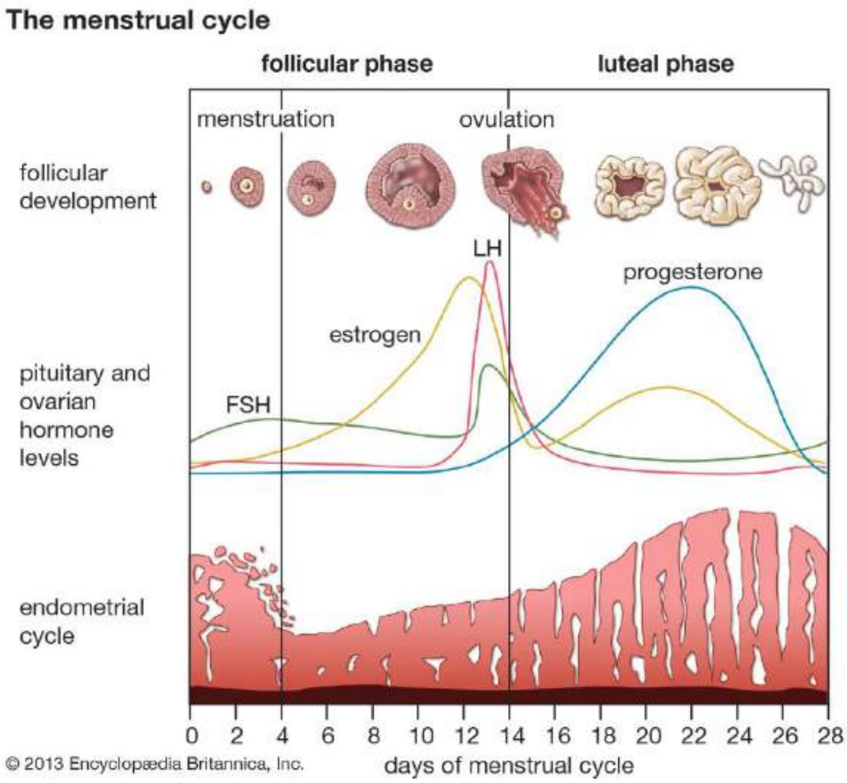
3. **Luteal Phase (Days 15-28)**
   - The ruptured follicle develops into a slowly degenerating corpus luteum

- The corpus luteum secretes high levels of progesterone, as well as lower levels of oestrogen
- Estrogen induces the thickening of the endometrium to prepare it for implantation, while progesterogn, preserves the thickened endometrium by inhibiting further release of LH and FSH preventing any follicles from developing




**4. Menstruation**

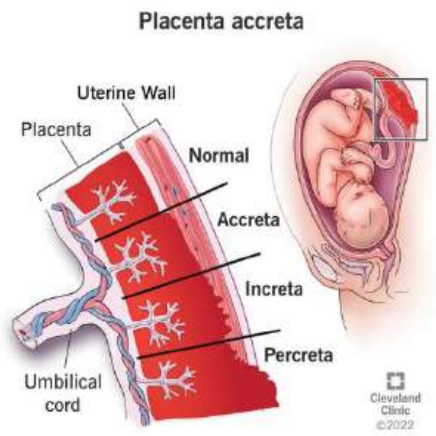
- If fertilisation occurs, the blastocyst will implant in the endometrium and release hormones to sustain the corpus luteum
    - ➢ human chorionic gonadotropin (hCG) prevents the corpus luteum from degrading, thus stimulating progesterone and maintaining a thickened endometrium.
    - ➢ After placenta develops it takes over the production of progesterone and oestrogen, so the corpus luteum degrades.
- If fertilisation doesn't occur, the corpus luteum eventually degenerates (forming a corpus albicans after ~ 2 weeks)
- When the corpus luteum degenerates, estrogen and progesterone levels drop and the endometrium can no longer be maintained
- The endometrial layer is sloughed away and eliminated from the body as menstrual blood
- As estrogen and progesterone levels are too now low to inhibit the anterior pituitary, the cycle can now begin again

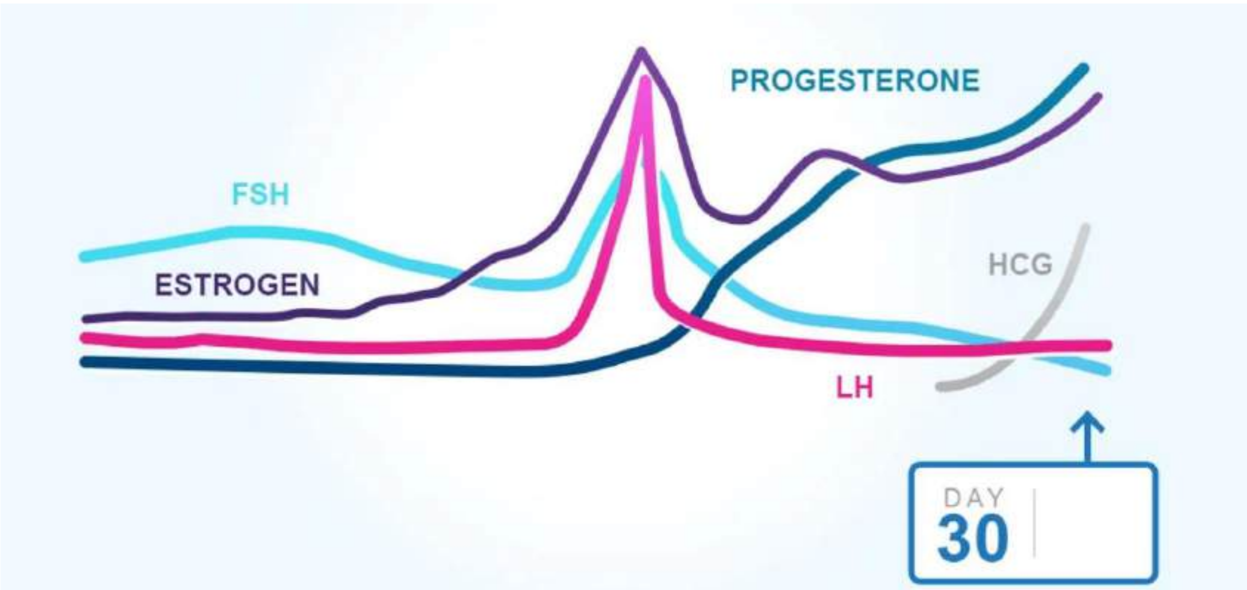



**During pregnancy**

- Embryos produce human chorionic gonadotropin (hCG)
    - hCG prevents the corpus luteum from degrading, thus stimulating progesterone and maintaining a thickened endometrium.

- After placenta develops it takes over the production of progesterone and oestrogen, so the corpus luteum degrades.
- Placenta
  - formed by maternal and foetal tissue
  - each side produces villi (increases the SA)
    - embedded in the uterine wall and connected to the foetus by an umbilical cord.
  - placenta transports blood and nutrients from mother to foetus and removes $CO_2$ and urea and passes them to the mother to excrete.
    - these diffuse across the narrow space between the villi - no blood comes in contact ∴ no blood transfer between mother and child




**During birth**

"For birth to occur the uterus must contract and the cervix must soften."

- **Prostaglandins** are secreted by the uterine wall and initiate labour
  - eg: **oxytocin** allows for contractions of the uterus & softening of cervix for baby to come out
- **Estrogen** and **progesterone** decrease → allows for more intense contractions
- Contractions lead to release of **beta-endorphins** to give mother natural pain relief

- As baby is born **adrenaline** is released to push it out
- Placenta expelled once baby is born through secretion of **oxytocin**
- Decline in **Prolactin** post birth produces milk

evaluate the impact of scientific knowledge on the manipulation of plant and animal reproduction in agriculture

| **Scientific knowledge** | **Before science knowledge was acquired** | **Impact of new scientific knowledge** |
|---|---|---|
| Chromosomal theory of inheritance: Sutton and Boveri proposed that chromosomes exist in pairs, inherited from each parent (1900s) | Physical 'factors' inherited from parents were unknown | Provided basis for understanding and control of outcomes of **artificial pollination** and **artificial insemination** |
| Structure of DNA discovered by Waston and Crick (1953) | Little understanding of DNA replication and how genetic code is 'read' | > Understanding of transcription and translation, and how mutations causes disease and allow evolution<br>> Basis of **cloning** |
| DNA sequencing invented (1970s) | Not possible to 'see' a DNA sequence | > Helped development of **recombinant DNA** technology<br>>Identification of alleles responsible for traits |
| Development of recombinant DNA technology (1970s) | DNA could not be transferred between species | > Genes can be transferred between species to create a **transgenic organism**.<br>> Widespread in plant agriculture |

**Selective Breeding (Animals)**

1. Farmer selects strongest male (bull) in the herd
2. This male will inseminate the females

| **Advantages** | **Disadvantages** |
|---|---|

| | |
|---|---|
| • increased yield<br>• cost-efficient<br>• does not involve biotechnology | • herd susceptible to environmental pressures and disease<br>• limits gene pool → lower diversity<br>    ○ same bull used |

**Cloning (Animals)**

1. Somatic cell removed from donor sheep (this is sheep that will be cloned)
2. Egg cell removed from a different donor sheep
3. Nucleus is removed from both cells
4. Both nucleus' are fused together in a culture through electric shocks
    a. egg cell has some DNA from somatic cell containing full set of chromosomes
5. Egg is inserted into a third surrogate
    a. The sheep that was born was identical to the first donor sheep.

| Advantages | Disadvantages |
|---|---|
| • guaranteed expression of desired trait<br>• cloned individuals have identical requirements | • expensive<br>• ethical concerns<br>• no adaptability<br>• low success rate in animals |

**Cloning (Plants)**

1. Cut on rootstock and matching cut on scion
2. Cut surfaces are fit together
3. The grafted area is held together
4. The scion grows producing desired plant characteristics

| Advantages | Disadvantages |
|---|---|
| • Guaranteed to express desired trait<br>• Can produce fruit more quickly | • Offspring genetically identical<br>• Requires high maintenance |

**Transgenesis (Animals)**

1. Desired gene is cut from DNA of selected organism
2. This transgene is held in recombinant plasmid
3. The gene is inject into the zygote of another species via gene guns
4. Zygote + transgene is implanted into surrogate mother
5. Offspring expresses the transgene in their phenotype

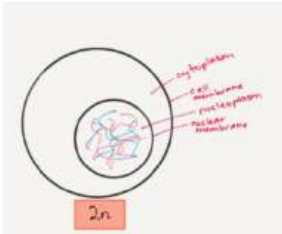
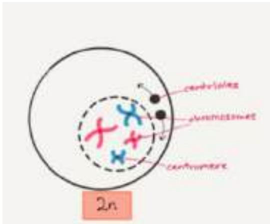
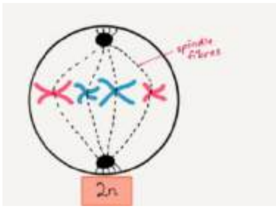
| Advantages | Disadvantages |
|---|---|
| • guaranteed expression of desired trait<br>• increases yield<br>• increases genetic diversity<br>  ○ if transgenic organism is not cloned | • unknown side effects on humans<br>• mixing of modified organism with wild animals → possible harm<br>• Trade issues |

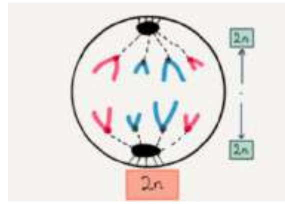
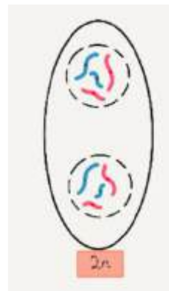
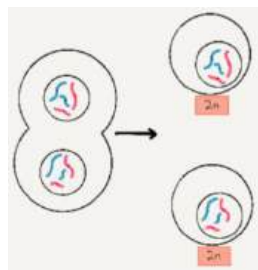
**Cell replication**
**Inquiry question: How important is it for genetic material to be replicated exactly?**

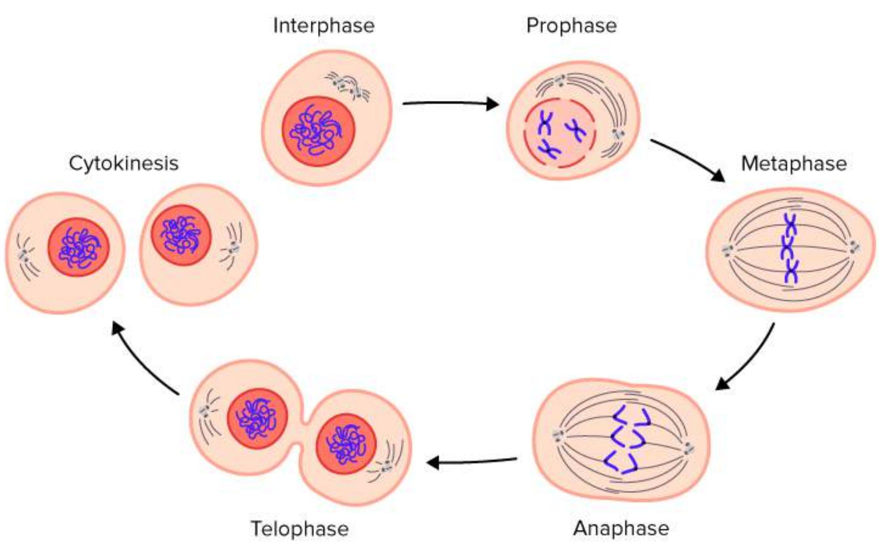
model the processes involved in cell replication, including but not limited to:

- ➔ mitosis and meiosis

**Mitosis:**
**Location:** Somatic (body) cells.
**Result:** Diploid (2n) parent cell creates two diploid daughter cells.
**Function:** Growth, repair, maintenance.

| | |
|---|---|
| **Interphase**<br>• DNA replicates forming chromatin (mixture of DNA and proteins that form the chromosomes)<br>  ○ thin and coiled together |  |
| **Prophase**<br>• Duplicated chromosomes condense<br>• Spindle fibres form at either end of the dividing cell. These fibres lengthen and shorten to pull chromotids apart<br>• Nuclear envelope breaks down |  |
| **Metaphase**<br>• Pairs of condensed chromosomes (sister chromosomes) line up at the equator of the cell |  |

| | |
|---|---|
| **Anaphase**<br>● Spindle fibres draw chromatids to opposite ends of the cell |  |
| **Telophase/Cytokinesis**<br>● Two new nuclear envelopes begin to form around the separated sister chromatids |  |
| **Cytokinesis**<br>● Cytoplasm constricts at the centre of the cell to divide the daughter nuclei<br>○ cleavage occurs producing two genetically identical diploid daughter cells |  |

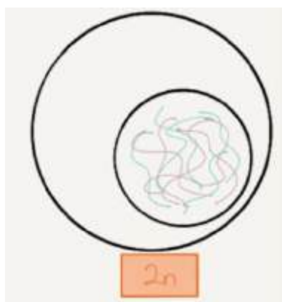
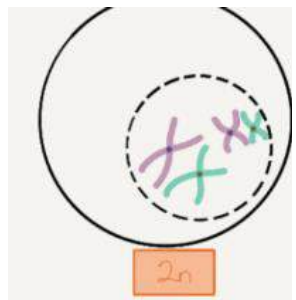
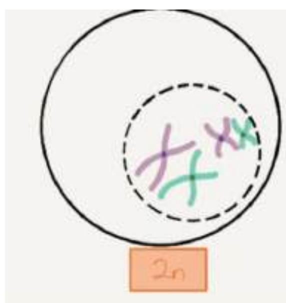
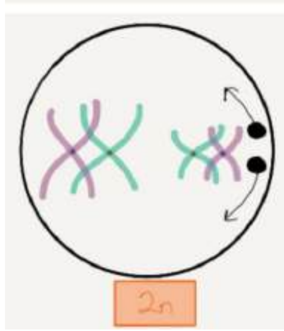
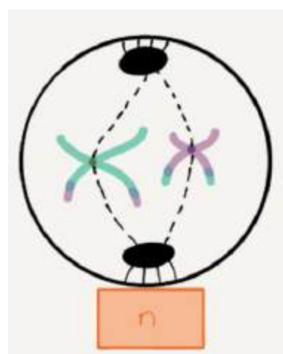


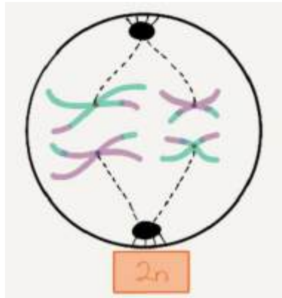
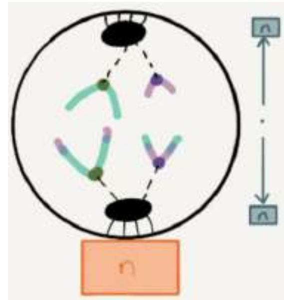
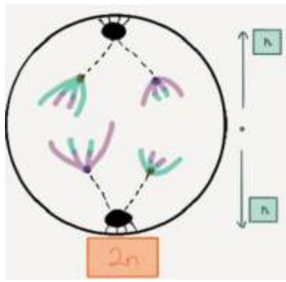
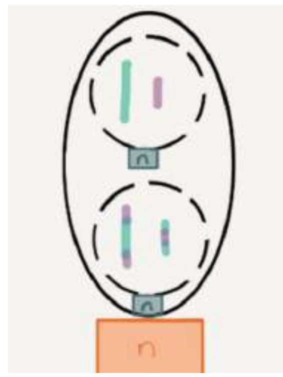
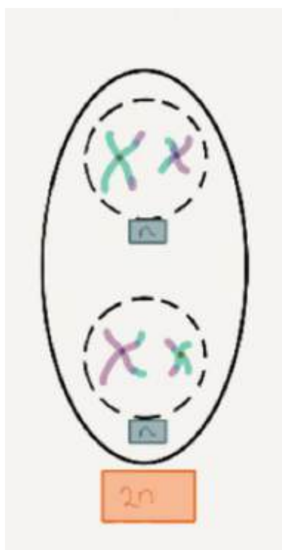
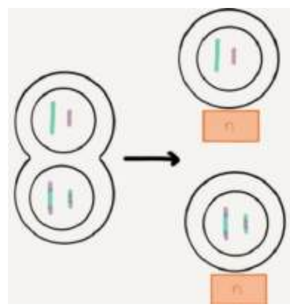

**Meiosis:**
**Location:** Germline (sex) cells.

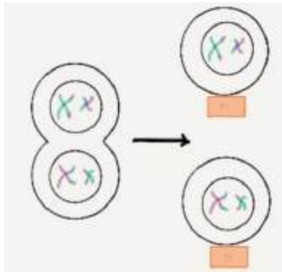
**Result:** Diploid parent cell creates 4 haploid (n) daughter cells.

- **Meiosis I:** Reduction division. Creates haploid cells from diploid cells.
- **Meiosis II:** Maintains haploid number in daughter cells.

**Function:** Gamete production for sexual reproduction.

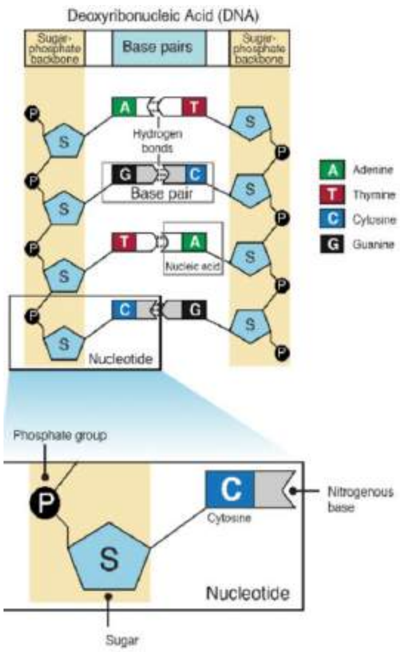
| **Meiosis I** | | | **Meiosis II** |
|---|---|---|---|
| **Interphase**<br>• DNA replicates forming chromatin (mixture of DNA and proteins that form the chromosomes)<br>○ thin and coiled together |  |  | **Prophase**<br>• Chromosomes condense and nuclear envelope breaks down |
| **Prophase**<br>• Duplicated chromosomes condense and nuclear envelope breaks down<br>• Homologous chromosomes pair up, aligning next to eachother<br>• Crossing over occurs where segments of DNA at the same locus swap to create new gene combinations |  |  | **Metaphase**<br>• Chromosomes line up at equator of the dividing cells |

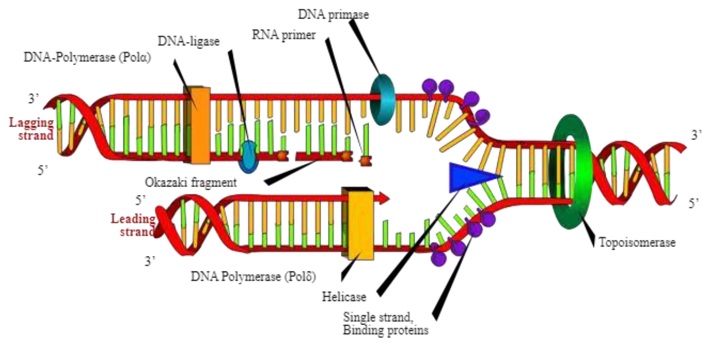
| | | | |
|---|---|---|---|
| **Metaphase**<br>• Homologous pairs line up at the equator of the cell |  |  | **Anaphase**<br>• Sister chromotids are separated by the spindle fibres and pulled to either ends of the cell |
| **Anaphase**<br>• Homologous pairs are separated to either ends of the cell |  |  | **Telophase**<br>• Nuclear division completed<br>• 2 haploid non-identical nuclei are formed per daughter cell. |
| **Telophase**<br>• Nuclear division completed<br>• 2 haploid non-identical nuclei are formed |  |  | **Cytokinesis**<br>• Cell pinched in half and cells cleave<br>  ○ 4 haploid, non-identical gametes are formed |

| | | | |
|---|---|---|---|
| **Cytokinesis**<br>• Cell pinched in half and cells cleave<br>  ○ 2 haploid non-identical cells formed |  | | |

➔ DNA replication using the Watson and Crick DNA model, including nucleotide composition, pairing and bonding

- DNA is a double stranded, helical molecule
- There are two sugar phosphate backbones on the outside, which are held together by hydrogen bonds between pairs of complementary nitrogenous bases on the inside
    - A=T (2 Hydrogen bonds)
    - G≡C (3 Hydrogen bonds)
- Each base combines with a sugar molecule (deoxyribose) and a phosphate group to form a nucleotide
- The sugar phosphate backbones are antiparallel




- DNA replication is the process by which DNA is copied and occurs in interphase
- DNA replication is a semi-conservative process, because when a new double-stranded molecule is formed:
    - ➢ One strand will be from the original molecule
    - ➢ One strand will be newly synthesised
- Each strand has a 5’ prime end and a 3’ prime end
    - ➢ Two strands run in opposite directions
    - ➢ Determinants how each DNA strand is replicated

1. Topiomerase uncoils the double stranded DNA always working ahead of the replication fork preventing supercoiling
2. Helicase separates the two strands by breaking hydrogen bonds creating a replication fork and single-stranded binding proteins prevent the two separated strands from reannealing
    - ➢ The separated strands provide a template for creating a new strand of DNA
3. Primase starts the process by adding an RNA primer (starting point) at the 5’ prime end to the 3’ end
4. DNA polymerase III binds to the primer and moves from the 5’ to the 3’ end adding floating nucleotides from the nucleolus to the strand (leading strand)

- ➢ The lagging strand runs in the opposite direction hence the strand is made in Okazaki fragments
- ➢ Each fragment starts with an RNA primer where DNA polymerase III adds a short row of nucleotides at the 5’ to the 3’ prime end
- ➢ The next primer is then added further down the lagging strand where another Okazaki fragment is made
- ➢ This is repeated until the strand is complete

1. Once the DNA is made, DNA polymerase I removes all primers filling in the gaps with bases.
2. The gaps are sealed with ligase
3. DNA polymerase II has an editing function, but no exonuclease activity
4. Synthesised leading and lagging strands twist into their helical forms as replication fork continues down strand

assess the effect of the cell replication processes on the continuity of species

| **Mitosis: Growth & Repair** | **Meiosis: Gamete production** |
| --- | --- |
| • Cell replication needs to result in two daughter cells with identical information<br>• This is because the genetic information contains genes that code for proteins<br>• Proteins play important structural, enzymatic and hormonal roles in the body<br>• If the genetic sequence is altered during replication, the structure and function of the proteins may be affected | Gametes are genetically unique, increasing genetic variation and the likelihood that offspring will reach sexual maturity. Subsequently, offspring with hybrid vigour are produced, who are far more likely to reach sexual maturity and hence ensure species continuity. |

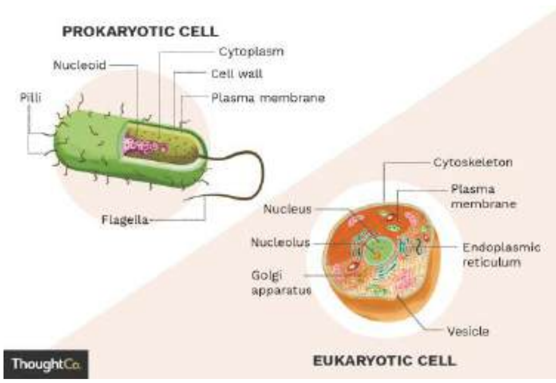
**DNA and Polypeptide Synthesis**
**Inquiry question: Why is polypeptide synthesis important?**

Construct appropriate representations to model and compare the forms in which DNA exists in eukaryotes and prokaryotes

**General comparison**

| | **Prokaryotes** | **Eukaryotes** |
| --- | --- | --- |

| | | |
|---|---|---|
| Outer cell structures | Cell membrane<br>Cell wall always present | Cell membrane<br>Cell wall depends on the kingdom |
| Internal structures | Cytosol<br>No membrane-bound organellelles<br>Lack a nucleus, has nucleoid | Cytoplasm<br>Membrane bound organelles<br>Nucleus |
| Size | Small, 1-10 micrometres | Large, 10-100 micrometres |
| Single or multi-celled | Unicellular | Unicellular or multicellular |




**DNA in prokaryotes and eukaryotes**
The nucleotides and their functions are the same in all organisms
Differences include its packaging, quantity, replication…

| | **Prokaryotes** | **Eukaryotes** |
|---|---|---|
| **Location of chromosomal DNA** | Nucleoid | Nucleus |
| **Chromosomal quantity** | One circular chromosome | Multiple linear chromosomes; number depends on species<br>Humans: 46 |
| DNA packaging | DNA packaged into a single circular chromosome<br><br>No histone proteins | DNA packaged into linear chromosomes<br><br>DNA tightly wrapped around histone proteins to form nucleotides |
| **Additional DNA** | Plasmids | No plasmids<br><br>Mitochondrial and chloroplast |
| DNA replication | Single replication origin | Multiple replication origins |

| | | |
|---|---|---|
| | 2000 base pairs per second<br><br>Chromosomal and plasmid DNA replicate independently | 100 base pairs per second<br><br>Chromosomal, mitochondrial and chloroplast DNA replicate independently |
| Coding DNA | Fewer genes than eukaryotes<br><br>No introns (non-coding DNA)<br><br>Coding DNA is organised into operons (gene clusters) | More genes than prokaryotes<br><br>Exons (coding DNA) separated by introns, allowing for gene splicing and multiple protein products per gene |

Model the process of polypeptide synthesis, including:

- ➔ transcription and translation

- **Peptides** are short chains of 2-50 amino acids
- **Polypeptides** are linear molecules, made up of multiple peptides
- **Proteins** are the functional unit, made up of one or more polypeptides

Amino acids are the building blocks of proteins and can be obtained by food or produced by our bodies

**Transcription**: The process in which genetic code is copied into a single-stranded RNA molecule called messenger RNA (mRNA) in the nucleus.

- DNA > mRNA

**Translation:** The process of converting mRNA sequence into a specific sequence of amino acids (polypeptide) where it's carried by tRNA in the ribosomes.

- mRNA > protein

**mRNA:** single-stranded nucleic acid, consisting of ribose sugar, phosphate backbone and nitrogen bases

**tRNA:** small RNA molecule that transfers specific amino acids to the ribosome during the formation of polypeptides

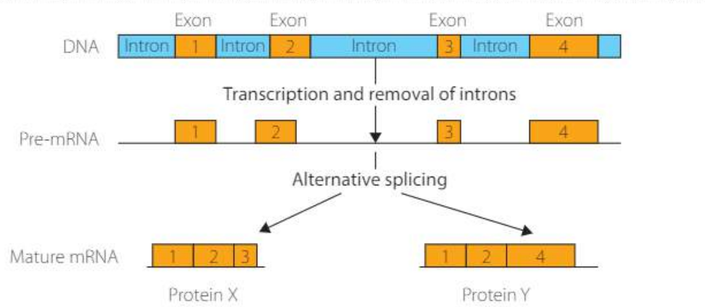
**Codon:** set of three nitrogen bases in mRNA

**Anticodon:** complementary set of three nitrogen bases on tRNA

**Transcription (nucleus)**

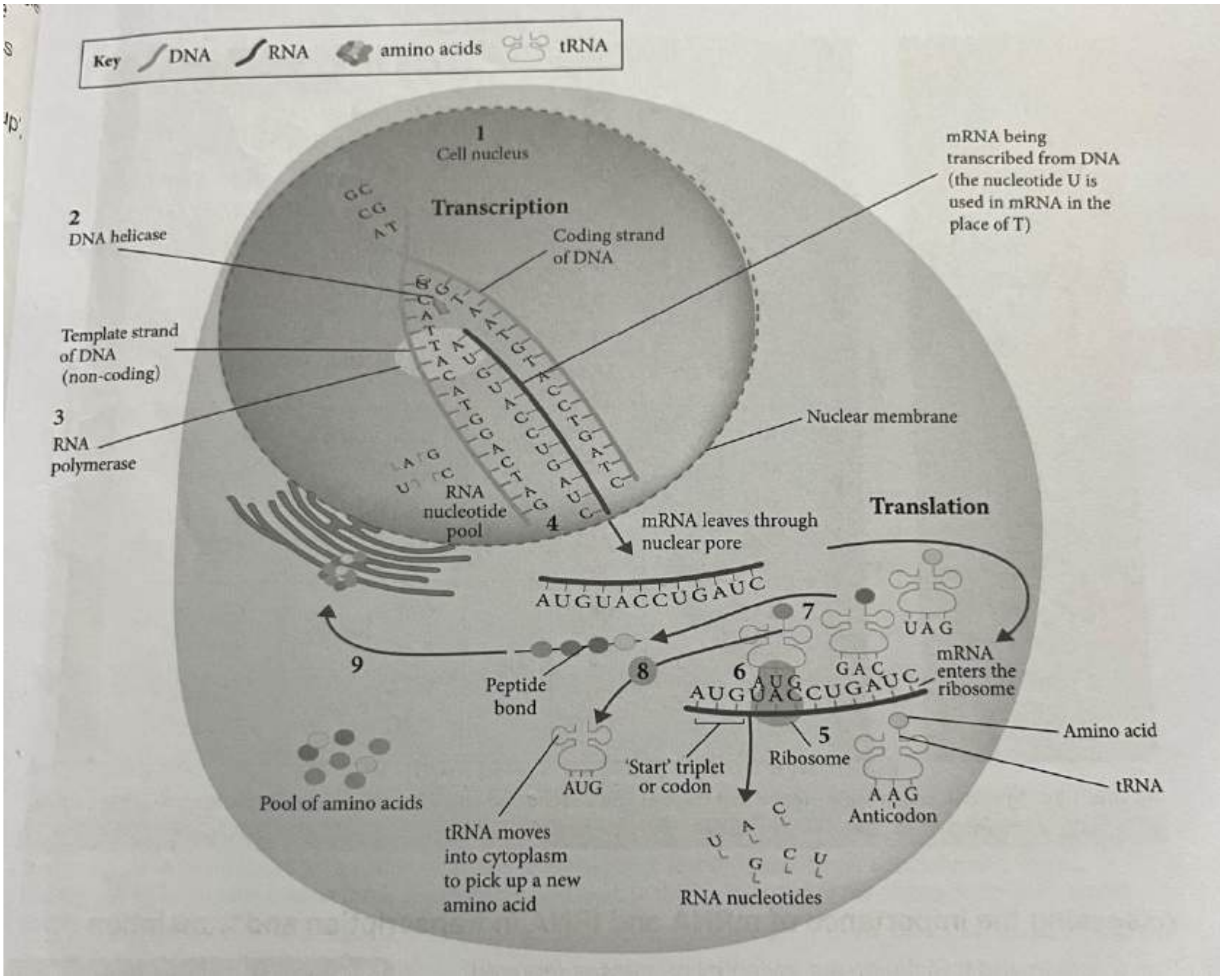
1. A cell is stimulated to produce a specific protein

2. DNA helicase unwinds and unzips the gene responsible for coding the protein. This exposes the bases of the coding strand and non-coding strand
3. RNA polymerase binds to the promotor at the start of the gene on the non-coding strand and runs the length of the gene, adding free nucleotides as it proceeds, until it reaches a stop codon. Both exons and introns are transcribed into a single-stranded pre-mRNA sequence. The pre-mRNA contains uracil in place of thymine.
4. **RNA processing:** The introns are removed from the pre-mRNA and the exons are bound back together to form the final mRNA, in a process called splicing. Alternate splicing, which can also remove exons, produces combinations of proteins from the same gene, The final mRNA sequence exits the nucleus through a nuclear pore into the cytoplasm.




**Translation (ribosomes)**

5. The mRNA transfers to the ribosome. It passes through the ribosome, and the nucleotides are read in sets of three (codons), always starting with a start codon (AUG).
6. Transfer RNA (tRNA) binds with the condon via a complementary anticodon sequence.
7. The other end of each tRNA is bound by a specific amino acid, which will be incorporated into the growing polypeptide chain. The amino acids are linked by a peptide bond.
8. Each tRNA molecule can be recycled after it has delivered its amino acid.
9. When a stop codon is reached, the polypeptide chain terminates and the polypeptide is folded into a specific 3D shape to form a functional protein. The mRNA can then be used to produce more of the same protein.




➔ assessing the importance of mRNA and tRNA in transcription and translation

| mRNA | tRNA |
|---|---|
| I - mRNA is a single-stranded nucleic acid<br>D - mRNA consists of a ribose sugar, phosphate backbone and nitrogen bases (A, U, G, C)<br>E - DNA does not leave the nucleus. As a result, a message must be sent from the nucleus to the ribosomes where protein synthesis occurs. This message is sent in the form of mRNA<br>A - Protein synthesis would therefore not be possible without mRNA | I - tRNA is a small single-stranded nucleic acid<br>D - The tRNA molecule has a distinctive folded three-loop structure. One of the loops contains an anticodon; a sequence of 3 nucleotides, complementary to a corresponding sequence on the mRNA. Each tRNA has a corresponding amino acid attached to the opposite end of the anticodon<br>E - As a result of the complementary nature of the mRNA and tRNA codon-anticodon complex the specific sequence of amino acids required for |

| | |
|---|---|
| | protein synthesis occurs<br>A - Protein synthesis would therefore not be possible without tRNA. |

**ChatGPT good answer:**

Messenger RNA (mRNA) is a single-stranded molecule that carries complementary bases to a specific DNA sequence, coding for a protein. Since DNA remains in the nucleus, mRNA serves as a crucial messenger that transmits the genetic information needed for protein synthesis to the ribosomes. Without mRNA, the synthesis of proteins would be impossible, leading to the failure of essential biological processes. Additionally, transfer RNA (tRNA) plays a vital role by bringing the appropriate amino acids to the ribosome based on the mRNA codons. Without these processes, organisms would not be able to produce proteins necessary for growth, repair, and overall function, ultimately meaning that life as we know it would not exist.

➔ analysing the function and importance of polypeptide synthesis

It is important because it requires the accurate translation of the DNA code into a functional protein. These proteins have important structural, enzymatic and hormonal roles within cells. In the context of disease, the process will incorporate a genetic mutation into a polypeptide, even if it has the potential to produce a non-functional protein.

➔ assessing how genes and environment affect phenotypic expression

**Genotype:** the genetic makeup of an organism
**Phenotype:** the observable physical properties

Genes carry the set of instructions for all proteins necessary for growth and development.
**Gene expression:** the process by which information from a gene is used in the synthesis of a functional gene product (protein)

**Not influenced by environment:**
- Blood type
- Cystic fibrosis
- Haemophilia

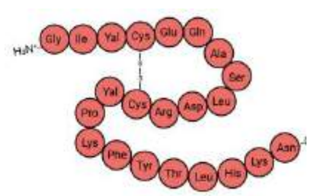
**Influenced by environment**
- Breast cancer
- Schizophrenia
- Hypertension

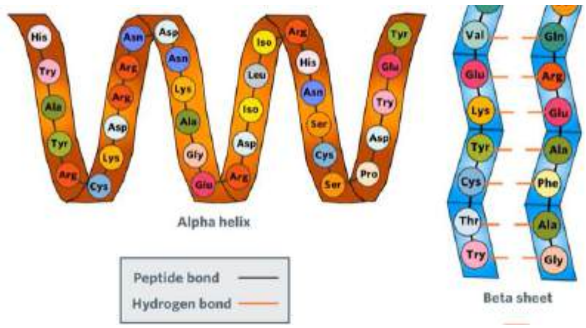
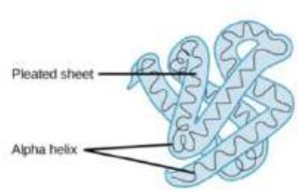
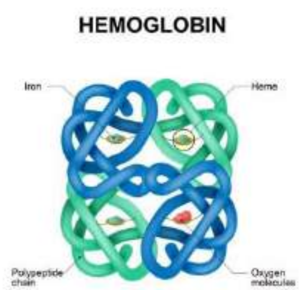
★ Flower colour is determined by soil pH. The pH effects the availability of other ions in the soil and these ions are responsible for the colour change.
  - ➢ pH<5 = acidic soil = blue flowers
  - ➢ pH> or equal to = 7 = neutral & alkaline soil = pink flowers

**I -** The phenotype is the physical expression of a trait, as a protein or physically observable characteristic.
**D -** The phenotype is determined by the genotype and can be influenced by the environment, depending on the gene in question. For example, blood type is determined solely by the genotype, while height and weight is influenced by both genotype and environment.
E/A **-** In conclusion, the genotype always plays a role in determining the phenotype while the environment also influences the expression of many genes.

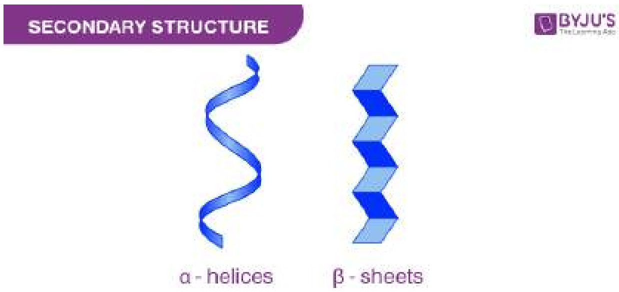
investigate the structure and function of proteins in living things

**Levels of structure**

| Protein Shape | Features |
| --- | --- |
| Primary | ● Linear sequence of amino acids that make up a polypeptide chain<br>○ different R chain for each amino acid<br>➢ Each amino acid has a unique R group, distinguishing it from other amino acids<br>  |
| Secondary | ● Regular, repeated patterns of folding of the protein backbone<br>● Due to interactions between amino acids<br>● Hydrogen bonds form between oxygen atoms and hydrogen atoms within the backbone, causing the polypeptide chain to coil or pleat<br>● Bonds cause polypeptide to fold into:<br>➢ Alpha helix: forms when the polypeptide backbone coils around a helical axis in a clockwise direction<br>➢ Beta sheet: segments of the polypeptide chain line up next to each other and hydrogen bonds form between closely aligned amino acids |

| | |
|---|---|
| |  |
| Tertiary | ● pleated sheets and helix bond through **disulphide bonds**<br>● r chains may form **ionic bonds** if they have opposite charges<br>● bonds cause folding, giving the protein its 3D structure<br> |
| Quaternary | ● combination of 2 or more polypeptides to form a single protein<br>○ eg: haemoglobin → 2 alpha and 2 beta chains<br> |

| | |
|---|---|
| *Primary* | ● single chain of amino acid |
| *Secondary* | ● Polypeptide chain arranged in helix or pleats |
| *Tertiary* | ● 3D shape formed by folding the secondary structure |
| *Quartanery* | ● Two or more polypeptide chains combines |




**Protein function**
**The structure of the protein is always essential to its function**

- Proteins can be classified as globular (have an irregular amino acid sequence and a spherical shape determined by the tertiary or quaternary structure) or fibrosis (thin, elongated molecules made from repetitive amino acid sequences that give rise to a secondary structure only)
- **Globular:**
    - transport: involved in either the active or passive movement of substances across membranes or between cells - haemoglobin - transport of 02 around the body
    - enzymes: speeds up reactions by lowering activation energy and folds at active site to match shape of substrate - lactase - breaks down milk
    - hormones: globular shape to bind target cells - insulin - insulin travels through blood stimulating muscle, fat and liver cells to take up and store the glucose as glycogen

**Fibrosis:**

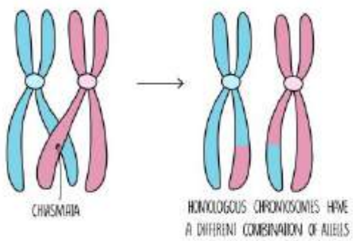
- Actin and myosin which are involved in muscular contractions, and collagen, the most abundant protein in the human body, which gives skin its elasticity

**Genetic Variation**
**Inquiry question: How can the genetic similarities and differences within and between species be compared?**

Conduct practical investigations to predict variations in the genotype of offspring by modelling meiosis, including the crossing over of homologous chromosomes, fertilisation and mutations

**Genes:** sections of chromosomes that code for the same trait
**Allele:** different versions of the same gene
**Homozygous:** the genes from both parent have the same allele
**Heterozygous:** the gene from each parent has a different allele

| **Process** | **How genotype variation is produced** |
|---|---|

| | |
|---|---|
| **Meiosis - crossing over - prophase**<br> | > alleles are exchanged between non-sister chromotids on homologous chromosomes<br>> this results in new combinations of alleles on a chromosome<br>> this produces new variation of genotypes upon fertilisation |
| Independent assortment - metaphase I | > Genes and alleles for different traits are inherited independently of each other.<br>> Chromosomes align randomly at the metaphase plate, leading to different combinations.<br>> This creates genetic variation in offspring after fertilization. |
| Random segregation - anaphase II | > alleles of the same gene separate randomly and equaly into daughter cells<br>> this produces new variation of genotypes at loci across chromosomes upon fertilisation |
| **Fertilisation** | > the union of a male and female gamete<br>> in a human ejaculate there is approximately 300 million sperm<br>> each of these sperm is genetically unique: Crossing over, independent assortment, random segregation<br>> only one sperm will fertilise the egg<br>> the successfully sperm is primarily determined by chance |
| **Mutation** | > a change in the DNA sequence<br>> a mutation can occur during meiosis where it can be inherited, leading to the formation of new alleles<br>> mutations are essential for evolution |

Model the formation of new combinations of genotypes produced during meiosis, including but not limited to:

- ➔ interpreting examples of autosomal, sex-linkage, co-dominance, incomplete dominance and multiple alleles
- ➔ constructing and interpreting information and data from pedigrees and Punnett squares

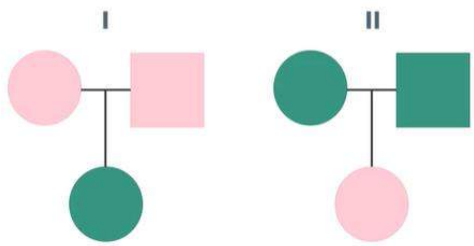
TT = Homozygous dominant

tt = homozygous recessive
Tt = Heterozygous

Autosomal inheritance: transmission of genes located on the autosomes

**Pedigrees**
- Determine the type of inheritance shown in the lineage
- Determine the genotypes of individuals in the lineage
- Predict the likelihood of offspring having the trait

I: If the offspring has condition but parents don't → recessive
II: If the offspring does not have it but parents do → dominant




**Sex linkage**
- Refers to genes that are located on the sex chromosomes
- Most sex-linked characteristics are found on the X chromosome and are recessive
- Normal 3:1 Mendelian ratio is not observed

**Example:** Haemophilia - a recessive sex-linked condition found on the X chromosome
Possibilities:
Females → $X^HX^H$ - no haemophilia, $X^HX^h$ - no haemophilia, but a carrier, $X^hX^h$ - haemophilia
Males → $X^HY$ - no haemophilia, $X^hY$ - haemophilia (cannot be carrier as only 1X chromosome)

Female carrier & unaffected male

| | $X^H$ | $X^h$ |
|---|---|---|
| $X^H$ | $X^HX^H$ | $X^HX^h$ |
| Y | $X^HY$ | $X^hY$ |

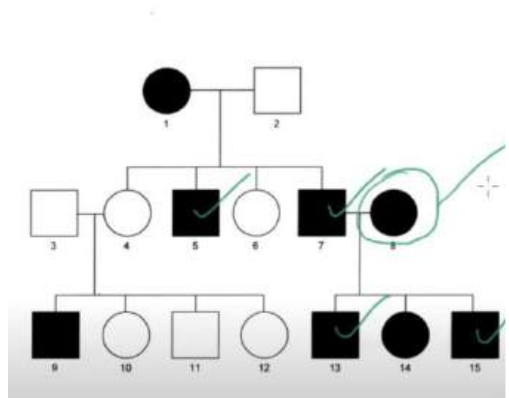
**How to determine if a trait is autosomal or sex-linked**
1. Determine if trait is dominant or recessive
2. Use these rules:
- Definitely sex linked if
    - males are affected more

- if mother has trait then the sons should have it
- There is no male to male transmission

3. If these rules don't work, then it is autosomal

**Example 1:**
- recessive trait
- males are affected more
- sons of mother with trait have it
  - male to male transmission dismissed
- female only has trait because father has it

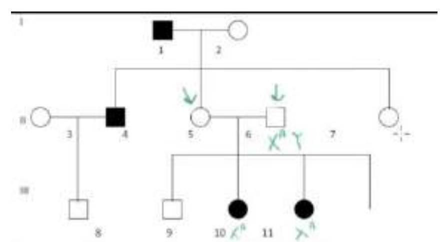
∴ trait is sex-linked



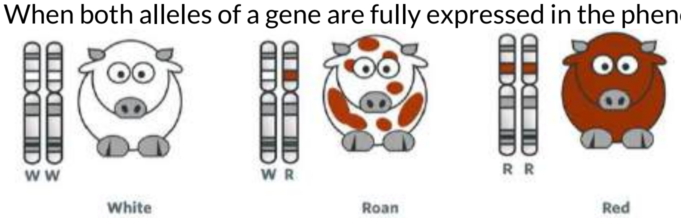
**Example 2:**
- recessive trait

Let A = healthy & a = affected
- father is healthy but daughters have trait
  (cannot happen for sex-linked as daughter will be passed the healthy trait)

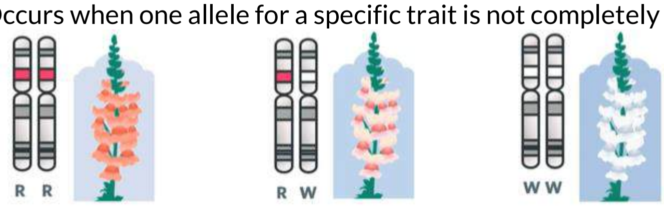
∴ trait is autosomal



**Codominance**
When both alleles of a gene are fully expressed in the phenotype of a heterozygote




**Incomplete dominance**
Occurs when one allele for a specific trait is not completely expressed over its paired allele




**How to determine between codominance and incomplete dominance**
- Three different phenotypes present
- If both parents are heterozygous, offspring can have any one of three phenotypes

    - If each parent is homozygous for the different alleles, then all offspring will be heterozygous and show the intermediate phenotype

**Multiple alleles**

- Occurs when there are three or more possible alleles for a gene. While the genotype of an individual will only ever have two alleles at a locus, there is a higher number of possible genotypes (hence phenotypes) at the locus
- For example, the ABO blood group is also an example of multiple allele inheritance
    - ★ $I^A, I^B$, i

Collect, record and present data to represent frequencies of characteristics in a population, in order to identify trends, patterns, relationships and limitations in data, for example:

- ➔ examining frequency data

**Calculating allele frequency from alleles**

**Allele frequency** is the relative proportion of a particular allele in a population. So, it can be represented using the following equation:

$$\text{Allele frequency} = \frac{\text{Number of an allele in a population}}{\text{Total number of alleles in the population}}$$

**Calculating allele frequency from phenotypes**

- Taking a large enough sample, will be likely to give accurate representation of the rest of the population.
- So, we can just look at a proportion of individuals from a population, and use them to work out the allele frequencies for the whole population.
- We can avoid genome sequencing and work out allele frequencies by measuring the frequencies of observable traits
- This can save a lot of work, but it only works for traits which are affected by one gene−−like feather colour in Andalusian chickens.
- Also, we need to know how each allele affects an observable trait−−like how having two b alleles makes Andalusian chickens white, but two B alleles make them black and so on.

| | Black | Blue | White |
|---|---|---|---|
| **Genotype** | BB | Bb | bb |
| **Number of chickens with genotype** | 36 | 48 | 16 |
| **Number of B alleles** | 72 | 48 | 0 |
| **Number of b alleles** | 0 | 48 | 32 |

**Well, the Hardy-Weinberg model describes and predicts allele frequencies in a non-evolving population. It does so through two components:**

1. The Hardy-Weinberg equilibrium
    - states that allele frequencies in a population will stay stable from one generation to the next, in the absence of evolutionary processes
2. The Hardy-Weinberg equation
    - used to calculate genotype frequencies in a population, based on given allele frequencies

The Hardy-Weinberg equation is given by $p^2 + 2pq + q^2 = 1$, where $p$ is the frequency of the dominant allele and $q$ is the frequency of the recessive allele.

The three terms in the Hardy-Weinberg equation represent the frequencies of the three genotypes:

- $p^2$ = frequency of the **homozygous** dominant genotype.
- $2pq$ = frequency of the **heterozygous** genotype.
- $q^2$ = frequency of the homozygous recessive genotype.

Imagine a population of 100 Andalusian chickens, where the allele frequency for B is 0.6 and the allele frequency for b is 0.4. From this information, we can calculate the frequency of each of the three genotypes (and so, phenotypes).

The **homozygous dominant phenotype** is black feathers, which is given by the genotype BB:

- To determine the genotype frequency, $p^2 = (0.6)^2 = 0.36$.
- From this, we can determine that the proportion of the population with black feathers is $0.36 \times 100 = 36$ chickens.
- Based on similar logic, the genotype frequency of bb is $q^2 = (0.4)^2 = 0.16$, and the number of white chickens is $0.16 \times 100 = 16$ chickens.

**Predicting speciation**

- **Natural selection** is a mechanism for evolution, which explains how individuals with traits best suited to their environment are most likely to survive, reproduce and pass on their alleles to the next generation. The Hardy-Weinberg equation doesn't consider the environment at all
- The Hardy-Weinberg model doesn't account for the fact that a **mutation** might occur. When a mutation occurs in the gene of interest, it essentially creates a new allele which doesn't fit into the equation!
- When allele frequencies in a population aren't following the Hardy-Weinberg equilibrium, we can assume that evolutionary processes are occurring.
- On top of this, by comparing the allele frequencies for a particular gene between two populations, we might notice that they're different. This would suggest that the populations are in the process of speciation

- analysing single nucleotide polymorphism (SNP)
    - Is a point mutation present in more than 1% of the pop
    - Just like alleles of a gene, each individual has two copies of each SNP, one on each homologous chromosome
    - SNPs are the most common type of genetic variation between indiviudals
    - The human genome contains a SNP every 1000 nucleotides, which means there are about 4-5 million SNPs in a single persons genome
    - Most SNPs are found in the introns therefore having little effect on cellular function
    - Exon SNPs have a more significant impact, including the development of diseases
    - These SNPs are biological markers that help scientists to identify alleles associated with specific diseases
    - They can also be used to determine ancestry by comparing the frequency of every SNP in a persons genome with a database of SNPs from geographical populations around the world
        - For example, a particular SNP associated with curly hair might be shared among individuals with curly hair.

**Genome-wide association studies (GWAS)**

- SNPs act as chromosomal tags to specifc regions of DNA, and these regions can be scanned for variations that may be linked to diseases or disorders
- GWAS rapidly scans for SNP markers across the genomes of individuals with a known disease or disorder and compare them to 'control' individuals
- Significant differences in allele frequency caused by SNPs are the first step in identifying cause and effect relationship between SNP and a disease or disorder. Once a link is made, scientists look to develop better treatments and diagnostic and prevention strategies
- GWAS have identified SNPs related to conditions including cancer, diabetes, heart disease…

**Inheritance Patterns in a Population**
**Inquiry question: Can population genetic patterns be predicted with any accuracy?**

investigate the use of technologies to determine inheritance patterns in a population using, for example:

➔ DNA sequencing and profiling

**DNA sequencing** is the process of determining the precise order of nucleotides within a segment of DNA. It includes any method or tech used to determine the specifc order of A, T, G, C in a strand of DNA

**DNA profiling** Involves analysing specific regions of the DNA that vary greatly between individuals DNA that contains short tandem repeats (STRs), located in the introns for the purpose of identification

- If focus on a single STR, even though everyone has the same repeating units (e.g. CGA), the number of units varies between people resulting in different lengths
- Every person has 2 alleles of each STR
- These different lengths are useful for making profiles

The exact number of STRs varies from person to person, because we inherit our DNA from our parents. DNA profiles can be used to:

- Confirm how closely related individuals are
- Trace inheritance patterns
- Solve crimes

PCR → technique used to exponentially amplify large numbers of copies of a specific sequence of DNA

- The sample can be used for gene cloning, DNA sequencing or DNA profiling
- PCR consists of a DNA sample, primers (short single-stranded pieces of DNA, usually 18-22 bases long) that are complmenetary to either end of the DNA region to be amplified, free nucleotides, Taq polymerase, magnesium chloride which taq requires to be an active enzyme

1. The ingredients are placed into a PCR tube, which is then inserted into a thermocycler machine
2. Denaturation - DNA strands separate as hydrogen bonds are broken due to heat (95 degree 1 min)
3. Annealing - DNA primers aneal to the 3' ends of the target sequence (55 degree for 1min)
4. Elongnation - Taq polymerase binds to the primer and synthesises new DNA strands (72 degree for 2 min)
5. This is repeated 30-40 times resulting in the exponential amplification of a specific region of DNA where it can be used as an insert to clone a gene, used as a marker in DNA profiling studies, or sequenced to identify SNP or genetic mutations

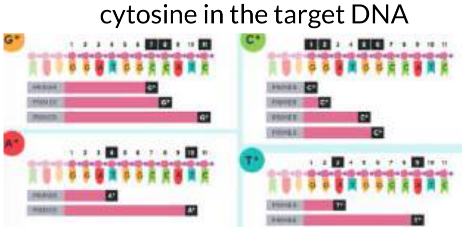
(picture)

**(Agarose) Gel electrophoresis** → technique that separates segments of DNA based on their size (length)

- The molecules (which contain a charge) are pushed by an electrical field through a gel
- The smaller the molecule, the faster and further they move
- Its used to confirm that the correctly sized PCR product has been amplified, by comparing it to a size standard or marker, or it can be used as a diagnostic tool
    1. DNA is extracted
    2. PCR isolation and amplification of DNA
    3. Liquid agarose, which contains dye that binds to DNA, is poured into a casting chamber setting as a gel
    4. The gel is submerged in a buffer solution and the DNA samples are loaded into the wells at the negative electrode end
    5. An electric charge is applied, the buffer conducts electricity and the negatively charged DNA molecules are pulled through the pores in the gel towards the positive terminal
    6. The dye binds to the DNA as it migrates which is seen through blue or UV light

**DNA sequencing using PCR and gel electrophoresis**

- The main method used in single molecule DNA sequencing is the sanger method
- Sanger sequencing
    1. Collect DNA sample
    2. Extract DNA from sample
        - Chemicals (i.e detergents) are added which break open the cells
        - DNA is separated from the other cell components, e.g. proteins
    3. Amplify DNA
        - Involves using PCR to make lots of copies of the DNA
        - Optional: only if the sample was too small to provide enough DNA
    4. Perform Sanger sequencing reaction
        - Goal: separately identify the position of each nucleotide
        - Use 4 'special' PCRs = normal PCRs with a chain terminating nucleotide
            - ★ Normal guanine has the 3'-OH on the sugar but the chain terminating nucleotide doesn't hence it can't bind to the next nucleotide
- Over enough rounds, DNA polymerase will incorporate a chain-terminating guanine at every single cytosine in the target DNA



    5. Determine DNA sequence
        - Run all 4 PCR reactions on an electrophoresis gel, to determine the lengths of the fragments in each reaction
        - Works from the smallest to the largest fragment where the next nucleotide is complementary to the chain terminating nucleotide in that reaction

**Application - sickle cell anaema**

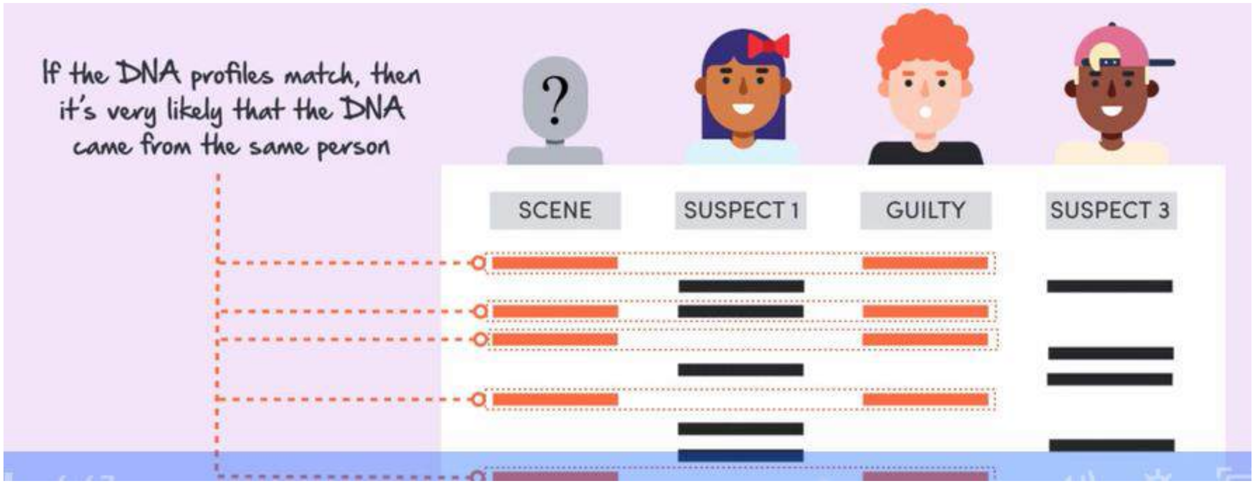
- In medicine, DNA sequencing can be used to determine if a patient is at risk of or affected by a genetic disease
- The genetic disorder where people have misshapen red blood cells is caused by the substiution of a single nucleotide in the gene that creates haemoglobin
- The normal DNA code is GAG, and the mutated code is GTG. This changes the amino acid from glutamic acid to valine, so that haemoglobin folds into an abnormal shape
- This causes the haemboglobin molecules to clump together resulting in a crescent shaped red blood cell which cant carry oxygen as efficiently
- This mutation is identified by using DNA sequencing

**DNA profiling using PCR and gel electrophoresis**
**Gel - fingerprinting**

1. Collect DNA sample
2. Extract DNA from sample
    - Chemicals (i.e detergents) are added which break open the cells
    - DNA is separated from the other cell components, e.g. proteins
3. Amplify STR fragments
    - Involves using PCR to make lots of copies of <13 different STR regions (e.g. CGA)
    - This is achieved by using primers which flank the region by binding to the DNA on each side of the STR
    - Since each STR is found at the same place on a chromosome, the same primer will flank both of the alleles and amplify them, since the DNA on either side is the same
4. Determine length of the STRs
    - Amplified STR fragments are separated by gel electrophoresis
    - Larger fragments move through the gel slower, compared to smaller fragments
5. Interpreting the electrophoresis gel
    - Pattern of bands on the electrophoresis gel (i.e. the number of bands and their locations) creates a DNA profile
    - Can determine the number of repeats in the STR by referring to the DNA ladder (mixture of known DNA fragments, each of which has a specific size)

**Application - identification of an unknown person (forensics)**

- Determine the identity of the:
    - ➢ Suspect of the crime
    - ➢ Victim (if they're recognisable)
- A sample taken from the scene is used to generate a DNA profile
- The victim/suspect is the person whose DNA profile is a complete match




investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns and relationships, for example:

- ➔ the use of population genetics data in conservation management

- Conservation genetics is a subfield of population genetics
- Aim: Avoid extinction of species by applying conservation methods that ensure the maintained of biodiversity
- Scientists analyse alleles of multiple genes to examine a species' genetic diversity
- The numerous types of DNA analysis (e.g SNPS, DNA profiling) → better understanding of microevolution (natural selection and mutation)

**Measuring extinction risk**

- 1982-1983 - heavy rain to cause cactus to produce smaller seeds
- Change in seeds has knock-on effects for species feeding on the seeds, such as the Common Cactus Finch
- If the finch population has high genetic diversity:
  - ➢ Variation in observable traits exists (including beak size)
  - ➢ Finches with smaller beaks can easily eat the small seeds
  - ➢ Finches with small beaks more likely to survive than finches with larger beaks
- If the finch population has low genetic diversity:
  - ➢ Less variation in genes contributing to beak size
  - ➢ Lower number of finches with favoured small beaks
  - ➢ Population could be significantly reduced or eradicated

Populations with greater genetic diversity withstand enivironmental changes better
Conversation scientists therefore must measure population genetics to calculate the risk of extinction

**Monitoring inbreeding**

- In 2016, researchers found the population in scotland of choughs were decreasing because closely related individuals were breeding together
- Closely related individuals have similar ancestors, so are more likely to inherit the same unhealthy alleles
    - ➢ Unhealthy alleles are often recessive: which means that individuals can be healthy with one copy of the allele, but having two copies gives them a disease
    - ➢ Hence when two choughs with the recessive allele breed they 1/4 choughs inherit two copies of the allele → led newborn choughs to die soon after birth. The high death rate among young birds led the Scottish Chough population to fall rapidly
- Inbreeding occurs when individuals only have access to a small range of potential mates (e.g. may occur if a pop size suddenly falls due to a environmental catastrophe)
- Or if a population is split into smaller groups (e.g. due to habitat fragmentation)

Comparing the genomes of a random population sample can allow us to monitor inbreeding by:

1. Allowing us to determine how closely related populatio members are
2. Allowing us to measure the allele frequencies for unhealthy mutations

Conservastion scientists then can look at population genetics to monitor inbreeding

**Investigating speciation**

- Species → group of organisms that interbreed to produce fertile offspring
- Cryptic species → are groups of species originally thought to be the same, but are different
    - ➢ Usually evolve from two populations of the same species
        - During speciation, populations stop interbreeding, preventing DNA from being shared between populations, creating a barrier to gene flow
            1. Alleles may be lost in one pop and not the other
            2. New alleles may arise in one pop and not the other
- This means that the allele frequencies will grow more different → speciation
- Coversation scientists look at population genetics to better understand ecosystems

**Cheetah**

- The cheetah is a vulnerable species, with a population about 7000 individuals, with numbers continuing to decline due to reduced prey and habitat and conflict with humans
- The use of a variety of genetic markers and techniques (e.g. STR markers, SNPs and whole-genome sequencing) in large-scale collaborative projects over the past 40 years has shown that cheetahs have a very low level of genetic variation, which makes them more vulnerable to diseases, environmental change and extinction → why?

- Cheetahs experienced a population bottleneck about 10 000-12 000 years ago that resulted in a reduced population and fragmentation of the species into six subspecies across Africa and Iran
- The bottleneck combined with reduced gene flow and genetic drift, resulted in a low level of genetic variation within the species
- Genetic information is central to the success of conservation management steragies, for example a shared global stud bookm with DNA profiles of each captive cheetah is used for controlled breeding programs to increase gene flow and maintain genetic diversity

➔ population genetics studies used to determine the inheritance of a disease or disorder

    - Many genetic disorders are affected by more than one gene
    - Polygenic inheritance - refers to when the inheritance of an observable trait is determined by many genes
    - Genes showing polygenic inheritance often have environmental
    - Identifying the regions of DNA associated with a genetic disorder shows:
        - Who is likely to be affected
        - What goes wrong in the cell to cause the disease
        - How can we treat the disease

GWAS

- Genome wide assoication studies (GWAS) involves taking the DNA of an individual, and looking for certain variations
- Rather than looking through the entire genome, GWAS involves a targeted search for bases that are known to be highly variable in the population (SNPs)
- By comparing the SNPs of people with a disease to those without, we can see which SNPs are associated with a disorder
- SNPs are more common than mutations causing rare genetic diseases as they occur in 1% of the population compared to a fraction of a percent in rare diseases

Limiations

- Population stratification refers to when disorder are more common in people with a particular ancestry or ethnicity
- In developed countries people with a South Asian background are more likely to develop type 2 diabetes than other ethnicities
- People are shared ethnicity are more likely to share certain SNPs
- However these shared SNPs may not be associated with having a disease and may just be associated with ancestries
- To solve this we can't just take random samples from the with and without the disease population to compare. When sampling we need to control for ancestry

- → population genetics relating to human evolution

We can base our ideas about the evolutionary relationships of different species on their observable characteristics as well as DNA evidence

DNA and evolutionary relationships

- When we compare the DNA of the primates we can see that humans share 99% of their DNA with chimpanzees and 98% of their DNA with gorillas
- This allows us to tell that the last common ancestor of humans and chimpanzees lived more recently than those of humans and gorillas
- The more we collect and analyse DNA samples, the more accurate our understanding of primate evolution is
- In 2012 we realised humans and gorillas are more closely related than we thought
- Previously weve only been able to compare the human genome of 1% of the gorilla genome
- Increased collaboration, funding, and technological improvements have led to the ability to compare whole human and gorilla genomes
- In order to estimate how long ago the last common ancestor of different primate species lived, scientists can use the knowledge that on average each base pair of primate DNA has a one in a billion chance of mutating per year

Studying human migration

- The first homosapiens evolved in central africa
- From there, humans began to migrate all around the world
- The first human to leave Africa did so around 60, 000 years ago
- We can work out how long ago different groups of humans migrated using population genetics
- We start off by looking at how genetic markers, like SNPs vary across people in different populations by using databases such as the human genome project
- The amount of difference between the genetics of two people increases with the distance apart that they live
- So, someones ancestors who lived mainly in Africa will have more genetic markers in common with someone whos ancestors who lived in the middle east compared to someones ancestry in india
- Knowing the rate of DNA mutation means we can work out the rate that the new genetic markers arise, and how long ago populatiosn last shared a common ancestor
    - ★ Last common ancestor in india and africa was 50, 000 yrs ago → this means that the ancestors of all the Indian population left Africa 50, 000 yrs ago. For instance the last common ancestor of the Indian population and a middle easten population lived 30, 000 years ago. → since they have a more recent common ancestor, it is likely that the indian population originated from humans who first migrated to the middle east

**Module 6 Genetic Change**

**Mutation**
**Inquiry question: How does mutation introduce new alleles into a population?**

Explain how a range of mutagens operate, including but not limited to:

**Mutagen:** an agent that causes mutation
**Mutation:** change to DNA base sequence

- ➔ Electromagnetic radiation sources
  - Exists as a spectrum of wavelengths with different amounts of energy
  - The spectrum includes ultraviolet light, X-rays and gamma rays…etc.
  - Three types of EM radiation are physical mutagens (UV light, X-rays, gamma rays)
    - ➢ Emitted by the sun where most is absorbed into the atmosphere, but some reach the earth's surface

**UV light**
- UV light is a type of non-ionising radiation (low-energy radiation that does not have enough energy to remove an electron) that causes covalent bonds to form between neighbouring thymines and/or cytosines on the same strand
- This structure known as a pyrimidine dimer, causes DNA to bend or kink
- As a result, DNA polymerase may not accurately 'read' the bases in the dimer during replication, and incorrect bases can be incorporated into the new synthesised DNA strand

**X-rays and gamma rays**
- Type of ionising radiation
- When these waves enter a cell and hit DNA, their high level of energy is absorbed, which knocks electrons out of atoms and breaks the chemical bonds within DNA
- Gamma rays are produced from the radioactive decay of certain radioisotopes used as fuel in nuclear power plants for instance
  - ➢ Only on rare occasions humans are exposed to high doses such as a reactor accident

- ➔ Chemicals
  - Compounds that cause mutations by altering the chemical structure of DNA bases
  - Classified based on how they interact with DNA
  - Three main types: modifiers, base analogues, intercalating agents

1. Modifiers
   - Change the chemical structure of a base

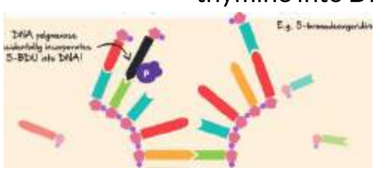
- DNA polymerase then incorporates a base that is complementary to the modified base during mitosis or meiosis
  - ★ Alcohol
- Cytosine to Uracil: A common modification is the deamination of cytosine, where a chemical reaction removes an amino group from cytosine, converting it into uracil. Normally, uracil is not found in DNA (it is found in RNA), and during DNA replication, DNA polymerase may pair uracil with adenine instead of the correct pair for cytosine, which is guanine

2. Base analogues
   - Molecule that resembles or mimic a base that is incorporated into DNA
   - In the next cell division, DNA polymerase incorrectly matches a base with an analogue
     - ★ Some chemotherapy drugs
3. Intercalating agents
   - Molecule that inserts between adjacent bases on the same DNA strand
   - This causes the DNA to bend or kink where DNA polymerase then deletes or adds an extra base during mitosis or meiosis
     - ★ Some chemicals in cigarettes

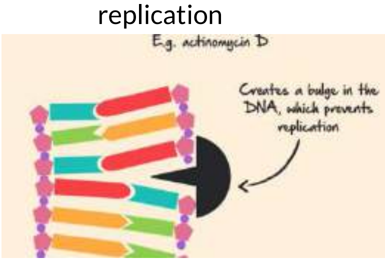
**EASY ATOMI VERSION - FBI —> MBI**

Different chemicals cause mutations in different ways

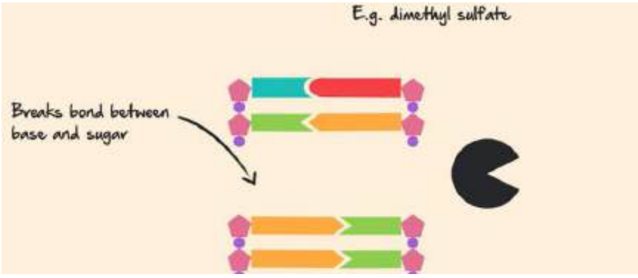
1. **Modifiers -** Chemical is accidentally incorporated into DNA, instead of proper nucleotides during replication
   - ★ DNA polymerase accidentally incorporates 5-bromodeoxyuridine (5-BDU) instead of thymine into DNA during replication as its similar to thymine




2. **Base** analogues **-** Chemical inserts itself into the DNA creating a bulge in the helix preventing DNA replication




3. **Intercalculating** agents **-** Chemical makes gaps in the DNA




➔ Naturally occuring mutagens
  - Agents found in the environment that can cause mutations in DNA where each operate in their own certain way
  - Can include: transposons, viruses, chemical compounds produced by other organisms and reactive oxygen species

**Transposons**

- DNA sequences that are able to move and insert themselves at a new location in the genome, which can disrupt genes
  ★ Variegated colouring of maize cobs is caused by transposons elements that insert themselves near a gene that encodes kernel colour

**Retroviruses**

- Can directly causes mutations in the cell of an infected host when they insert into DNA
- After entering the cell, these viruses convert their RNA genome into DNA, in a process called reverse transcription.
- This copied DNA inserts into the hosts genome, changes the DNA sequence and can inactivate a gene
  ★ Human Immunodeficiency Virus (HIV)

**Mycotoxins**

- Chemicals produced by some fungi which can contaminate food
- When they are ingested, they enter cells, where they are metabolised, and the chemical products bind to DNA to cause point mutations
  ★ The breakdown product of aflatoxin B1, causes guanine to mutate to thymine

Compare the causes, processes and effects of different types of mutation, including but not limited to:

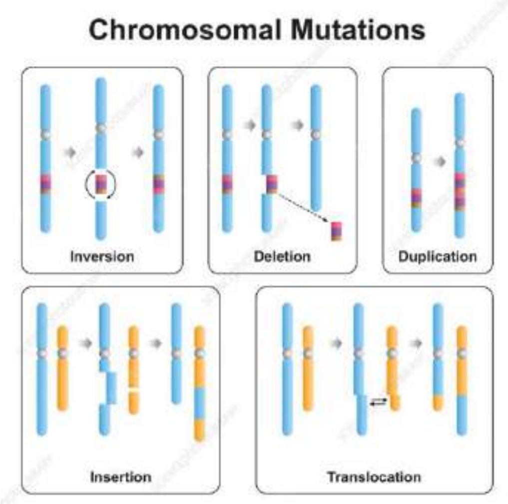
➔ point mutation: Change to a single base in DNA

| Types | Effects | Causes |
|---|---|---|
| **Substitution:** swapping one base for another<br><br>**Insersion:** base added to the | **Silent:** No change, as the codons code for the same amino acid<br><br>**Missense:** One amino acid altered | > Physical, chemical or naturally occuring causes<br>> DNA polymerase inserting the incorrect base into a new DNA |

| | | |
|---|---|---|
| sequence<br>**Deletion:** removing base from sequence<br>> causes frameshift mutation which shifts other bases creating a complete change in the translated polypeptide sequence from the mutation site onwards | due to different codon<br><br>**Nonsense:** Premature stop codon forms, shortening the chain | strand during DNA replication |

➔ chromosomal mutation: involve changes in the number or structure of chromosomes (IITDD)

**Insertion**: section is added
**Inversion**: section is inverted + re-inserted
**Translocation**: section moved to non-homologous chromosome, sometimes pieces from two chromosomes will swap
**Duplication**: section is doubled
**Deletion:** entire section of DNA is removed from the chromosome




**Changing chromosome numbers**

- Occur during meiosis, as the cell divides, the chromosomes do not separate completely. This is called non-disjunction.
- Can be found in somatic cells and gametes
- Result: some of the gametes produced contain additional chromosomes while others contain fewer chromosomes
- Two main forms: aneuploidy and polyploidy

1. **Aneuploidy**
    - The presence of an abnormal number of a particular chromosome
    - Can be an extra or missing chromosome
    - Most known is trisomy - having three homologous chromosomes instead of two, usually fatal in humans
        - ★ Down Syndrome- extra chromosome 21 or Klinefelter syndrome – XXY

**CHATGPT - EASY**
**Causes of Down Syndrome**

- Chromosomal Abnormality: Usually caused by an extra copy of chromosome 21 (trisomy 21).
- Maternal Age: Higher risk associated with older maternal age (over 35 years).
- Translocation: In some cases, part of chromosome 21 attaches to another chromosome.

**Processes**

- Nondisjunction: During meiosis, chromosomes fail to separate properly, leading to an egg or sperm with an extra chromosome.
- Fertilization: When this abnormal gamete fuses with a normal one, the resulting zygote has three copies of chromosome 21.

**Effects**

- Physical Features: Distinctive facial appearance, slanted eyes, and a single palmar crease.
- Cognitive Impact: Mild to moderate intellectual disability.
- Health Issues: Higher risk of heart defects, gastrointestinal problems, and immune system disorders.
- Developmental Delays: Delayed speech, motor skills, and social skills development.

2. **Polyploidy**
    - organisms with extra sets of chromosomes, common among plants; usually the result of reproduction among hybrids
    - In humans polyploid zygotes do not survive, but you may get some somatic cells with this condition

★ Bread wheat, is a hexaploid plant (6n), it was formed from a hybrid cross of a cultivated wheat with a wild grass

| Condition | Causes | Processes | Effects |
|---|---|---|---|
| Down syndrome | Usually caused by segregation of the number of chromosomes in gametes. Can also occur in somatic cells. | Additional copy of chromosome 21 (trisomy 21) segregates in a daughter cell | A 2n=47 genotype. Cognitive impairment and a range of other disabilities |
| Turner syndrome | …. | X chromosomes fails to segregate into a daughter cell | A 2n=45 genotype (X) in females. Short stature, infertility |

Distinguish between somatic mutations and germ-line mutations and their effect on an organism

| **Type of mutation** | **Similarities** | **Differences** |
|---|---|---|
| **Somatic** | • Both are a type of mutation<br>• Can be caused by chemicals, UV radiation and mutagens<br>• Both can occur in plants and animals<br>• Occur in the DNA sequence | • Occurs in non-reproductive cells<br>• Cannot be passed on to offspring however, it can be passed onto daughter cells in mitosis<br>• Mistake in mitosis - localised to one part of an organism, does not affect whole organism - tumor in one part<br>• Can be caused by environmental/external factors |
| **Germ-line** | | • Occur in gametes - half carry the mutation<br>• **CAN BE** passed on to offspring, however is mostly doesn't have a phenotypic effect as it is recessive - generations later, if an individual is born with two copies of the mutant allele of the gene, they may possess a novel phenotype (new phenotype that wasn't present) - albinism<br>• Mistake in meiosis - affects every cell in the body |

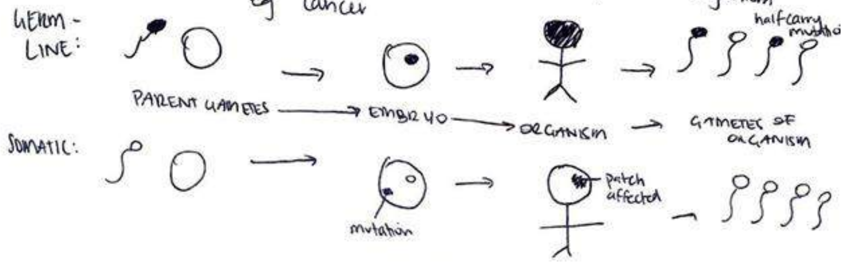
effect on organism

**Germ-line mutations:** (in sperm/egg)

- mistake in meiosis – effects every cell in the body
- passed onto offspring which inherit, effecting all cells of the organism
- cause of Diseases eg sickle cell anaemia, cystic fibrosis + colour blindness

**Somatic mutations:** (non-reproductive cells)

- mistake in mitosis – localised to one part of organism, does not effect whole organism
- Can be passed onto daughter cells in mitosis
  - ↳ not inherited by offspring
- Can be caused by environmental/external factors
- Localised effect, may lead to tumor in one part of organism eg cancer




Assess the significance of 'coding' and 'non-coding' DNA segments in the process of mutation

The base structure of a gene consists of a promoter, where transcription factors and DNA polymerase bind to stard transcription, following by alternating exons and introns and a transcription termination site

**Coding DNA (exons):** sequence of DNA that codes for protein

- Makes up 1% of the human genome
- Mutations have the most significant impact on the phenotype of an organism as they directly affect protein structure which is essential for their function

The rest of the genome is **non-coding DNA** as it is not translated into polypeptides**:** the portions of an organism's genome that do not code for amino acids

- Includes introns, intergenic regions, telomeres and centromeres
- Mutations can affect gene expression and cellular processes

Coding DNA segments are used for protein synthesis

Mutated segments will be copied to the mRNA and therefore will affect transcription and translation, affecting the resultant protein.

`protein synthesis → mutated genes → mutated aa → mutated protein → altered phenotypic traits`

**Non-coding - Introns**

- Located inside the genes of eukaryotic organisms
- Alternate with exons in a gene (introns-exons-introns-exons)
- Transcribed into pre-mRNA but then spliced out when mRNA is formed prior translation
- Mutation within an intron will have no impact on the final polypeptide unless it occurs in a **splice site**
- Splice sites are the sequences at the junctions between introns and exons
- They are recognised by the spliceosome, a complex of enzymes that cuts introns out of pre-mRNA and re-joins exons
- A mutation may cause the spliceosome to skip the splice site and remove an exon, which results in an incorrect mRNA sequence and polypeptide thus an incorrect protein

- Some non-coding DNA is used as a template thats transcribed to make functional RNA molecules such as tRNA and rRNA - issues in protein synthesises and cell functioning
- Regulatory sequences which are the promoters and operators - amount of protein produced may increase or decrease
- Repetitive sequences - no effects

**Intergenic regions**

- Long regions of DNA in between genes, making up most of the genome
- Intergenic regions and introns contain regulatory elements such as enhancers that 'talk' to genes and tell them when to be switched on and how much of a gene product to be made
- Mutations in enhancers affect the binding of transcription factors, which can increase or decrease the amount of gene product

**Telomeres and centromeres**

- **Telomeres** are located on the tips of chromosomes in eukaryotic organisms
- Consist of repeated stretches of short pieces of non-coding DNA
- Consist of 300-8000 repeats of the sequence CCCTAA
- They act as protective 'caps' that prevent the deletion of genes on the chromosomes
- Overtime they are deleted which causes our aging

- **Centromeres** are the specialised regions of non-coding DNA that separate the short and long arm of a chromosome
- Join sister chromatids together when a chromosome is copied during replication

- During cell division, they act as an anchor point for the spindle fibres to pull sister chromatids apart into the daughter cells
- It's essential that mutations do not occur as chromosomes may break down

Investigate the causes of genetic variation relating to the processes of fertilisation, meiosis and mutation

**Genetic variation** is the different combinations of alleles between individuals in a population or species
- Only concerned with germline cells as they can be passed onto offspring

**Meiosis**
- Crossing over: new combinations of alleles are created when genetic material is exchanged between non-sister chromatids of homologous chromosomes
- Random segregation of chromosomes: a parent randomly gives one allele for each gene to its offspring
- Independent assortment of chromosomes: the genotype of each gamete has an equal probability of occurring, because the allele a gamete receives for one gene has no influence on the alleles it receives for other genes

**Fertilisation**
- Brings together female and male gametes that have undergone the three processes.
- The mixing of haploid sets of chromosomes from two parents is an additional event that creates new allele combinations

**Mutation**
- Point and chromosomal mutations create new alleles that may be passed on to offspring
- Occur independently in individuals within a pop and over time this creates a mixture of new alleles
- Germ line mutations are the original source of genetic variation in populations and natural selection acting on these mutations has created diversity of life on Earth

Evaluate the effect of mutation, gene flow and genetic drift on the gene pool of populations

**Gene pool:** the total of all alleles of all genes in a population of a species at a time
- Can be changed by mutation, gene flow, genetic drift and natural selection
- Evolution occurs when changes in the gene pool happen over a long period of time

**How do geneticists calculate genotype and allele frequencies in a population?**
- The sum of the allele frequencies for two alleles at a given locus must be 1 (assuming there are 2 alleles)
- $p+q=1$ - where p is the frequency of one allele (T) and q is the frequency of the recessive allele (t)

- $p^2 + 2pq + q^2$
- Where $p^2$ is the frequency of one homozygous dominant genotype (TT), $q^2$ is the frequency of the homozygous recessive genotype (tt), and 2pq is the frequency of the heterozygous genotype (Tt)
- If there is a change over generations this indicates that one or more of these factors is affecting gene pool, indicating evolution

| | **Description** | **Effects on the gene pool of a population** | **Evaluation** |
|---|---|---|---|
| **Mutation** | Changes in the DNA sequence during meiosis that leads to the formation of new alleles<br>➢ New alleles can only be introduced into the gene pool of a species by mutation | The chance of a new mutation occuring in a gamete and being passed on to offspring is very low, and new mutations do not have immediate or significant effect on allele frequency in a population<br><br>Very few mutated alleles are advantageous and selected to increase in frequency - this is because if an environment is stable and an abnormal phenotype is introduced into the population, it is highly likely that the mutation will not benefit the organisms survival<br>→ deleterious mutations are usually acted upon by natural selection and removed from the population<br>→ neutral mutations are considered an 'evolutionary backup' as they can provide variations that have no immediate effect but may provide a selective advantage in the future if there are sudden changes<br><br>**Peppered moths**<br>• More white than black moths<br>• Industrial revolution caused | Creates new alleles and increases the gene pool of species = can be advantageous or disadvantageous |

| | | | |
|---|---|---|---|
| | | trees to turn black from white<br>● Dark colour was caused by mutation and was passed onto moths<br>● Due to natural selection dark coloured moths dominated | |
| **Gene flow** | The movement of genes into and out of a population when organisms migrate between areas | The mixing of individuals within a species may introduce new alleles to the population; or the proportion of an existing allele frequency to change, and it plays an important role in changing variation in the gene pool<br><br>Many endangered species, such as the **cheetah** exist in fragmented populations, which prevents gene flow, reduces genetic variation and places them at greater risk of extinction | Significant as can lead to evolution or even the extinction of a species |
| **Genetic drift** | Change in allele frequency due to random chance which may not necessarily be of benefit to the surviving alleles | Removing alleles by removing ones carrying the alles<br>1. Bottleneck effect - an example of genetic drift in which the frequency of alleles is changed due to a near extinction event<br>2. Founder effect - occurs when a small group of individuals separates from a larger population to establish a new colony. The small size of the new colony means it will experience strong genetic drift.<br><br>Genetic drift can have a strong effect on a gene pool if the population is small, but it has less effect in a large, randomly mating population | Significant as can lead to evolution and can have either no effect or a profound effect |

**Factors that cause changes in allele frequency within a Population**

| | Description | Effects on the Gene Pool of Population | Evaluation |
|---|---|---|---|
| **Mutation** | Changes in the DNA sequence during meiosis that leads to the formation of new alleles<br>- new alleles can only be introduced into the gene pool of a species by mutation | <u>Very few mutated alleles are advantageous and selected to increase in frequency</u> – this is because if an environment is stable and an abnormal phenotype is introduced into the population, it is highly likely that the mutation will not benefit the organism's survival.<br>- Deleterious mutations are usually acted upon by natural selection and removed from the population.<br>- Neutral mutations are considered an 'evolutionary back-up' as they can provide variations that have no immediate effect but may provide a selective advantage in the future if sudden changes to the environment were to ensue. | Creates new alleles and increases the gene pool of a species = can be advantageous or disadvantageous |
| **Gene flow** | **The transfer of genetic information from one population into another by migration**<br>- Involves existing individuals leaving and new ones entering the population by migration and immigration. Does not necessarily have to be of the same species. | - Add gametes/alleles from a population - increase variation<br>- Take away genes/alleles from a population - decrease variation<br>- Overall – change the allele frequency<br>- If individuals can enter and leave the population it can create a gene flow and in | It is very significant as it can lead to evolution |
| **Genetic Drift** | **Change in allele frequency due to random chance which may not necessarily be of benefit to the surviving alleles.**<br>- The remaining individuals in a population may not be an accurate representation of the allele and genotype frequencies of the original | Removing alleles by removing ones carrying the alleles<br>1. Bottleneck Effect – an example of genetic drift in which the frequency of alleles is changed due to a **near extinction event**<br>2. Founder Effect – individuals becoming geographically isolated from original population = not an accurate representation of the entire population<br>- The formation of a new population by a non-representative sample of individuals from the parent population | It is very significant as it is a mechanism for evolution, as it may have no effect but it may also have an irreversible, detrimental effect. |



**Biotechnology**

**Inquiry question: How do genetic techniques affect Earth's biodiversity?**

Investigate the uses and applications of biotechnology (past, present and future), including:

| | |
|---|---|
| **Past** | Aboriginals<ul><li>used aquaculture as evidenced by canals in western australia</li><li>canals would trap organisms → food</li></ul>Ancient Egypt<ul><li>bread making 10,000 years ago</li><li>noted that breaking down wheat would cause yeast to react with natural sugars to make the bread rise</li><li>used in today baker's yeast (Saccharomyces cerevisiae) since 1920s</li></ul> |
| **Present** | Industrial<ul><li>pollution prevention → biodegradable products</li><li>biofabrication → replace disease or injured tissue (eg 3D printers)</li></ul>Agricultural<ul><li>increases yield</li><li>increases quality</li><li>increases nutrition</li></ul>*eg: bt cotton injected with pest-resistant bacteria, golden rice developed with more vitamin A*<br>Medical<ul><li>gene therapy → CRISPR to give desired genes when other is not functional</li><li>IVF → help individuals conceive</li><li>insulin production → plasmid DNA cut & spliced with insulin</li></ul> |
| **Future** | <ul><li>synthetic biotechnology → production of a cell with a synthetic genome<ul><li>this cell may be able to communicate with normal cells to switch genes on/off (*eg: switching off cancerous cells)*</li></ul></li><li>CRISPR → cuts a gene with higher precision than restriction enzymes</li><li>Biofabrication → stem cells can be used to 3D print tissue and organs<ul><li>reduces death rate of people waiting for organ transplants</li></ul></li></ul> |

➔ analysing the social implications and ethical uses of biotechnology, including plant and animal examples

Biotechnology combines our understanding of biology with technology to create new useful products and processes that aim to improve our quality of life and the environment

| Organism | Examples | Social implications | Ethical implications |
|---|---|---|---|
| Plants | > Artificial pollination and recombinant DNA technology are used to express desirable traits<br>>> GMOs - golden rice | > Food labelling for consumer choice<br>> Patents to protect the intellectual property of companies that make a new GMO<br>> Testing to ensure the safety of GMOs for consumption and impacts on | > Modification of plants is considered more ethical than animals because plants don't have thoughts and feelings<br>> Humans may be interfering with the process of evolution by creating GMOs |

| | | | |
|---|---|---|---|
| | | ecosystems<br>> Reliable supplies of food and fibre<br>> Improvements on nutrition and health<br><br>Just plants -<br>> Reduced use of pesticides (less pollution of waterways)<br>> Increased yield of plants | |
| Animals | > Artificial insemination and recombinant DNA technology are used to express favourable traits<br>>> Transgenic salmon | …<br><br>Just animals -<br>> Health concerns (safety of meat, milk, or eggs)<br>> Economic impact (farmers upfront cost increase) | > Animal rights need to be considered<br>> Humans may be interfering with the process of evolution by creating GMOs |
| Humans | > products are developed to prevent, diagnose and treat diseases, e.g. genetic testing, genetic engineering and cloning | > consumer input needed to guide research priorities to beneift users<br>> better education needed so citizens can make informed decisions<br>> equity issues need to be addressed to make products accessible to all people<br>> privacy laws are required to prevent free access to a persons genetic information<br>> anxiety caused by knowledge of harmful mutations | > animal rights need to be considered in experiments on animals that test the effect of genetic mutations or new medicines<br>> therapeutic cloning requires robust debate about when life begins and whether the benefits of this research outweigh the cost of terminating a potential life |

➔ researching future directions of the use of biotechnology

New scientific discoveries, economics, the changing environment, ethical issues and the needs of society will influence the future directions of biotechnology

| **Future applications of biotechnology** | **Description** |
|---|---|
| **Medicine** | **> Personalised medicine** will use information about a person's own genes to tailor treatments to achieve the best outcome for them.<br>> An example is specific treatments for a subtype of cancer<br>> This is different from the ‘one size fits all’ approach to medical treatment, which benefits only |

| | |
|---|---|
| | some people<br>>> people with HER2 and breast cancer |
| **Agriculture & the Environment** | > **Genetic engineering** is used to introduce or edit a single gene to create a desirable trait in an organism<br>> Genetic engineering will work alongside synthetic biology, which uses online databases to combine multiple genes to redesign an organism to produce a substance for human use, or to create an organism with a new ability<br>> This might include new crop varieties that are resistant to a range of abiotic and biotic factors, or bacteria to clean up human pollution **(bioremediation)**<br>> to limit global warming to 1.5 degreesC **-** methane emissions from cattle can be reduced by growing 'clean meat' in the lab or by feeding cattle an asparagopsis species of seaweed, which prevents the formation of methane by inhibiting a specific enzyme in cows' guts |
| **Industry** | > a shift from chemical processes to bioprocesses would achieve a more sustainable future<br>>> biotechnology may help reduce plastics with biopolymers, such as polylactic acid, which is recyclable, biodegradable and compostable<br><br>> alternative sources of heat, electricity and fuel that do not emit greenhouse gases will minimise climate change<br>>> biofuels such as ethanol, produced by fermentation, will be used more as a fuel such as in aeroplanes |

- ➔ evaluating the potential benefits for society of research using genetic technologies

| Area | Genetic technology | Advantages | Disadvantages |
|---|---|---|---|
| Medicine | Vaccinations, genetic testing, genetic engineering | > vaccinations prevent infectious diseases<br>> early diagnosis of diseases (e.g. genetic testing) results in a better health outcome<br>> new treatments for diseases (e.g. gene therapy and recombinant proteins, like insulin for diabeteses) prevent illness and death | > rare allergic reaction to vaccinations<br>> ethical concerns about acess to personal genetic information<br>> genetic technologies are expensive |
| Agriculure | Transgenic plants and animals | > better food security<br>> more food variety, longer shelf life and better nutrional value<br>> healthy food is accessible to more people | > hybridisation with species in the wild, and possible decline in genetic variation, may have an impact on ecosystem services |
| Environment | Bioremediation, production of biofuels | > improve quality of ecosystem services e.g. cleaning up pollution in | > public concern about the environmental risk of uncontrolled |

| | | | |
|---|---|---|---|
| | | soil and the ocean<br>> Reducing $CO_2$ emissions to prevent climate change | GMOs |

➔ evaluating the changes to the Earth's biodiversity due to genetic technique

- DNA from GMOs can possibly escape into the wild
- It could impact biodiversity, if it hybridised with closely related species and outcompeted other species in a ecosystem
- More likely in plants as pollen can be distributed by various factors

- The risk of hybridisation also exists for GM animals
- GM salmon, called AquAdvantage salmon, were developed combining genes from the Chinook salmon and the ocean pout
- This results in the production of growth hormone all year round and the fish grow 25% faster than natural salmon
- They could have a selective advantage if they escaped into the wild, because they grow to adult size three times faster than wild salmon

- Conversation genetic is the use of genetic techniques to manage endangered species and to reduce the risk of extinction
- DNA sequencing, SNPs and STR markers are used to measure genetic diversity
- If a species has low genetic diversity, scientists may intervene and relocate animals or build habitat corridors between populations to increase gene flow
    - ★ Koala Kiss project aims to build uninterrupted forest habitats for koalas

**Genetic technologies**
**Inquiry question: Does artificial manipulation of DNA have the potential to change populations Forever?**
investigate the uses and advantages of current genetic technologies that induce genetic change

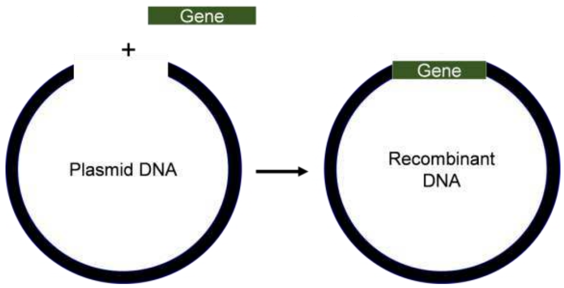
**Recombinant DNA technology**
- Used to manipulate and isolate DNA segments of interest
- Restriction enzymes cut DNA at specific sequences, creating "sticky ends" that can easily bond with other DNA fragments.
- The DNA of interest is inserted into a vector (e.g., a plasmid) which is also cut with the same restriction enzyme to ensure compatibility.
- The recombinant DNA is introduced into a host cell (e.g., bacteria) where is reproduces by binary fission to make many copies of the cloned gene

**Advantages**

Production of Therapeutics: Enables the **production of human insulin**, growth hormones, clotting factors, and vaccines

**Gene Therapy: Potential to correct genetic disorders by introducing healthy genes into patients' cells.**

**Genetically Modified Crops: Creation of crops with enhanced traits such as pest resistance, improved yield, drought tolerance, and increased nutritional value.**




**CRISPR-Cas9**

- Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) edit genes for the genome's benefit.
- CRISPR-Cas9 is a natural defence mechanism used by bacteria to attack viruses that infect them (bacteriophages)
- The genome of a bacterium contains a CRISPR locus, which consists of sequences called clustered regularly interspaced short palindromic repeats.
- A gene that encodes the Cas9 enzyme is also present
- When a phage infects a bacterium, the virus inserts its genetic material into it. The Cas9 enzyme locates the viral DNA in the cell, cuts it up into pieces and inserts it between the repeats in the CRISPR locus
- If infected again, it transcribes the viral DNA to make CRISPR RNA (crRNA)
- The crRNA, which is complementary to the viral DNA, combines with Cas9 and guides it to the virus to get cut up

- Scientists have engineered CRISPR-Cas9 to precisely edit specific locations in many organisms
- Guide RNA can be synthesised to target a specific gene to create a knock-out (inactivate gene), knock-in (correct mutation), or a transgenic organism (insert a gene)
    - ★ Bacillus thuringiensis (Bt) produces a protein toxic to specific pests
        - ➢ CRISPR-Cas9 is used to insert this Bt gene into corn DNA
        - ➢ The guide RNA directs CRISPR-Cas9 to the specific location in the plant's DNA where the Bt gene will be inserted
        - ➢ The plant now produces the Bt protein

        - ➢ When pests eat the plant, they ingest the Bt protein, which is toxic to them but safe for humans and other animals

CRISPR-Cas9 is a defense mechanism in bacteria to protect against viruses. In the bacterial genome, there is a CRISPR locus with repeated DNA sequences and viral DNA segments in between. Nearby, a gene encodes the Cas9 enzyme, which acts like molecular scissors.

When a virus infects the bacterium, it injects its DNA into the cell. The Cas9 enzyme cuts this viral DNA and inserts pieces of it into the CRISPR locus. This stored viral DNA serves as a "memory" of the infection.

If the same virus attacks again, the stored viral DNA is transcribed into CRISPR RNA (crRNA). The crRNA, matching the viral DNA, combines with Cas9 and guides it to the virus. Cas9 then cuts the viral DNA, destroying the virus. This system is used in gene editing to precisely cut and modify DNA sequences.

- Scientists have engineered CRISPR-Cas9 to precisely edit specific locations in many organisms
- Guide RNA can be synthesised to target a specific gene to create a knock-out (inactivate gene), knock-in (correct mutation), or a transgenic organism (insert a gene)
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        - ➢ The plant now produces the Bt protein

**Advantages**
Precision: CRISPR-Cas9 can target specific DNA sequences, allowing for precise edits
Versatility: It can be used in a wide range of organisms, from bacteria to plants and animals
Multiple Edits: It can edit multiple genes simultaneously, making complex modifications possible

compare the processes and outcomes of reproductive technologies, including but not limited to:

- ➔ artificial insemination
- ➔ artificial pollination

**Artificial insemination** is a type of assisted reproduction technology that uses any method other than sexual intercourse to introduce semen into the reproductive tract of a female
**Artificial pollination** is the intentional human transfer of pollen from the male to the female reproductive organ of a plant
**In Vitro fertilisation (IVF)** is a process where eggs are fertilised by sperm outside the body, and the resulting embryos are then transferred to the woman's uterus to achieve pregnancy

**Artificial Pollination (Plants)**

1. Pollen from the anther is lightly transferred onto a brush
2. Pollen is brushed onto stigma of another flower of the same species
3. Pollen fertilises ovules

4. Pollinated flower produces offspring with desirable traits

| Advantages | Disadvantages |
|---|---|
| • cross-breeding of favourable traits<br>• increases yield due to successful pollination<br>• decreases likelihood of self-pollination | • cannot guarantee desired trait is passed on<br>• decreases biodiversity as same species is being produced |

**Artificial Insemination (Animals)**

1. Semen is collected from a male
2. Insemination gun shoots semen into the cervix up to the uterus of the female

| Advantages | Disadvantages |
|---|---|
| • Used to inseminate large numbers of females<br>• Transport of semen is easier than transporting whole animal and can be stored for long periods | • cannot guarantee desired trait is passed on<br>• Reduced genetic variations make populations susceptible to environmental changes |

| | Processes | Outcomes |
|---|---|---|
| **Similarities** | > human intervention - sperm (animals) and pollen (plants) are collected from a male and transferred to a female<br>> fusion of genetically unique male and female gametes<br>> sperm and pollen need to be cryopreserved for use in future | > offspring with desirable traits<br>> increases the chance of fertilisation when animals don't mate or if there is a lack of pollinators<br>> reduction in variation, because one male may be used reproduce with many female<br>> used to save endangered plants and animals<br>> better control of passing on a particular trait to offspring compared to random mating |
| **Differences** | > sperm is collected with an artificial vagina or by electroejaculation. Pollen is collected with a brush<br>> in flowering plants, pollen is delivered from anther to stigma. In animals, sperm is introduced into the vagina<br>> the male reproductive organs of a plant may be removed to prevent self-pollination. Animal reproductive organs are not affected | > much higher yields in plants than animals because pollen can be quickly transferred<br>> artificial insemination is usually restricted to fewer individuals (e.g. passing on traits of winning race horse) Pollen can be transferred between many plants in a large crop |

**IVF**

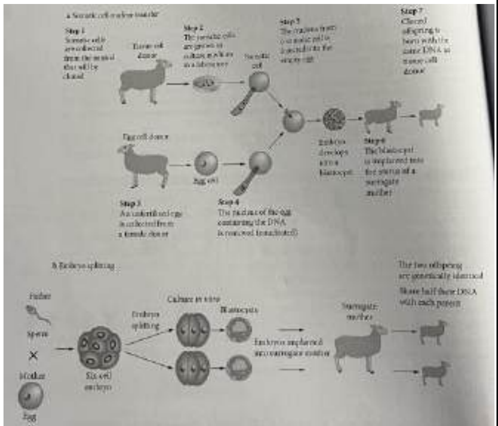
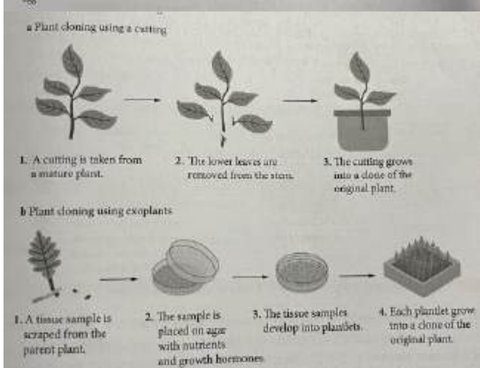
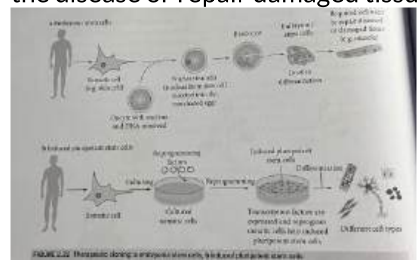
1. eggs are fertilised by selected sperm outside the womb (in vitro)
    a. usually in petri dish
2. zygote is cultured (grown) until the early stages of development
3. embryo is inserted into the reproductive tract of mother using catheter
4. attaches to endometrial lining or frozen in liquid nitrogen for future use

| **Similarities** | **Differences** | **Outcomes** |
|---|---|---|
| > involves the fusion of gametes<br>> AI and AP enable breeders and farmers to select specific genetic traits in animals and plants, respectively. IVF allows for the selection and manipulation of embryos before transfer, enabling control over genetic traits in offspring<br>> genetic material (sperm, pollen, embryos) can be cryopreserved for future use | > Eggs are fertilised outside the body in a laboratory dish before embryo transfer<br>> Often used in cases of infertility, genetic disorders, or to preserve fertility before medical treatments | > increases chances of fertilisation for females unable to conceive naturally<br>> help save endangered animals |

investigate and assess the effectiveness of cloning, including but not limited to:

- ➔ whole organism cloning (reproductive cloning)
- ➔ gene cloning

| | **Whole organism cloning** | **Gene cloning** | **Therapeutic cloning** |
|---|---|---|---|
| **Definition** | Creation of a new molecular organism that is genetically identical to its parent | The process of making identical copies of a segment of DNA that codes for a polypeptide | The creation of a cloned embryo for the purpose of producing embryonic stem cells |
| **Process** | > Whole organisms can be cloned using two techniques, Somatic cell nuclear transfer (SCNT) or Embryo splitting<br><br>1. Somatic cell is extracted from animal that will be cloned and starved of nutrients to stop cell division in a laboratory<br>2. Egg cell from female donar is enucleated using a micropipette (nucleus is removed)<br>3. Somatic cell is inserted into the enucleated egg cell and treated with electricity, forcing them to fuse together to form a ‘fertilised’ egg cell<br>4. Electric shock triggers cell division and | • Gene is isolated from a cell<br>• Plasmid is removed from bacteria<br>1. Gene fragment is cut from a restriction enzyme creating sticky or blunt ends<br>2. Plasmid DNA (vector) is cut open by same restriction enzyme creating complementary ends to the DNA of interest<br>3. Gene fragment is inserted into the plasmid where ligase or topiomerase joins them together | 1. A somatic cell is extracted from the patient who needs the therapy.<br>2. An egg cell from a donor organism is enucleated using a micropipette<br>3. The patient’s somatic cell is inserted into the enucleated egg cell. They are then treated with electricity, which causes them to fuse together to form a cell with the patient’s DNA.<br>4. An electric shock is applied to trigger cell division.<br>5. The inner cell mass of the blastocyst, which contains pluripotent stem cells, is extracted. These stem cells can differentiate into any cell type needed for therapy. |

| | | | |
|---|---|---|---|
| | embryo develops in vitro<br>5. Embryo is implanted into surrogate - Dolly the sheep in Scotland<br><br><br><br>FIGURE 2.31 Cloning a plant: a taking a cutting, b growing exoplants in tissue culture | 4. Plasmid is then introduced into bacteria (host cell) by a micro-injection (gene gun)<br>5. Plasmid will grow on agar containing an antibiotic that only allows bacteria that contains the plasmid to grow | 6. The extracted stem cells are cultured and induced to differentiate into the specific types of cells required for the patient's treatment (e.g., nerve cells, heart cells, etc.).<br>7. The differentiated cells are then transplanted into the patient to treat the disease or repair damaged tissues.<br> |
| **Effective** | Inefficient - clones have developmental abnormalities or die soon after birth<br>★ Dolly the sheep was the first after 276 attempts and died shortly after<br>Efficient - less expensive/time-consuming, higher success rate | Effective - scientists use commercially available enzymes, vectors and kits to do their experiments | Effective if done properly - Due to using pluripotent stem cells, they can grow into any kind of cell in the body and treat many diseases by replacing dysfunctional cells.<br><br>No risk of immune rejection as the cells are genetically identical to the patient. However, many attempts are often needed due to instability of the fused cells. |

describe techniques and applications used in recombinant DNA technology, for example:

**** ignore ******

- ➔ the development of transgenic organisms in agricultural and medical applications

- In order for a gene to be cloned it must be inserted into a vector which is usually a circular piece of DNA (plasmid)
- Plasmids are found naturally in the cytoplasm of bacteria
- They replicate independently, express genes they carry, and can be exchanged between bacteria
- Scientists take advantage of this to artificially introduce plasmids containing a desired gene into bacteria (transformation) and use them to produce many copies of a gene (e.g. to later introduce into an animal or plant to make a transgenic organism) or to express a cloned gene (e.g. to make a medically important protein such as insulin)

- Gene is isolated from a cell
- Plasmid is removed from bacteria

1. Plasmid DNA (vector) is cut open by restriction enzyme creating ‘sticky’ or ‘blunt’ ends
2. Gene fragment is cut from a source by same restriction enzyme so it has the same sticky or blunt ends
3. Gene fragment is inserted into the plasmid where ligase joins them together if it has sticky ends or topoisomerase to join them together
4. Plasmid is then introduced into bacteria (host cell) by a micro-injection (gene gun)

5. Plasmid will grow on agar containing an antibiotic that only allows bacteria that contains the plasmid to grow

(book picture)

**Medical applications of genetically modified bacteria**

- The production of insulin
- Type I diabetes cannot produce insulin to regulate their blood glucose level
    - ➢ The injection of insulin is the only available treatment
- The human insulin gene was cloned into plasmid that expresses the gene in bacteria or yeast
- The microrganism transcribes and translates large amounts of the insulin polypeptide in fermentation tanks
- Scientists than harvest and purify the insulin and use the hormone to treat diabetes

**Use of transgenic animals in medicine**

- Transgenic animals have been developed mainly to act as a convenient producers of recombinant proteins - biopharming
    1. Cloned gene is inserted into the DNA of the host organism

**** ignore ******

**transgenic organisms in agriculture** → insert genes from one organism into another

Example: Bt Corn
Trait: Pest resistance.
1. The bacterium Bacillus thuringiensis (Bt) naturally produces a protein that is toxic to specific insect pests, such as the European corn borer.
2. Scientists isolate the Bt gene responsible for producing the insecticidal protein.
3. The Bt gene is inserted into the DNA of corn plants using a plasmid vector.
4. The genetically modified corn plants are grown and tested. They now produce the Bt protein, which kills the insect pests when they eat the corn.
5. Bt corn is widely used by farmers because it reduces the need for chemical insecticides and increases crop yields.

**transgenic organisms in medical applications**
- production of insulin

- The production of insulin
- Type I diabetes cannot produce insulin to regulate their blood glucose level
  - The injection of insulin is the only available treatment
- The human insulin gene was cloned into plasmid that expresses the gene in bacteria or yeast
- The microrganism transcribes and translates large amounts of the insulin polypeptide in fermentation tanks
- Scientists than harvest and purify the insulin and use the hormone to treat diabetes

evaluate the benefits of using genetic technologies in agricultural, medical and industrial applications

| **Application** | **Example** | **Advantages** | **Disadvantages** |
|---|---|---|---|
| **Agriculture (crops)** | > Bollgard III cotton (insecticide resistance)<br>> 'Siren' canola (herbicide resistance)<br>>Transgenic crops Golden rice (vitamin A) | > reduced use of chemicals<br>> less likely to kill non-target pests<br>> reduced soil compaction because machinery is on the land for less time<br>> increased crop yield because of reduced competition from pests and weeds<br>> less tillage and soil degradation<br>> decrease in labour and fuel usage | > killing weeds reduces food supply for wild animals<br>> negative impacts on food chains<br>> potential cross-breeding and contamination of wild species<br>> development of resistance in target species<br>> non-target invertebrate species may also be killed<br>> possible decline in crop biodiversity<br>> high cost of seeds |
| **Agriculture** | Aquaculture - | > faster growth rates | > research is time consuming |

| | | | |
|---|---|---|---|
| **(animals)** | AquAdvantage salmon | > increased revenue for farmers | and expensive<br>> difficult to create because traits are often controlled by multiple genes<br>> potential side effects ushc as developmental abnormalities<br>> transgenic organisms may escape into the wild and introduce transgenes into environment |
| **Medical** | Genetic testing, vaccines, and personalised medicine | > more efficient testing and treatment procedures<br>> reduced cost to the healthcare system | > some treatments are too expensive for most people |
| **Industrial** | Production of chemicals, polymers and biofuels using bioprocessing<br><br>Food and fibre bioprocessing<br><br>Extraction of medals | > reduction of waste such as chemical solvents and heavy metals<br>> waste-by-product less likely to require treatment<br>> environmentally sustainable products (e.g. biofuels and recyclable polymers)<br>> less water used in production | > the use of biological processes may not be as efficient as chemical processes<br>> developing new techniques is expensive in the short term |

evaluate the effect on biodiversity of using biotechnology in agriculture

| |
|---|
| **Artificial pollination and insemination**<br>● same as selective breeding<br>● long term: decrease biodiversity and crops/populations of cattle susceptible to environmental pressures |
| **Transgenics**<br>● led to the production of genetically modified crops<br>● genetically more diverse as there are more alleles due to the transfer of genes from a totally different organism<br>    ○ increases gene pool and biodiversity<br>● overtime, as seen with Bt Cotton, crops without transgene will be killed off<br>● limitation: future crops are likely to be result of previous crops (might decrease biodiversity and have long term impacts on ecosystem, yet effect is still being studied and is unknown) |

| | Conventional agriculture | Agriculture biotechnology |
|---|---|---|
| **Crops** | > biodiversity is reduced because habitat is destroyed to access fertile land<br>> thousands of traditional crop varieties used to be grown<br>> biodiversity is reduced because herbicides are pesticides kill non-target species and pollute water ways<br>> aquatic biodiversity is reduced because large-scale irrigation drains rivers and waterways<br>> pollination depends on wind or animals, which maintains genetic diversity | > GM crops may limit loss of biodiversity by growing on marginal land<br>> GM crops are grown as monocultures. Over 1000 varieties face extinction<br>> GM crops require less herbicides and pesticides, which reduces negative effects on food chains<br>> Drought-resistant GM crops protect biodiversity because crops need less water<br>> Artifical pollination reduces genetic diversity becuase pollen from only one plant is used |
| **Livestock** | > animals mate randomly with each other, which maintains genetic diversity | > artificial insemination reduces genetic diversity because sperm from one male fertilises the eggs of many females |
| **Aquaculture** | > overfishing of natural fish stocks reduces genetic and species diversity<br>> genetic diversity is high because of gene flow between individuals in the wild | > genetic, species and ecosystem diversity is reduced because of diseasrs and nutrient pollution<br>> species diversity may decline if escaped GM fish out-compete wild species |

interpret a range of secondary sources to assess the influence of social, economic and cultural contexts on a range of biotechnologies

| Context | Influence |
|---|---|
| **Society** | **Education**<br>Individuals have different levels of education. This influences a persons ability to make decisions about the risks and benefits of a biotechnology<br>★ Vaccination hesitancy can be caused by limited understanding of how vaccines work<br>**Social media**<br>Information online is often a personal opinion rather than evidence based. Targeted digital advertising also reinforces a persons current belief system<br>★ A person may have a strong view about a transgenic organism without understanding the thorough safety tests involved in producing it<br>**Geography**<br>Countries have different priorities based on their environment. Australia has one of the highest rates of skin cancer, so most people would support skin cancer research |
| **Culture** | **Religion**<br>Many religions view fertilisation as the beginning of life, and therefore regard cloning and the use of embryos in biotechnology as destroying life. The dominant religious beliefs in a society may therefore limit the progress of research |

| | |
|---|---|
| | into therapeutic cloning.<br><br>Jehovahs Witnesses refuse blood transfusions based on biblical teachings. It is therefore important for doctors to determine individual preferences when treating individuals of some religious groups |
| **Economic** | **Personalised cancer therapy**<br>Personalised cancer therapy using antibiotics costs $100 000-$200 000 per year for each patient. Most people cannot afford this, and it is often too expensive to be subsidised by the public healthcare system. This may limit advances in this technology if costs do not come down.<br><br>**Public health and prevention**<br>Governments invest in screening and vaccination programs to reduce the long-term high costs of treating patients with a disease.<br>★ The national HPV vaccination program in Aus provides free vaccination to females and males aged up to 19, to prevent the development and spread of cervical cancer |

**Module 7: infectious disease**

**Causes of infectious disease**
**Inquiry question: How are diseases transmitted?**

describe a variety of infectious diseases caused by pathogens, including microorganisms, macroorganisms and non-cellular pathogens, and collect primary and secondary-sourced data and information relating to disease transmission, including:

➔ classifying different pathogens that cause disease in plants and animals

**Non-infectious** diseases are caused by factors other than a pathogen and cannot be transferrered between individuals

★ Down syndrome

**Infectious** diseases are caused by a pathogen and can be transferred from one individual to another

★ COVID-19

**Six main groups of pathogens:**

1. Prions
2. Viruses
3. Bacteria
4. Protozoa
5. Fungi
6. Macroorganisms

| Type of pathogen | Distinguishing features | Example |
|---|---|---|
| Prion | NON-LIVING<br>• Defective form of protein molecule<br>• **Does not contain DNA or RNA**<br>• Mostly attacks brain or nerve cells<br>• Molecular level-low nm<br>• Not visible with light microscope | • Bovine spongiform encephalopathy (a.k.a. mad cow disease)<br>• Creutzfeldt-Jacob disease |
| Virus | NON-LIVING<br>• Non-cellular<br>• **Contains DNA, RNA and protective coat**<br>• Requires a living host cell to replicate<br>• Less than 500nm<br>• Not visible with light microscope | • Hepatitis B - HBV<br>• AIDS - HIV<br>• Smallpox - Variola virus |
| Bacteria | LIVING<br>• Unicellular, **prokaryotic cell**<br>• Cell wall surrounding cell<br>• Up to 100um<br>• Visible with light microscope | • Tuberculosis - Mycobacterium tuberculosis<br>• Anthrax - Bacillus anthracis |

| Type of pathogen | Distinguishing features | Example |
|---|---|---|
| Protozoa | LIVING<br>• **Eukaryotic** unicellular organisms<br>• Usually complex life cycle<br>• 50-150um<br>• Visible with light microscope | • Malaria-Plasmodium sp.<br>• Giardiasis- Giardia lamblia |
| Fungi | LIVING<br>• **Eukaryotic** cells with **cell wall**<br>• Some unicellular, most are multicellular<br>• um to mm | • Tinea - Tinea corporis<br>• Thrush - Candida albicans |
| Macro-parasite | LIVING<br>• **Eukaryotic** cells - **multicellular** organism<br>• Mostly arthropods or worms<br>• External parasites = ectoparasites<br>• Internal parasites = endoparasites<br>• mm to metres<br>• **Visible with naked eye** | • Tapeworm - malnutrition/diarrhoea<br>• Paralysis tick - paralysis |

- ➔ investigating the transmission of a disease during an epidemic

**Epidemic:** an outbreak of an infectious disease that spreads rapidly among a large number of individuals within a defined geographical area

**Pandemic:** the spread of a new disease, or new variant, accross a wider geographical area, often worldwide

**Zika virus epidemic in brazil**

- Zika disease is caused by the zika virus

**Symptons:**

- Usually very mild and include fever, rash and headache, in many cases no symptoms at all
- However the disease is very problematic for pregnant females as the virus can pass through the placenta to the baby
- It can result in misscariage, preterm birth and microcephaly, a significant birth defect

**Transmission:**

- Is a zoonotic pathogen transmitted primarily by the bite of an infected Aedes aegypti mosquito
- Secondary modes of transmission include from mother to foetus during pregnacy, or sexual contact
- Was first identified in a sick monkey in the Zika Forest of Uganda in 1947
- In 1952, the first reported human case was confirmed, and it has subsequently been detected in 90 other countries
- In 2016 by WHO it was declared an epidemic in Brail when scientific studies demonstrated a link between a more virulent strain of the virus and increasing numbers of birth defects
- Estimated 1-1.5 million infections occured with several thousand confirmed causes of congenital Zika syndrome
- By mid-2017, cases dropped to zero in Brazil which was thought to be because of the combination of seasonal weather patterns that impaired the mosquito vector, vigilant control measures and a degree of herd immunity that resulted from high levels of infections

➔ design and conduct a practical investigation relating to the microbial testing of water or food samples

(photo)

➔ design and conduct a practical investigation relating to the microbial testing of water or food samples

(book)

➔ investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission

| **Mode of transmission** | Details | Examples |
|---|---|---|
| Direct contact | > transfer of pathogen upon physical contact<br>> exchange of bodily fluids during kissing or sexual intercourse<br>> inhaliation of airborne droplets that have been | > COVID-19 (SARS-CoV-2)<br>> aids (HIV)<br>> tuberculosis (Mycobacterium tuberculosis)<br>> measles (measles virus) |

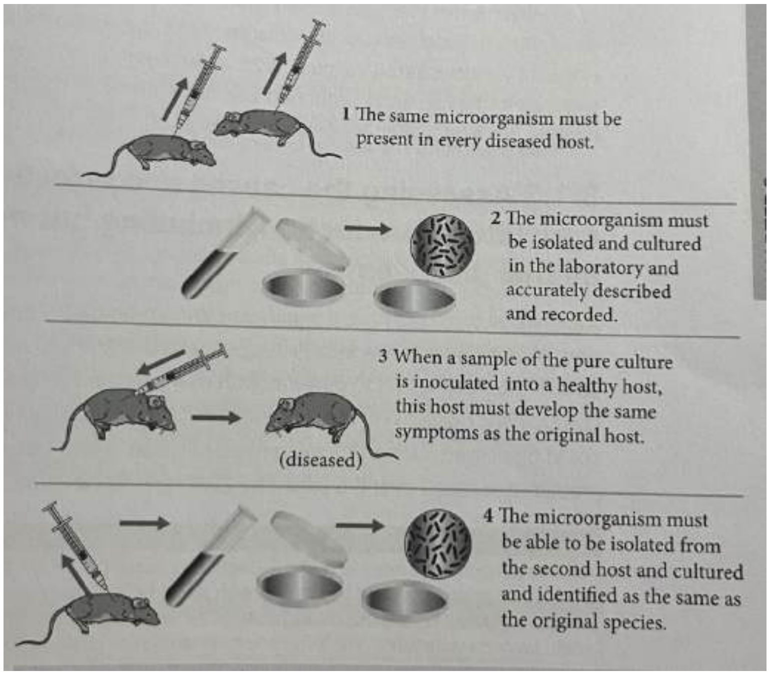
| | | |
|---|---|---|
| | coughed or sneezed into the surrounding environment by an infected person | > influenza (influenza virus) |
| Indirect contact | > transfer of pathogen that does not involve direction human-to-human contact<br>> can include transmission of airborne droplets, where pathogens remain viable outside the host body for many hours<br>> new individuals become infected when they touch a contaimined object and then transfer the pathogen to one of their mucous membranes<br>> consumption of contaminated food or water is also a common mode of indirect transmission | > cholera (Vibrio cholerae)<br>> food poisonining (Eschericha coli)<br>> measles (measles virus)<br>> influenza (influenza virus) |
| Vector-borne | > pathogen is transmitted typically by an arthropod vector<br>> responsible for nearly 20% of all infectious diseases, killing an estimated 700 000 people annually<br>> malaria causes an estimated 400 000 deaths annually<br>> dengue fever is the most significant viral vector-borne disease, with 96 million cases per annum, 40 000 deaths and over half the worlds population are at risk of contracting the disease | > malaria (plasmodium transmitted by the bite of an infected female Anopheles mosquito)<br>> dengue fever (Dengue virus transmitted by the bite of an infected female Aedes mosquito)<br><br>(check booklet) |

investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including:

- ➔ Koch's postulates

- Prior to the work of Pasteur and Koch, spontaneous generation was a widely held theory used to explain how new organisms came into existence directly from non-living matter
  - ★ It was believed that rats came from garbage and maggots came from rotting meat

- Robert Koch hypothesied that each infectious disease was caused by a specifc pathogen
- To test this, Koch worked with the bacterium Bacillus anthracis and demonstrated what is now known as Koch's postulates, the basis behind identifying a casual relationship between a specific pathogen and a disease
- Koch followed the same procedure to determine that Mycobacterium tuberculosis was the causative pathogen of tuberculosis and vibrio cholerae was responsible for cholera
- His work highlighted that transmission of an infectious disease occurs when an infected individual passes on the causative pathogen to a non-infected individual




➔ Pasteur's experiments on microbial contamination

- Pasteur hypothesised that the theory of spontaneous generation was incorrect
- He believed that microbes were airborne and that food and liquids spoil when these microbes land and become active
- In early 1860s, he performed swan-neck flask experiments, which disproved this theory demonstrating that microbes were airborne and therefore can be transmitted from an infected to a non-infected individual

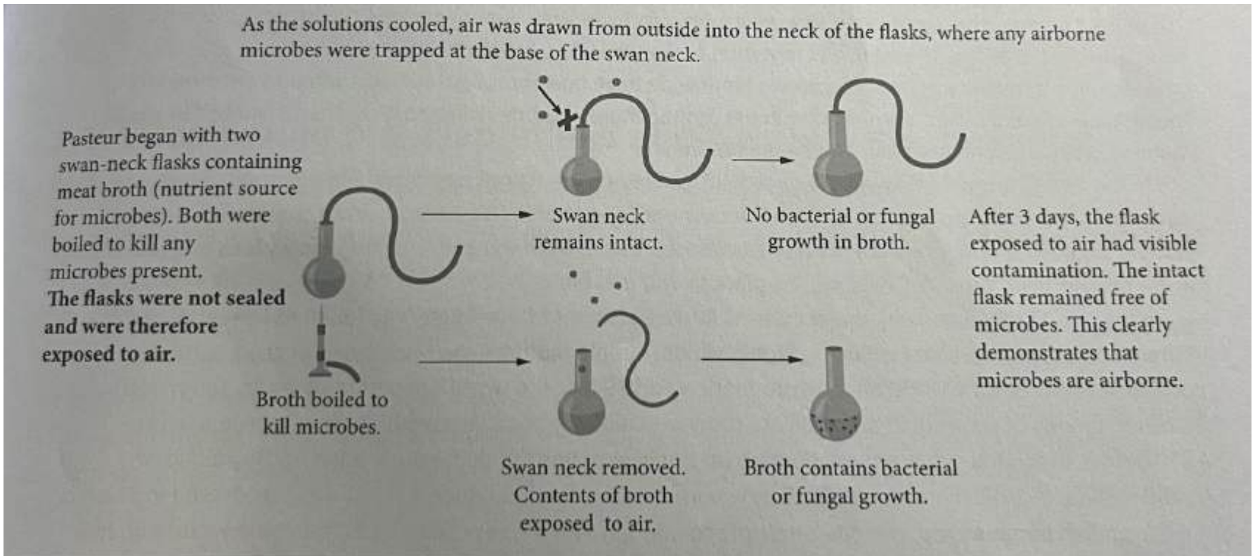


FIGURE 3.6 Pasteur's swan-neck flask experiment

Later on his combined areas of research led him to propose the germ theory, which states that specific microbes are the causative agents of infectious diseases

assess the causes and effects of diseases on agricultural production, including but not limited to:

- ➔ plant diseases

**Fire blight**

**Background:**

- Fire blight is a devastating plant disease of some fruit (apples, pears) worldwide. It is found in nearly every apple producing country, except Australia and Japan.
- Fire blight is excluded from Australia as a result of strict quarantine where all plant products are prevented from entering the country.
- Australia's pome fruit industry is worth $AU475 million per year.

**Cause:**

- Fire blight is a highly infectious disease caused by the bacterium Erwinia Amylovora

**Effect:**

- Fire blight attacks blossoms, leaves, shoots, branches, fruits, and roots. Infection results in tissue death and bacterial ooze droplets on infected tissue.
- It is spread to healthy plants through rain, wind, insects and pruning tools.
- It is difficult to contain once an outbreak occurs. Prevention is the best option for the management of Fire blight, such as:
  - ➢ on-farm biosecurity to prevent entry, establishment and spread of pests and diseases.
  - ➢ ensuring all staff and visitors are instructed in and adhere to on-farm hygiene practices.

**Impact:**

- All apple and pear growing areas in Australia are considered high risk areas for fire blight. In 1997, fire blight was found in Melbourne's Royal Botanic Gardens, costing the Australian pome and nursery industries an estimated $AU20 million in lost revenue.

➔ animal diseases

**Foot and mouth disease**

**Background:**

- Foot and mouth disease (FMD) is a highly contagious disease of cloven-hoofed animals including cattle, sheep, goats, and pigs. Australia is considered free of FMD.
- FMD is also excluded from Australia as a result of strict quarantine, where all animal products are prevented from entering the country.
- 29% of Australia's agricultural export commodities consist of meat and live animals with the total value of Australia's offfarm beef and sheep meat industry being AUD $17 billion, according to Meat & Livestock Australia.

**Cause:**

- FMD is caused by the foot and mouth disease virus (FMDV).

**Effect:**

- FMD causes fever and blisters in the mouth and hooves of cattle, sheep, goats, and pigs (and other cloven-hoofed animals).
- This leads to severe production losses. Although infected animals recover, the disease often leaves them weakened and debilitated.
- As a result, herds are destroyed.
-

**Impact:**

- A 2001 outbreak in the UK caused more than $AUD 19 billion in export losses.
- Many countries will only import from livestock and meat products from countries that are FMD free. As a result, if FMD entered Australia there would be severe economic consequences.
- It is estimated that a small FMD outbreak, controlled in 3 months, could cost around $AUD 7.1 billion, while a large 12 month outbreak would cost $AUD 16 billion.
- It is essential that Australia remains FMD free.

**Fungal diseases**

**Dry Bubble disease in mushrooms**

**Background:**

- Dry bubble disease is a fungal disease affecting commercially grown mushrooms, primarily white-button mushrooms.

- Caused by the fungus Lecanicillium fungicola.
- Symptoms vary depending on the infection time: late infections cause brown necrotic lesions on the mushroom's fruiting body, while early infections result in deformed masses of undifferentiated mushroom tissue.
- The disease is highly contagious, with spores easily spread through water and air.
- Mushroom crops are grown asexually and in monoculture, resulting in extremely low genetic variation, making them highly susceptible to disease outbreaks.
- Australian growers produce over 70,000 tonnes of mushrooms per annum, valued at over A$450 million, while the global market is valued at over A$65 billion.

**Cause:**

- Caused by the fungus Lecanicillium fungicola.

**Effect:**

- Causes brown necrotic lesions on the mushroom fruiting body in late infections.
- Leads to deformed masses of undifferentiated mushroom tissue in early infections.
- Reduces the quality and marketability of the mushrooms.
- Can lead to severe economic losses due to decreased yield and quality.

**Impact:**

- Estimated to cost the Australian mushroom industry 5% of its total annual revenue.
- Economic losses for mushroom growers due to reduced yield and quality.
- Increased costs for managing and controlling the disease.
- Potential for reduced supply of mushrooms in the market.
- Need for stringent hygiene practices to prevent the spread of the fungus.

compare the adaptations of different pathogens that facilitate their entry into and transmission between hosts

**Plasmodium Falciparum (Malaria)**
(photo from book)
**Adaptation of PF to facilitate entry into host**
1. To enter a new human host, PF first requires the infected female Anopheles mosquito to bite and penetrate the skin. To obtain a blood meal, the mosquito releases an anticoagulant protein that prevents the blood from clotting. It is at this point that PF exits the mosquitos salivary gland

**Adaptation of PF to facilitate transmission between hosts**
2. At each stage of the pathogens life cycle, different antigen molecules are produced, preventing the host from initaiting an effective immune response against the pathogen
3. Once is has entered the red blood cells, PF produces adhersion proteins, which are presented on the surface of the cell. The proteins change the shape of the RBCs, slowing their movement through blood vessels. This aids the merozoites in exiting the cell

**Influenza**
(photo)
**Adaptation of PF to facilitate entry into host**
1. The H and N surface proteins of the influenza virus have evolved to bind to specific receptor proteins on target epithelial cells that line the upper and lower respiratory tract. Once inhaled, the virus quickly binds to host cells, where it is endocytosed before hijacking the cells genetic machinery to replicate. While inside the host cell, it remains hidden from the host's innate and adaptive immune systems
**Adaptation of PF to facilitate transmission between hosts**
2. After the virus has reproduced within epithelial cells, it leaves the cell by pushing through the cell membrane. As it exits, the virus surrounded the RNA capsule with lipids and glycans from the host cell. By hijacking these host cell molecules, the virus is able to avoid detection by the immune system while outside host cells.
3. The influenza A virus has a high mutation rate, referred to as antigenic drift. Changes in the H and N surface proteins (antigens) mean the hosts immune system will not recognise the virus even if it has been encountered previously, as there is no 'memory' of the variant. This results in ongoing transmission of the virus. Antigenic drift makes it necessary to develop annual vaccines against the two primary variants (H1N1 and H3N2) responsible for seasonal flu outbreaks
4. Antigenic shift occurs when there is a new version of the H or N surface proteins introduced into the human population. Importantly, this can involve zoonotic strains exchanging genetic information with human influenza A strains, resulting in antigens not previously encountered by the human immune system. The lack of recognition by the immune system results in rapid transmission of the virus and is responsible for major influenza pandemics, including the 1918 pandemic and the more recent 2009 swine flu pandemic, both H1N1 variants.
5. The virus's main mode of transmission is through inhalation of shed virus in airborne droplets. The typical symptoms of coughing and sneezing are therefore highly effective at transmitting the virus. Furthermore,

once shed, the virus can remain viable on some surfaces for up to 2 days outside a host. It is therefore highly effective at indirect transmission.

**Responses to Pathogens**
**Inquiry question: How does a plant or animal respond to infection?**

investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-sourced investigation, for example:

- ➔ fungal pathogens
- ➔ viral pathogen
- Majority of plant diseases are caused by fungal pathogens
- Airborne fungal pathogens penetrate plant tissues via natural openings (stomata and lenticles), while some soil borne fungal pathogens enter via the plants root system
- All plants have a range of physical and chemical barriers that protect against pathogen entry and slow the spread if entry is achieved
- Physical barriers are structural obstructions that protect against microbe entry
- Chemical barriers are any of the chemical characteristics of the organism that oppose colonisation by microbes
- If a pathogen breaches the protective barriers of a plant, the plant will respond with a combination of cell and chemical mediated mechanisms

**General defences**

- Waxy cuticles on leaves and stems made up of lignin and cutin which makes it hard for pathogens to bypass as they 'slide' off
- Bark - an external layer of dead cells
- Cell wall physically prevents pathogens entering the cell
- Thorns, hairs, bristles
- Antibiotics which kill bacteria specifically
- Toxins target insects that feed on plants

**Active Defence / Barriers**

1.Gene for Gene resistance: some plants possess resistance genes which produce proteins that are able to disable specific pathogens - in response to entry of a pathogen

2.Basal Resistance: When a pathogen is detected, the gaps between cells shut to reduce the chance of a pathogen spreading throughout the tissue

3.Hypersensitive response: if basal resistance fails, cells surrounding the pathogen die, cutting off nutrient source for a pathogen - causes dying / yellowing leaves

4.Systemic (throughout) acquired resistance: produces salicylic acid (signally molecule) to alarm the rest of the plant tissues that a pathogen has invaded - communicates the initiation of other chemical barriers

1. **Gene for gene resistance**
    - Plants have resistant genes (R genes) that code for R proteins
    - Each R protein can bind directly with a specific avirulence (AVR) protein that is coded for by an AVR gene
    - AVR proteins are required to infect host cells
    - Once bound, the R-AVR complex prevents the pathogens, from infecting new cells
    - Becasue closely related species of pathogens have conserved AVR proteins, an R gene offers broad-spectrum protection against any pathogen that produces the same AVR protein
    - However if a plant does not have the relevant R gene, it is significantly at risk of infection

2. **Basal resistance**
    - Triggered by the recognition of pathogen-associated molecular patterns (PAMPs), which are highly conserved molecules common to different groups of pathogens
    - They include chitin, a cell wall component in fungal pathogens
    - In addition, plants can sense 'self' damage-associated molecular patterns (DAMPs), which include the release of cell wall components from infected cells
    - The PAMPs and DAMPs are recognised by host pattern recognition receptors (PRRs), which results in fortification of plant tissues.
    - Cell junctions are closed, restricting access to invading pathogens

3. **Hypersensitive response**
    - Many pathogens have developed mechanisms to inhibit basal resistance
    - If basal resistance fails, the localised hypersensitive response (HR) is activated to restrict and stop the pathogen from spreading to other parts of the plant
    - The plant cells in the region alter the cell wall chemistry, and the pathogen is trapped inside the host and dies with the cell during apoptosis
    - The HR is more effective if the R gene-AVR gene complex is present

4. **Systemic acquired resistance**
    - Systemic acquired resistance (SAR) is a non-specifc, whole plant response that occurs once the HR has been triggered
    - It results in plant tissues becoming highly resistant to a wide range of pathogens for an extended period
    - This systematic resistance is activated by the accumulation of salystic acid (SA), which initiates a signalling pathway to other parts of the plant, causing them to produce a range of broad-spectrum pathogenesis-related proteins

- These ‘prep’ the plant for any pathogens that bypass the localised HR

**Example:**

- Eucalyptus and Acacia form barrier zones in new tissue.
- It is produced by the vascular cambium and prevents pathogen spread from adjacent tissue.
- Acacia binervia(coastal wattle) and Acacia pycnantha (golden wattle) are able to produce cyanogenic glycosides (cyanide-generating chemicals) that are cytotoxic to both insect grazers and invading pathogens.
- The chemical blocks enzymes involved in cellular respiration deterring or killing the insect pest or pathogen.
- Acacia species are susceptible to powdery mildew, a fungal disease caused by one of several fungi.
- The plant will respond as required via the gene-forgene resistance, basal resistance, HR and SAR

- Fungus uses this nutrients to grow more fungi

**Plant Response:**

- Fungus releases small proteins molecules called effectors into the cell
- Plant cell can detects fungus receptors by releasing proteins that act as defence against the fungus by binding to the effector proteins
- This binding informs the plant that there is an infection taking place
- When the plant knows that there are infected cells, it will then sacrifice the cells surrounding by killing them to cut off nutrient supply to the fungus
- After cell death the pathogen will eventually starve from having no nutrients and die
- *Gene for Gene response: produces chitin a chemical which breaks the fungus's cell wall*

- viral pathogens

**Disease:** Banana bunch disease - Banana bunchy top virus

**Causes:** prevents plant producing fruit, reduced growth with emerging leaves becoming choked and bunched

**How:**

- Infected Aphids and other insects that tap into the phloem to feed act as a vector, creating a pathway for the transmission of the virus
- Leaf munching insects create a wound site for viruses to enter into a plant
- Virus invades plant cells in order to replicate more viral particles
- The virus can inhibit cellular processes such as water and nutrient uptake and even photosynthesis

**Plant Response:**

- Systemic acquired response
- Hypersensitive response (evidence is yellowing leaves)

analyse responses to the presence of pathogens by assessing the physical and chemical changes that occur in the host animals cells and tissues

The innate immune system can be divided into two categories that prevent the entry of pathogens into the body; and the cellular and chemical responses that occur if a pathogen breaches these barriers and enters the body. Both are **non-specifc** (don’t target a specific pathogen).

1. Physical barriers
2. Chemical barriers

Any orifice provides an access point for pathogens and possible infection

**1st line of defence**

| **Physical** | **Description and location** | **Image** |
|---|---|---|
| Skin | Made up of a tough layer of tightly packed cells covering all external surfaces, the skin can be penetrated only through wounds. Dead cells forming its outermost layers create an environment too dry for microbes to thrive, and are constantly flaking off, carrying pathogens on the surface | |
| Mucous membrane | These line the bodys openings and internal surfaces, including the mouth and nose, and the tracts of the digestive, respiratory, urinary and reproductive systems. They secrete mucous, a sticky fluid that traps pathogens entering the body through these openings | |
| Cillia | These are microscophic hair-like projections covering the surfaces of the mucous membranes lining the lungs. Beating constantly and rhythmically, they sweep mucous and any trapped dust and pathogens out of the lungs and upwards to the throat where it leaves via coughing or sneezing | |
| Eyelashes | Eyelashes trap dust, dirt, and other small particles that could potentially harm the eye or introduce pathogens. | |
| Urine flow | Urine is passed under pressure during urination, which removes any microbes present in the urinary tracts. | |

| Chemical | Description and location | Image |
|---|---|---|
| Secretions of the digestive tract | Hydrochloric acid, released into the stomach, and alkaline bile, released just below the stomach into the small intestine, help fortify the digestive tract. Most microbes entering with food are killed by stomach acid, and almost all that can survive the stomaches acidic conditions are killed when its contents move into the small intestine, where bile changes the pH from strongly acidic to slightly alkaline | |
| Tears | Tears protecting the eyes surfaces contain an antibacterial enzyme called lysozyme, capable of digesting bacterial cell walls. Bacteria are killed by lysozyme, then tears carry the dead bacteria from the eyes surfaces | |
| Saliva and breast milk | These contain lysozyme, which digests bacteria entering with food, reducing the numbers reaching the stomach | |
| Secretions of sweat glands and hair follicles | These oily secretions containing antimicrobial chemicals create a favourable environment for microflora living on the skin and scalp in a mutualistic relationship with their host. | |
| Cerumen (earwax) | Earwax is high in fatty acids, which lowers the pH to 4, which kills most microbes | |
| Urine flow | Urine is slightly acidic (pH 6) which kills many microbes - chemical as it flows out of the body | |

**2nd line of defence (innate immune system)**

- If a microbe breaches the physical and chemical barriers of the body, antigen molecules on the surface of the microbes are recognised by the host as 'non-self' (antigens), which initiates the innate immune response.
- The activation of the innate immune system in turn initiates the adaptive immune response as required (e.g. production of B and T lymphocytes and antibodies)

1. Inflammation response
2. Phagocytosis
3. The lymph system
    - Lymph is fluid in which white blood (immune) cells are found. Lymph nodes are small, round masses of tissue that are found in certain areas (such as the neck, groin and armpits). They filter bacteria and other foreign materials out of lymph and expose them to B and T cells and macrophages that can engulf them
4. Cell death to seal off pathogens (apoptosis)
    - If the infected cells are surrounded by a wall of dead cells it prevents the infection from spreading to other areas and infecting them
    - This wall of dead cells forms a capsule (granuloma)
    - The cells inside will then die, causing the destruction of the pathogens that are infecting them
5. Fever – caused by histamines. The fever (high temp) kills invaders by denaturing their proteins.

**Innate cellular responses**

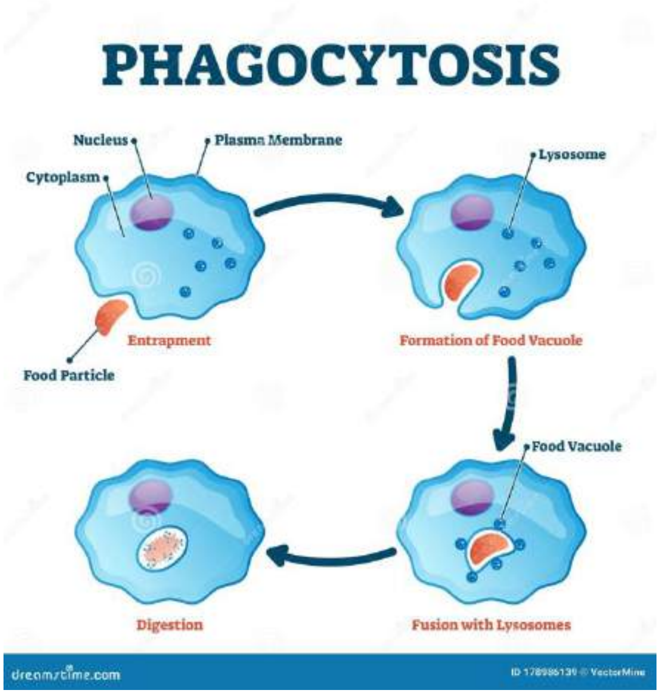
Leukocytes (WBC) are involved in both the innate and adaptive immune responses. Collectively there function is to fight infection. Leukocytes in the innate immune response are categorised as either:

1. granulocyes (neutrophils, eosinophils and basophils) - contain enzyme-filled granules that damage or digest pathogens after phagocytosis
2. monocytes differentiate into either macrophages or dendritic cells both which are phagocytotic

**Three main types of phagocytes are - macrophages, dendritic cells, neutrophils**

| **Leukocyte** | **Category** | **Role** | **Image** |
|---|---|---|---|
| **Macrophage** | Monocyte | > act primarily as phagocytes. Also release chemoattractants, which activate other immune cells and increase their migration to an infected site<br>> fixed macrophages are located in tissues susceptible to pathogen invasions (skin, lungs, intestine)<br>> mobile macrophages circulate the blood stream and lymph system, and are attracted to an infected site by the chemoattractant molecules released by mast cells | |

| | | | |
|---|---|---|---|
| | | > long-lasting phagocytes<br>> once destroyed pathogen, parts of the antigen are displayed on their surface | |
| **Dendritic cell** | Monocyte | > phagocytic antigen-presenting cells (APCs) that express both major histocompatibility complex (MHC) class I and II molecules on their surface. Bridge the innate immune system with the specific adaptive immune<br>system<br>> circulate in the bloodstream where they capture and phagocytose pathogens directly; or are attracted to an infection site upon detecting chemoattractants released by other immune cells | |
| Mast cell | Granulocyte | > in all vascularised tissue. Stimulate vasodilation and vessel permeability through release of histamine, which leads to increased migration of macrophages and neutrophils into infected area | |
| **Neutrophil** | Granulocyte | > phagocytic cells that make up 50-60% of all leukocytes<br>> defend against smaller pathogens, including bacteria, viruses and fungi. The death of neutrophils and tissue surrounding an infection is visible pus<br>> self destruct after couple of days to prevent excessive inflammation and tissue damage<br>> short- lasting | |
| Eosinophil | Granulocyte | > defend against parasitic infections that are too large to be phagocytosed directly. Instead they attach en masse to the parasite and release digestive enzymes from ther granules to destroy the pathogen | |
| Basophil | Granulocyte | > can phagocytose, but their primary role is the release of histamine and heparin. Histamine causes further vasodilation and subsequent inflammation. Heparin prevents clotting from occurring too quickly, which would otherwise inhibit the immune response | |
| Natural killer (NK) cell | Granulocyte (lymphocyte) | > do not require prior activation and therefore play a central role in the innate immune system<br>> continually patrol the body and can distinguish, via the MHC I receptor pathway, between healthy cells and pathogen-infected cells | |

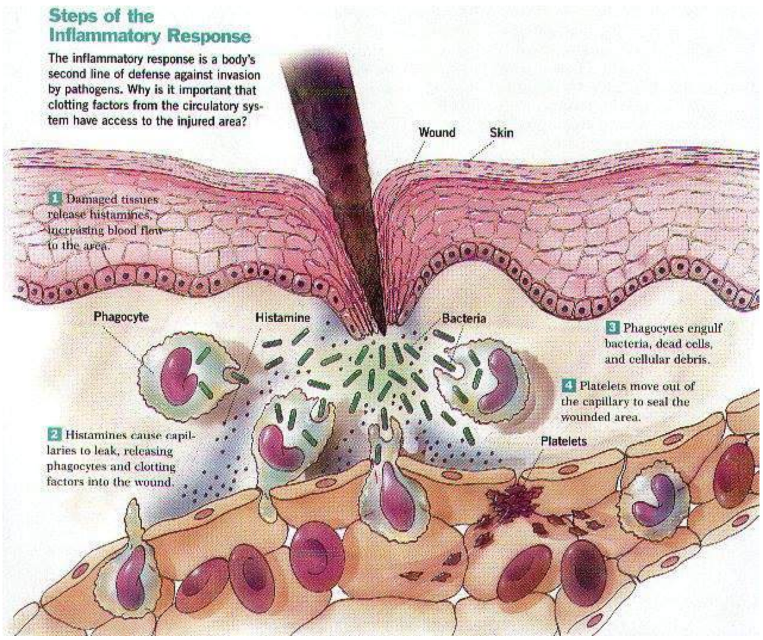



**(picture of MHC class cells)**

**Innate chemical responses**

1. Inflamation response
    - Occurs at site of infection
    - Mast cells are activated when their pattern recognition receptors (PRRs) detect the presence of pathogen-associated molecular patterns (PAMPs) on the pathogens surface, which causes degranulation of the mast cells and the release of proteases and histamine
    - The mast cells and fixed macrophages also release cytokines, which attract macrophages to the infection site
    - Cytokines, including interleukins and interfersons, coordinate the innate inflammation response and also the adaptive immune response. Interleukins act as messenger molecules between immune cells, while interferons are released by infected cells, triggering surrounding cells to become more resistant to the invading pathogen
    - Interferons also target phagocytes and NK cells to target and destroy infected cells

(pciture of interferons)




2. Complement system
    - Includes a group of up to 50 proteins that become activated when they encounter a pathogen
    - Activation triggers a cascade that coats a pathogen in complement proteins, which complements the binding of antibodies to the pathogen and subsequent phagocytosis
    - Also increases inflammation and some proteins act directly on the cell membrane of pathogens to destroy them

**Immunity**
**Inquiry question: How does the human immune system respond to exposure to a pathogen?**
investigate and model the innate and adaptive immune systems in the human body

(draw a diagram when practicing) once ones good paste it in here)

explain how the immune system responds after primary exposure to a pathogen, including innate and acquired immunity

**3rd line of defence - Adaptive immune system -**

- Specific and produces an immunological memory of antigens that have been encountered
- Offers a rapid highly effective response to specific antigens that have been encountered previously
- Antigens are any molecules, usually peptides or proteins that a host recognises as foreign or 'non-self' when the physical and chemical barriers have been breached

**Two groups of lymphocytes are involved in the adaptive immune system**

1. **B cells** are produced and mature in the bone marrow. They are responsible for **antibody-mediated immunity (or humoral immunity)**
2. **T cells** mature in the thymus. They are responsible for **cell-mediated immunity**.

B and T cells can exist in three states

1. **Naive cells** have not been exposed to antigen
2. **Effector cells** have encountered a specific antigen and are actively involved in its removal. Once the pathogen is defeated, proliferation (reproduction) stops and the cells die
3. **Memory cells** continue to circulate after the antigen has been eliminated. It is the memory cells that initiate a quick immune response if the antigen is encountered again

**Activating the adaptive immune system**

- All nucleated cells in the body express proteins on their surface, known as major histocompatibility complexes class I (MHC I) molecules
- Each person has unique MHC I molecules allowing the cells of both the innate and adaptive immune system to identify 'self' from 'non-self' antigens
- If a 'self' cell has been infected by a pathogen, it presents the foreign antigen on its surface via the MHC I molecules
- When an immune cell binds with the MHC I molecule, the foreign antigen is recognised and an immune response will result

(photo)

- In addition to MHC I molecules, antigen presenting cells (APCs: macrophages, dendritic cells and B cells) also express MHC class II (MHC II) molecules
- After phagocytosis, a pathogen is digested and a portion of the pathogen is expressed on the surface of the phagocyte via the MHC II molecule
- The APC then migrates to the lymph system, where it presents the specific antigen to helper T cells via the MHC II molecule or to natural killer cells (cytoxic T cells) via the MHC I molecule
- It is at this point that the adaptive immune system is activated
- Once activated T helper cells and T cytoxic cells mature and proliferate

(photo)

**Antibody-mediated immunity**

- Helper T cells activate the antibody-mediated response
- The T helper cells bind (via their T cell receptors) to the MHC II molecule of naive B cells that are expressing the same specific antigen that activated the T helper cells

- This, combined with the release of cytokines by the T helper cells, promotes rapid clonal expansion of the B cell population specifically linked to the identified antigen

(photo)

Each naive B cell contains a unique B cell receptor (BCR) capable of recognising a specific antigen. The antigen is processed and presented on its cell membrane via MHC II molecules. The activated B cells differentiate into memory B cells or plasma cells, which act as antibody factories. The circulating antibodies bind directly with antigens to form an antibody-antigen complex, preventing them from causing further infection and making them easier to recognise and be destroyed by phagocytes. The memory B cells (and memory T cells) are the basis of acquired immunity. They provide ongoing immunity due to their receptors, which are specific to a previously encountered antigen.

**Antibodies**

- Antibodies (immunoglobulins) are quaternary proteins consisting of four polypeptides: two heavy chains and two light chains joined to form a Y-shaped molecule.
- Most of the heavy chain and the lower end of the light chain are conserved in all antibodies.
- In contrast, the ends of the heavy and light chains form variable regions that are unique to each antibody and act as the antigen-binding site.
- Antibodies are naturally produced by B cells in response to antigen exposure and therefore play a critical role in the immune system's defence against infection and disease.
- Antibodies function in one of several ways, including neutralisation, aggulination, opsonisation and complement activation
- In all cases, pathogen mobility and reproduction are inhibited, allowing other components of the immune system to destroy the pathogen

(photo)

**Cell-mediated immunity**

- Antibodies are unable to bind antigens within a cell. As a result, the cell-mediated immune response is required to target pathogens that have already infected host cells. This process is regulated by T cells, which only recognise antigens when they are expressed on the surface of an APC (via MHC II molecules) or an infected cell (via MHC I molecules).
- All T cells express a T cell receptor - cluster of differentiation 3 (TCR-CD3) receptor complex, which is required to bind both MHC I and MHC II molecules. The CD3 receptor is conserved between the cell types, while the TCR is highly variable, to account for different antigens. The different types of T cells are classified based on their additional surface CD receptors. For example, T helper cells express CD4, and T cytotoxic cells express CD8, which are also required to bind MHC II and MHC I molecules of APCs, respectively.

**(photo)**

- The cell-mediated response is activated when an APC presents a specific antigen via an MHC II molecule to a complementary T helper  cell. The T helper cells then release cytokines to activate the mass cloning of T cytotoxic cells and memory T cells with receptors specific to the antigen.

- Activated T cytotoxic cells exit the lymph system and migrate to the site of infection, where they bind via their specific receptor to infected cells presenting the antigen via their MHC I receptors. The T cytotoxic cells degranulate, releasing cytotoxins including perforin, which lyse the cell, killing any pathogens within it. T cytotoxic cells perform a similar function to NK cells, with one key difference: they require activation and, as such, form part of the adaptive immune response.
- Once an infection has been defeated, regulatory T (T reg) cells are produced. T reg cells release cytokines that prevent further production of T cytotoxic cells and B cells. Most circulating immune cells linked to the defeated pathogen die off, except for a population of memory T cytotoxic and B cells.
- These memory cells can remain circulating or accumulate in the spleen and lymph system, where they will rapidly proliferate to very large numbers if they are exposed to an antigen they have encountered before. Acquired immunity has been achieved.

(photo)

**Prevention, Treatment and Control**
**Inquiry question: How can the spread of infectious diseases be controlled?**
investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease

**Key considerations about what measures need to be implemented to reduce the spread include:**
- Virulence (harmfulness)
  - ➢ Highly virulent pathogen can cause serious disease, even in low numbers
  - ➢ Less virulent pathogen only becomes serious whe large numbers are present
- Infectiousness
  - ➢ Highly infectious pathogens are easily transmitted
- Ability to survive outside host
  - ➢ Some pathogens quickly perish outside the body and can only be transmitted through close contact
  - ➢ Others that are more robust are more likely to spread widely to cause an epidemic

**Covid-19**

| **Level of intervention** | **Details of intervention strategies** |
|---|---|
| Local | > forced two week quarantine for indiviudals returning from overseas; fines for breaches<br>> forced lockdowns, mandatory mask wearing in all public spaces, nighttime curfews and travel bubbles where enforceable<br>> ‘pop up’ testing and tracking facilities, central to limiting the spread of COVID-19<br>> vaccine targets linked to eased restrictions with mandatory vaccination required for certain professions. Proof of vaccination required to enter businesses and public facilities<br>While introduced at the regional level, most of these measures were monitored and enforced at the local level |

| | |
|---|---|
| Regional | > ‘hard’ border lockdowns to restrict movement within and between states<br>> extended lockdowns, with heavy restrictions on the liberties and movement of residents. Government approval and permits required to travel into regional areas<br>> national cabinet formed to address the specific health and economic needs of each state and territory. Included a staged approach to the vaccination rollout, with elderly and people forced into unemployment by the lockdowns<br>> ‘no jab, no play’ and ‘no jab, no pay’ legislation |
| Global | The WHO is responsible for the global governance of health and disease. It communicates directly with policymakers at the national and regional levels to inform them of trending disease patterns. It also recommends and supports specific strategies to limit disease spread.<br>FOR COVID-19 this included:<br>> mathematical modelling to track and predict spread of the different virus variants. This allowed regions to be proactive rather than reactive to outbreaks<br>> a coordinated approach involving thousands of scientists and public health authorities, to produce and distribute vaccines. Within 18 months of the disease first being identifies, multiple companies had produced vaccines.<br>> where necessary, the planning and building of emergency operation centres, training of health workers and financing of vaccination programs |

investigate procedures that can be employed to prevent the spread of disease, including but not limited to: hygiene practices

**Hygeine practices**

| **Level of hygiene** | **Details** | **Examples** |
|---|---|---|
| **Personal** | Practices that an individual can do to maintain good health and minimise the risk of becoming infected or infecting others when sick | > handwashing with soap and water after using the toilet, before eating or handling food<br>> frequent bathing<br>> correct respiratory hygiene when coughing and sneezing<br>> using protection during sexual intercourse |
| **Domestic** | Practices done at the household level to minimise disease spread and focussing on food storage and preparation | > keeping raw and cooked foods separate<br>> cooking food for the appropriate time and at the appropriate temp<br>> storing food at the recommended temperature<br>> using clean water for cooking |

| | | |
|---|---|---|
| **Community** | Large-scale infrastructure programs that focus on providing society with access to clean water and food, and sanitary conditions | > water treatment and purification (filtration and chlorination)<br>> sewage removal and treatment<br>> correct animal husbandry techniques<br>> maintiancence of hygiene in markets |

**Herd immunity**

- Some individuals cannot be vaccinated, usually for age-related or health reasons
- When a significant majority of individuals are vaccinated (around 90%), unvaccinated individuals are also protected as there are fewer hosts available for the disease to spread between

➔ Quarantine

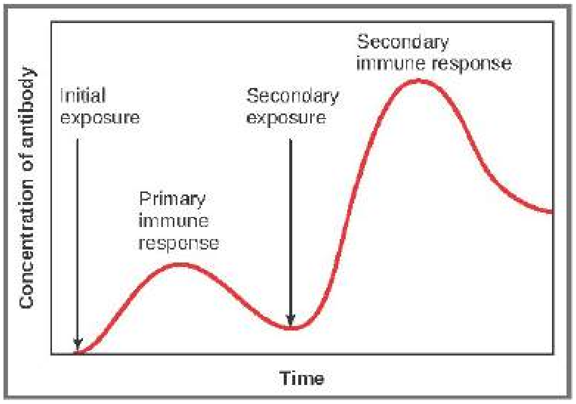
- Period of strict isolation to prevent the introduction and spreaf of disease or unwanted animals, plants and pests into the country or across state borders
- Quaratine period is typically for the period in which a pathogen is communicable
- Quarantine, if effective prevents, unwanted animals, plants and pests into the country or across state borders.
- The quarantine period prevents infected individuals from encountering non-infected individuals.
- Other biosecurity strategies that complement quarantine include border inspections, enforced destruction of diseased animals and plants, and banning the importation of organic products from most countries.

➔ vaccination, including passive and active immunity

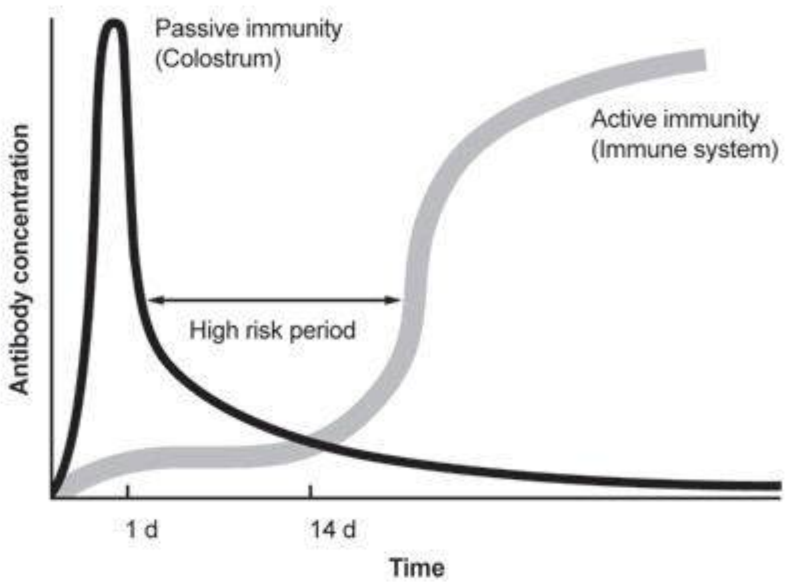
- **Vaccination** is the process of making people (or animals) resistant to diseases caused by specific pathogens.
- The immunity provided is either active or passive, depending on the source of the immunity.

- **Active immunity** occurs when activated B cells differentiate into plasma cells and antibodies are produced. This can occur naturally when an individual is exposed to a pathogen or artificially through vaccination.
- Vaccines activate the adaptive immune system, leading to the production of antibodies and memory B and T cells specific to the antigen. They work by exposing the immune system to a compromised version of a pathogen, which induces a primary immune response without the associated symptoms of the disease. This exposure provides active immunity with long-term protection from the disease.
- If a secondary exposure occurs, through either a booster shot or exposure to the actual pathogen, the immune response is faster and stronger, due to the presence of circulating antibodies and memory B and T cells generated from the primary exposure. This leads to more effective

elimination of the pathogen and results in higher levels of antibodies, which remain circulating for longer periods.




- **Passive immunity** involves an individual receiving antibodies not produced by their own immune system.
- This can occur naturally, from mother to foetus via the placenta, or from mother to newborn during breastfeeding. It can also be achieved artificially through vaccination, where antibodies are injected directly into an individual (e.g. certain diphtheria and tetanus vaccines).
- Passive immunity bypasses the adaptive immune system and provides immediate protection against a disease.
- However, because no memory B or T cells are produced, the protection is short- lived, and booster shots must be given if an individual is deemed to have been at risk of exposure.




- ➔ public health campaigns
  - Seek to provide community members with the necessary information to have them engage in behaviours that improve their health and prevent the spread of infectious disease
  - Campaigns use media platforms to spread information and reach target audiences
    - ★ No Jab, No Play - Mandated that children must be fully vaccinated to attend childcare and kindergarten unless they have a medical exemption.
    - ★ Media Platforms: Television, radio, social media, and government websites to spread information about the importance of vaccinations
- ➔ use of pesticides
  - Pesticides (insecticides, herbicides and fungicides) are another strategy to prevent the spread of infectious animal and plant diseases.
  - These chemicals target pathogens and insect vectors that carry pathogens
  - While there is still widespread dependence on pesticides, there is also growing awareness of the impact they can have on people's health and the environment.
  - Furthermore, because of the widespread use of pesticides and the rapid life cycle of target species, genetic resistance has developed in most cases.

- ➔ genetic engineering
  - Genetic engineering involves the intentional modification of an organism's genome to alter its phenotype.
  - While there are legitimate concerns associated with genetic engineering, it is particularly valuable in agriculture because of the growing resistance of pathogens and insect vectors to traditional chemical treatments.
    - ★ A familiar example is Bt crops (e.g. cotton, corn, potato), which are created by inserting a trans-gene from the soil bacterium Bacillus thuringiensis (Bt) into a

plant's genome. When the trans-gene is expressed, the plant produces a protein toxin that kills any insect pests that attempt to eat it.

investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example:

- ➔ antivirals
  - Because viruses require a host cell to reproduce, there are significant challenges to developing antiviral drugs
  - Any attempt to target the virus inevitably causes death or damage to the host cells
  - As a result, they never provide a cure, instead they, suppress the rate of viral replication by a range of mechanisms including:
    - ➢ Inactivation of the capsid or envelope proteins, which inhibits viral attachment to the host cell
    - ➢ Inhibition of transcription and translation during viral protein synthesis
    - ➢ Inhibition of new viruses being released from host cells
- The **effectiveness** of antivirals is further challenge by the high mutation rate of viruses, particullarly that of RNA viruses, which can be up to 1 million times higher than that of the host
- This results in rapid development of drug resistance
- In summary antiviral medications are effective at reducing symptoms and in some cases extending the life of infected individuals
- However, the specificity of antiviral drugs and the high mutation rate of viruses make most drug development options economically non-viable

(photo of virus)

- ➔ antibiotics
  - Categorised as either, bactericidal or bacteriostatic
  - Bactericidal antibiotics kill bacteria by interfering with the formation of the cell membrane cell wall or cell contents
    - ★ Penicillin and vancomycin
- Bacteriostatic antibiotics inhibit bacterial growth by interfering with DNA replication and protein synthesis
- They do not kill the bacteria and therefore rely on the immune system of the host to clear the infection
  - ★ Amoxicillin and clindamycin

- Both antibiotic groups have broad-spectrum and narrow-spectrum examples
- Broad-spectrum antibiotics are effective against a wide range of bacterial species and are the most commonly prescribed as they can be used when the causative pathogen is suspected of being bacterial without the actual species being known
- In contrast narrow-spectrum antibiotic is highly specific and acts only on a very small number of bacterial species

- Key strength compared to antivirals is that they target cell structures and molecules not found in humans, such as the cell wall or prokaryotic ribosomes
- They do not affect host cells and therefore have very little side effects
- However, the overuse and misuse of antibiotics has led to the evolution of ‘superbugs’ - bacteria that are resistant to antibiotics
    - ★ Drug resistant tuberculosis, gonorrhoea
- Without the development of new drugs, diseases that have been long controlled by antibiotics will evolve to cause higher mortalities

(photo of bacteria)

investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic

- The primary principle of controlling disease spread during epidemics and pandemics is the same; prevent infected individuals coming in contact with non-infected individuals
    - ➢ To do this, environmental and quaratine methods can be employed
- Environmental management: vaccination, provision of clean water and food, appropriate sanitation and PPE, control pop movement, maintaining good air quality
- Quaratine: inspection, regulation, restriction of movement, enforced destruction of diseased organism

**COVID-19 (SARS-COV-2)**

| **Method** | **Control** | **Evaluate** | **Cons** |
|---|---|---|---|
| **Quarantine** | | | |
| Isolation of travelers | Prevented initial spread by isolating potential cases. | Initially effective in preventing the virus from entering the community. | High economic impact on travel and tourism industries. Social and psychological impact on individuals in quarantine. |

| | | | |
|---|---|---|---|
| Hotel quarantine | Contained travelers in controlled environments to monitor for symptoms. | Effective in containing potential cases initially; however, breaches led to significant outbreaks. | Breaches in quarantine protocols led to outbreaks in New South Wales and Victoria. High cost and resource-intensive. |
| **Environmental Management** | | | |
| Pop-up testing facilities | Enabled widespread testing to identify and isolate cases quickly. | Highly effective in tracking and isolating infected individuals. Essential in managing outbreaks and reducing community spread. | Logistical challenges in setting up and maintaining facilities. Limited access in remote areas. |
| Lockdowns | Forced people to stay at home, reducing virus transmission opportunities. | Extremely effective in reducing new cases, as seen in Victoria. Drastic drops in cases post-lockdowns 1 and 2. | Significant economic impact. Mental health issues due to prolonged isolation. |
| Mandatory mask wearing | Required masks in public spaces to reduce aerosol transmission. | Scientifically proven to reduce spread by up to 80%. Combined with social distancing, effectiveness increased to 90%. | Public resistance and compliance issues. Discomfort and difficulty in enforcing in all areas. |
| Social distancing | Reduced close contact between individuals to minimize virus transmission. | Effective in combination with other measures. Reduced transmission rates significantly. | Difficult to maintain in crowded areas and certain work environments. Impact on social interactions and businesses. |

| | | | |
|---|---|---|---|
| Vaccination rollout | Immunized population to reduce severity and spread of the virus. | Highly effective in preventing severe illness and death. Reduced ICU admissions among double vaccinated individuals to 1.9%. Cumulative protection with full vaccination. | Vaccine hesitancy and misinformation. Logistical challenges in distribution and ensuring equitable access. Time lag in achieving high vaccination coverage. |

interpret data relating to the incidence and prevalence of infectious disease in populations, for example:

➔ mobility of individuals and the portion that are immune or immunised

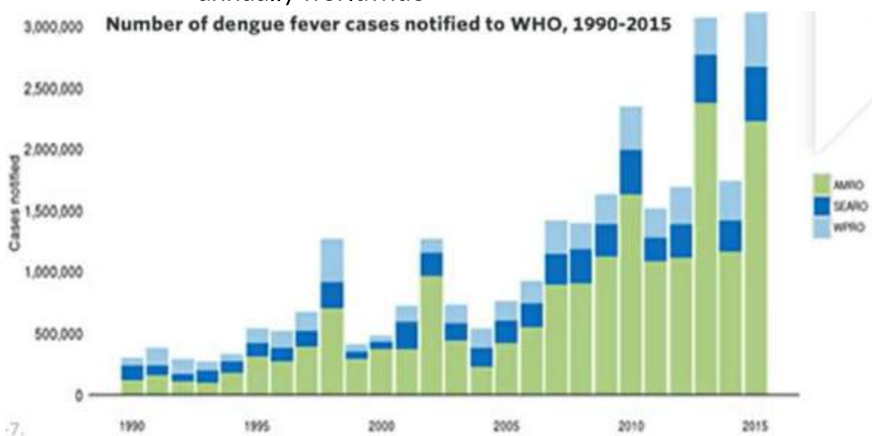
| | |
|---|---|
| Prevalence<br><br>Total number of current cases during specific time/<br>Size of pop at start of monitoring period<br>x 100 | The total number of cases of disease in a population at a specific time |
| Incidence<br><br>Number of new cases during specifc time/<br>Size of pop at start of monitoring period<br>x 100 | The number of new cases of a disease occurring in a population at a specific time |

- **Mobility** of a population is an important consideration when assessing the risk of disease transmission
    - ➢ Includes movement within a region and between states, and international travel
- Humans act as a carrier for pathogens therefore population mobility increases the chance of the spread of disease. Although some diseases that are infectious, we have immunity against them (vaccinations). For some diseases there are no vaccines
- All must be considered in combination with incidence and prevelance rates to determine what limiatios need to be imposed on a pop to reduce transmission
- Herd immunity

➔ Malaria or **Dengue Fever** in South East Asia

- Dengue fever is a mosquito-borne viral infection found in tropical and sub-tropical climates worldwide
- Causes flu-like symptoms and can be potentially lethal

- Due to increased urbanisation in these regions, aprox half of the world pop is now at risk of contracting it
- Incidence has continued to grow in recent decades
- Due to increased diagnosis, the WHO belives the actual number of reported cases is low, with some estimates suggesting there are 390 million new infections per year
- Of these, 500 000 cases develop into severe dengue, which results in up to 25 000 deaths annually worldwide




- Increased incidence over course of time

evaluate historical, culturally diverse and current strategies to predict and control the spread of disease

- Prior germ theory, predicting and controlling disease spread was largely dependent on observed correlations between specific behaviours or environmental conditions and disease outbreaks
- In response, practices were introduced to minimise contact between infected and non-infected people
- After development of germ theory, specific pathogens and their mode of transmission could be linked to a particular disease hence creating improvements in strategies used to predict and control disease spread

| | |
|---|---|
| **Historical strategies** | > Lack of scientific knowledge meant that religion influenced cultural values and beliefs<br>★ Bubonic plague: killed min 25 million during middle ages, was attributed to the wrath of God punishing sinful behaviour<br>>> this early historical view of disease spread resulted in limited sucess in predicting and controlling outbreaks<br><br>> late 14th century, link was made between the plague and movement of symptomatic people between towns and cities. In response, the concept of quarantine was first introduced in European ports to reduce transmission. Ships were required to remain anchor for 40 days before crew were permitted to leave<br><br>> in mid 1840s Ignaz Semmelweis pioneered antiseptic procedures, including handwashing before performing medical procedures. Despite limited knowledge of pathogens, semmelweis made the connection between |

| | |
|---|---|
| | women dying during childbirth from puerpal sepsis and doctors delivering babies after performing autopsis on cadavers. By introducing mandatory handwashing and antiseptic procedures prior to child birth, it would reduce mortality rates. After procedures were installed women had a 2% chance of dying from 20% proving it had been controlled<br><br>> in 1854, John Snow proposed that cholera was water borne through identifying the trend that higher prevalence of cases occurred around specific water pumps<br>> Removed water pump that was responsible for harbouring contaminated water<br>> Effective as educated people on the importance of water sanitation |
| **Cultural strategies** | ★ over 4000 years greek, roman and chinese cultures disinfected water by either boiling or charcoal filtration. While they had no concept of germ theory, the link had been made between consumption of dirty water and disease.<br>> In contrast some cultural practices can promote spread of infectious disease.<br>★ West african countries, burial ceremonies involve direct contact with deceased. This practice proved to be one of the main modes of transmission through bodily fluids and is highly contagious with mortality rate at 90%. The epidemics were bought under control through a range of initiatives including improved public health and education campaigns and enforcement of safe burials |
| **Current strategies** | > Global Surveillance of health cases (similar to John Snow)<br>- Healthcare providers are required to report all cases of specific diseases to the Australian Department of health (has been made easier through technology)<br>- This allows for early detection, identification of the source and data analysis of health trends to create effective responses to outbreaks<br>★ during COVID tracking movement of people through sign in helped to identify close contacts to prevent further spread<br><br>> recent control measure focus on R-nought value, which is an indicator of how contagious an infectious disease is. The goal is to reduce r-nought to less than 1.<br><br>> Health Campaigns<br>- Educate people on the importance of hygiene<br>★ during COVID wearing masks, washing hands and getting tested if signs and symptoms appear<br><br>> Quarantine & Isolation<br>- Individuals who have the disease are isolated from society until they no longer harbor the pathogen<br>- This prevents the transmission of the pathogen, controlling the spread<br><br>> Antibiotics & Anti-viral drugs.<br>- Help to prevent the growth and kill bacteria + viruses therefore preventing transmission to control spread ><br><br>>Vaccines<br>- Strengthens the immune system through activating memory B and T cells<br>- Prevents pathogens infecting new hosts, reducing transmission and controlling spread of pathogen<br>- Herd immunity can protect individuals not vaccinated<br><br>>Viruses being used as vectors |

| | |
|---|---|
| | - Genetically engineering viruses to inject beneficial DNA into organisms can provide organisms DNA to fight of infections that require this gene |

investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example:

- ➔ bush medicine
    - Historical and traditional use of native Australian plants for both physical and spiritual healing by Indigenous Australians
    - Research into this has discovered a range of active ingredients, including some with anti-cancer and antiviral properties

**Tea tree oil (Melaleuca alternifolia)**
The tree is endemic to Australia. Indigenous Australians use various parts of the tree to create a paste that contains the oil. It is used as an antiseptic to treat wounds and infections.

Pharmaceutical companies have determined the active ingredients to be: terpinen-4-ol and 1,8-cineole, alphaterpineol and gamma-terpinene.

**Eucalyptus oil (Eucalyptus sp.)**
There are over 700 Eucalyptus species across Australia. Indigenous Australians have long used the oils for their natural antimicrobial and antiseptic properties.

Pharmaceutical companies have determined the active ingredient to be 1.8-cineole. Now used commercially in mouthwash, throat lozenges and cough suppressants.

**Emu bush (Eremophilaglabra)**
Indigenous peoples of the Northern Territory used the leaves to make a antiseptic solution that was used to treat wounds.

Pharmaceutical companies have found Eremophila glabra has the same antibacterial qualities as established antibiotics. Researchers have proposed using the plant for to sterilise artificial joints prior to surgery.

- ➔ smoke bush in Western Australia

**Smoke bush (Conospermum stoechadis)** is located in South Western Western Australia (WA), between Geraldton and Esperance. It has been used by Aboriginal people for thousands of years for its natural healing properties.

Researchers have long been interested in the plant as a potential source of treatment for various diseases, including cancer and HIV.

Between 1960-1980 extensive research by the USA National Cancer Institute found no application for the plant in the treatment of cancer.

In the late 1980s, smoke bush was one of 7,000 plants screened as a possible source of treatment against HIV. It was one of four plants found to contain the active ingredient conocurovone which can destroy the HIV virus in low concentrations.

**Debate**
In response to the isolation of conocurovone from smoke bush, the USA government filed for a patent in both the USA and Australia. This provided the USA government with exclusive licensing rights to the use of smoke bush in the development of any medicinal treatments.

The WA Government has the legislative power to grant licensing rights and commercial deals of its choice to pharmaceutical companies. For example, AMRAD paid over $1.5 million for exclusive research rights to smoke bush. If successful, royalties are estimated at over $100 million p/a.

At the time of issuing the license, the WA government had not acknowledged or included the Aboriginal people of WA in any royalties or compensation agreements.

**Considerations**
- Aboriginal peoples have used smoke bush for thousands of years for its healing properties and passed on this knowledge from generation to generation.
- Government agencies and pharmaceutical companies have invested significant time and millions of dollars in researching smoke bush (and other bush medicines).

**Questions**
- Are the pharmaceutical companies entitled to a return on their investment? If so, how much?
- Are the Aboriginal peoples of WA entitled to a return on their investment? If so, how much?
- Could a patent restrict the use smoke bush by Aboriginal peoples?

**Module 8: Non-infectious Disease and Disorders**

**Homeostasis**
**Inquiry question: How is an organism's internal environment maintained in response to a changing external environment?**
construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to:

- temperature
- glucose

- **Homeostasis** is the process by which organisms maintain a relatively stable internal environment. This stable state of environment is essential for efficient metabolic activity
- **Homeostasis** is regulated by the nervous and endocrine systems
- When changes from the normal state are detected, the body responds to return each system to the normal state - negative feedback loop
- Homeostasis consists of two stages
    1. Detecting changes from the stables state
    2. Counteracting these changes from the stable state
- The process by which this occurs is the stimulus- response pathway, which can be represented by negative feedback loops.
- For each 'system' such as body temp, blood glucose and osmoreregulation, two negative feedback loops operate to counter movement in either direction away from the 'normal state'

- Any change that induces a response is called a **stimulus**
- The stimulus is detected by a **receptor** which receives and transmits an impulse to a **control centre**
- The control centre interprets the message and sends a signal to the appropriate **effector** which brings about a **response** to restore the normal state

(picture)

(picture of table)

**Thermoregulation** is the ability of an organism to maintain its body temperature within a certain range independently of its external environment.

★ In humans, thermoregulation maintains the body temp at 37 to ensure the structural integrity of enzymes that catalyse cellular reactions

**Glucose** is an essential requirment for the production of adenosine triphosphate (ATP) during cellular respiration. ATP is a molecule that carries energy for most chemical reactions in a cell.

**Osmoregulation** involves the maintenance of osmotic pressure in an organism. An abnormally low or high concentration of sodium in the blood can lead to health problems ranigng from nausea, muscle weakness and confusion to even death

investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including:

**Endotherms** is an organism that uses energy to regulate its internal body temperature

- Range of adaptations in maintaining homeostasis (behavioural, structural and physiological)

➔ trends and patterns in behavioural, structural and physiological adaptations in endotherms that assist in maintaining homeostasis

**Behavioural adaptations**

- Concious actions organims take to warm up or cool down, to avoid hypothermia and hyperthermia

| Type of thermoregulation | Function | Behavioural adaptation |
|---|---|---|
| • Hypothermia avoidance | • To keep body heat<br>——————————<br>• To enhance body heat | • Physical postures (basking, rolling into a ball)<br>• Nest building<br>• Grouping strategies (huddling, nest-sharing)<br>——————————<br>• Increased movement<br>• Increased energy intake (i.e. more eating) |
| Hyperthermia avoidance | • To dissipate body heat<br>——————————<br>• To decrease body heat production | • Habitat selection (places with shade and water)<br>• Physical postures (stretching out body)<br>• Panting<br>——————————<br>• Reduced movement<br>• Decreased energy intake (i.e. less eating) |

Animals will follow patterns of behaviour to regulate their body temp.

★ Koalas adopt different body postures in trees, depending on the heat of the day. This pattern of behaviour can be presented as a histogram, with explanations of the different body positions in relation to thermoregulation

(photo)

**Structural adaptations**

- Physical features of an organism that assist in maintaining homeostasis (e.g. body shape, colour, thickness of fur)
- Allens rule states that there is a trend in the length of limbs and appendages (part of an organism) in relation to the climate
- The SA of an endotherms body, relative to its body volume, affects the amount of heat lost to the surrounding air.
- Differences in body shape, limb length and ear size can be seen in hares living in habitats with a cold versus a hot climate

(picture)

**Physiological adaptations**

- Occur within an organism to regulate and maintain homeostasis (e.g. changes in heart rate, oxygen uptake, hormone levels)
- Antidiuric hormone (ADH) regulates the amount of water in the body
- When released from pituitary gland, it prevents the kidneys from making too much urine and results in water retention
- The spectacled wallaby (Lagorchestes conspicillatus) and Rothschilds rock wallaby (Petrogale rothchildi) both live in arid environments
- The hare wallaby shelters in spinifex bushes during the heat of the day, when the air is dry and the temp reachers 40+
- The rock wallaby shelters in cooler caves with a temp of 30 and higher humidity
- The level of ADH in the blood of each species shows very different trends
- The hare wallaby uses ADH as a physiological adaptation to reduce urine production, to retain water in the body
- The rock wallaby has a different physiological response
- It relies on reduced blood flow to the kidnets instead of hormonal control
- This combined, with its ability to seek out cool rock shelters in its habitat allows this species to regulate its internal water levels

(phtoo)

➔ internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways

**Hormones in animals**

- Hormones are signalling molecules secreted by special groups of cells, called endocrine glands, to maintain homeostasis
- They travel in the blood and in the fluid between cells

- There are three types: steroid hormones, peptide hormones and amine hormones
- The chemical structure of a hormone determines on its solubility, the way it travels through the body and how it interacts with cells

| Hormone group | Solubility | Chemical Structure | Mode of action | Examples |
|---|---|---|---|---|
| Steroid | Fat soluble | Made from cholestral | > Slow onset but long lasting<br>> travel in the blood attached to a water-soluble carrier protein<br>> freely diffuse across the plasma membrane<br>> bind to protein receptors in the cytoplasm and nucleus to activate gene transcription | Testosterone, estrogen, and progesterone |
| Peptide | Water soluble | Small chain of amino acids | > fast acting and transient<br>> travel freely in the blood<br>> cannot move accross the plasma membrane<br>> bind to receptors on the cell surface to activate signal transduction and gene transcription | Insulin and oxytocin |
| Amine | Fat or water soluble | Derivatives of the amino acid tyrosine | > share properties with steroid and peptide hormones | Adrenaline |

- Hormones can signal to the same cell (autocrine), cells that are nearby (paracrine) or cells at distant sites in the body (endocrine)
- Each type of hormone affects cells at different distances and their effects can last for varying periods of time
- Some hormones bind to a specific receptor on many different types of cells, while others only bind to a receptor on specific cells

(photo)

| Forms of hormone signalling | Description | Example |
|---|---|---|
| Photo | > localised effect<br>> cell releases a hormone that is recognised by itself<br>> occurrs during early development and infections | T-lymphocytes release a hormone in response to a viral infection, which stimulates the production of more lymphocytes to kill infected cells |

| | | |
|---|---|---|
| | > causes a quick response<br>> last a short time<br>> hormone is quickly degraded | Blood clotting. When a blood vessel is damaged, broken endothelial cells release von Willebrand factor (vWF), which stimulates the production of platelets. These blood cells clot the blood and ‘patch up’ the wound |
| | > slow onset but long-lasting effects<br>> produced by cells in endocrine glands | Menstual cycle. LH and FSH hormone are produced in the pituitary gland and target the ovary to produce and release mature eggs |

**Hormones in plants**

- Hormones are produced in different parts of a plant..e.g. Stem tip, buds or root tops
- Travel through the phloem and xylem and pass between cells through plasmodesmata
- There are five main groups of plant hormones

| **Plant hormone** | **Function** |
|---|---|
| Auxins | > control the primary elongation of stems and inhibit the lateral growth of branches<br>> control the growth of stems towards light (phototropism) |
| Cytokinin | > produced in the tips of roots and travels upward through the xylem<br>> promotes cell division in growth areas<br>> balance between auxins and cytokinins determines whether the plant produces roots or shoots |
| Gibberellins | > trigger seed germination and growth of the stem region between internodes (places from which leaves grow) |
| Ethylene | > gas produced by plants that stimulates ripening of fruit, loss of leaves (abscission) and flower death |
| Abscisic acid | > synthesised in chloroplasts of leaves and is a signal of dehydration<br>> regulates water loss by controlling the opening and closing of stomata<br>> also promotes seed dormancy |

(photo)

**Neural pathways**

- All animals have a nervous system
- The vertebrate nervous system has two main parts
    1. The central nervous system (CNS)
        - Brain and spinal chord
        - Brain is the processing centre of the body
        - Spinal cord is an extension of the brain that carries electrical and chemical signals between the brain and the PNS

- Three main parts of the brain are: cerebrum, cerebellum and brainstem and each controls different functions in the body
  (photo)

2. The peripheral nervous system (PNS)
   - Consists of all the nerves that are not part of the CNS, that branch out to the organs, glands and muscles
   - The PNS is divided into the somatic and autonomic nervous systems, based on the parts of the body and the responses they control
   - Some actions, such as breathing, are done without thinking others such as moving an arm are done voluntarily

(photo)

- The basic building block of the nervous system is the neuron
- A specialised type of cell that has three main parts: dendrites, an axon and a cell body (soma)
- The dendrites are where a neuron receives a signal from another neuron
- The axon is where a neuron sends a signal to a neigboring neuron
- The soma is where the cell nucleus is located
- Transcription and translation occur here, and proteins are transported for use in dendrites and axons
- Multiple axons bundle together to form a fibre, and fibres group together to form a nerve
- Signals travel through neurons as an action potential
- The tips of neighboring neurons do not touch each other; they are separated by a small gap called a synapse
- When an action potential reaches the end of a neuron, it triggers the release of neurotransmitters
- These chemical molecules travel across the synapse, bind to receptors on the next neuron and the message continues along the nerve

(phpto)

- **Motor neurons** in the PNS transmit signals from the CNS to organs, glands and limbs
- **Sensory neurons** provide feedback on the status of the action from the periphery of the body to the CNS
- Nerves are classified as afferent or efferent based on the diffraction of the information flow
- Afferent nerves are made up of sensory neurons, they receive an input from the environment, such as light or sound, and transmit it to the CNS
- Afferent nerves are also responsible for homeostasis in the body
- Efferent nerves consist of motor neurons that transmit signals in the opposite direction from the CNS to organs and muscles to initiate an action
- The nervous system of invertebrates varies in its structure and complexity
- Even though some invertebrates have a brain, they do not have a centralised nervous system like vertebrates
- This means that nerves throughout the body can control actions independently of the brain

(phtoo)

To summarise, the neural system is a control centre in negative feedback loops that maintain homeostasis
A change detected by a receptor is transmitted as an electrochemical signal along nerves to the brain, which results in a signal that is transmitted back along the nerves to cause a response

- ➔ mechanisms in plants that allow water balance to be maintained
    - Stomata play a critical role in transpiration and water balance in plants
    - Stomata are openings (pores) on the underside of leaves
    - Each pore is surrounded by two guard cells, which can change the size of the pores opening by expanding and contracting
    - The two main functions of stomata are gas exchange and regulation of weather movement by transpiration
    - Plants need to balance the instake of $CO_2$, needed for photosynthesis and growth, with the loss of water by transpiration
    - Stomata size varies depending on the time of day and environmentaal conditions
    - Their size is regulated by the active transport of potassium ions and osmosis, which changes the turgor of guard cells and the size of pore opening
    - Stomato open during the day when photosynthesis occurs, and close at night to reduce water loss
    - The closure of pores is also a physiological adaption to converse water during stress
    - When water is scarce or salinity is high, the hormone absicisic acid binds to proteins in guard cells, which results in the exit of water, shrinkage of the guard cells and closure of the pore
    - This disrupts photosynthesis and growth but will only change when the stress signal is reduced

(phpto)

- Plants also have a range of structural adaptations to minimise water stress
- Plants in arid climates (e.g. cacti) have a reduced number of stomata, leaves with a wavy (hydrophobic) surface and a reduced SA to minimise water loss via transpiration
- Succulent plants can store water in their leaves and stems, while white mangroves secrete excess salt through glands near the tip of each leaf stalk

**Causes and Effects**
**Inquiry question: Do non-infectious diseases cause more deaths than infectious diseases?**

investigate the causes and effects of non-infectious diseases in humans, including but not limited to:

- ➔ genetic diseases

**Non-infectious disease:** Diseases that are not caused by a pathogen and therefore cannot be spread via human to human

**Genetic diseases** can be caused by germ-line and somatic mutations as well as mutations in mitochondria

- Autosomal recessive condition and a mitochondrial condition

★ Cystic fibrosis
  - An autosomal recessive disease caused by germ-line mutations in the cystic fibrosis transmembrance conductance regulator (CFTR) gene
  - This gene encodes a protein that maintains salt and water levels various organs of the body
  - The disease is most commonly caused by the deletion of a single amino acid, called phenylalanine, at amino acid 508 position (F508del)
  - The gene is transcribed but the polypeptide doesn't fold correctly, and it is degraded before it reaches the correct position in the cell membrane
  - The effect of CFTR mutations is the accumulation of thick mucus in the lungs, which leads to persistent coughing and difficulty breathing
  - Life expectancy = 40-50 yrs

(picture)

★ Mitochondrial diseases
  - Each mitochondrian in a human cell has one small, circular chromosome
  - Genetic mutations in the mitochondrial genome can also cause diseases
  - A cell contains mitochondria, so there will be a mixture of normal and mutant genomes in any one cell
  - An egg contains thousands of mitochondria but the small number in sperm are degraded after fertilisation
  - Offspring therefore only inherit mitochondria, and any mutations from the mother
  - Mitochondrial mutations generally cause muscular and neurological problems, because these cells need high amounts of energy
  - The severity of a disease will depend on the proportion of mitochondria with the mutation

(picture)

➔ diseases caused by environmental exposure
  - Diseases can be caused by exposure to the environment such as toxic substances in contaminated food or water, inhalation of chemicals or exposure to radiation

★ Smoking and lung cancer
  - Lung cancers normally start in the cells that line the bronchi, broncioles and alveoli
  - Smoking tobacco causes 85% of lung cancers

- The link between smoking and lung cancer was first demonstrated in epidemiological studies in the 1940s
- Aniaml experiments in the 1950s showed that mice developed cancer when the tar that forms from smoke inhalation was injected into their bodies
- In the 1960s molecular researchers discovered that chemicals in cigarettes bind to and mutate DNA
- In cigarettes there are about 7000 chemicals and 70 of these are carinogens
- Benzo[a]pyrene (BP) is the most studied and is produced when tobacco is burned and inhaled into the lungs
- When it enters the body, enzymes metabolise BP into a compound that binds to guanine in DNA
- This causes DNA to bend, resulting in G to T transversions, a signature feature of lung cancer genomes caused by smoking
- The effect of the mutations is the uncontrolled cell division in lungs that metasasises to other parts of the body

➔ nutritional diseases
- Any type of non-infectious disease that is caused by deficienceis in the diet, absorption problems, overnutrition or ED such as obesity and type 2 diabeties
  - ★ Obesity and type 2 diabetes

- Obesity is an excessive accumulation of fat in the body
- It is defined using a calculation called the body mass index (BMI)
- BMI is mass in kilograms divided by height in metres squared
- A BMI below 25 is classified as healthy, 25-30 as overweight, and greater than 30 as obese
- Doctors also take other measurements such as weist circumference to help diagnose the condition
- Obesity is caused by the consumption of more calroeis than are burned by the body
- Due to increasing obesity more people are being diagnosed with type 2 diabeteies
- Type 2 diabeties accounts for 90-95% of all diabeties
- People with the disorder produce insulin but the body does not use it effectively, so glucose does not enter the cells, resulting in the accumulation of glucose in the bloodstream
- The body may eventually stop producing insulin, and the effects of the disease are similar to type 1
- There is no cure but the symptons can be managed through lifestyle choices, drugs and or insulin therapy

➔ cancer (is also a genetic disease)
- Cancer is a disease caused by the uncontrolled growth of abnormal somatic cells
- This results in the formation of a tumour or, in the case of blood cancers, the accumulation of abnormal cells in the circulatory system
- Tumours can be benign or malignant
- Benign tumours remain localised and can be surgically removed (non-cancerous)

- Malignant tumours invade nearby tissues and spread to other parts of the body through the circulatory system in a process known as metastasis (cancerous)
- Cancer can begin in almost any type of cell in the body
- It is caused by mutations in genes that play an important role in regulating normal cell divsion and DNA repair - tumour suppressors and proto-oncogenes
- Tumour suppressor genes slow down cell division, repair errors in DNA or activate protein signalling pathways that instruct cells to die (apoptosis)
- Proto-oncogenes express protein receptor proteins, signalling proteins and transcription factors that promote cell division - they are called oncogenes in their mutated form
- A combination of mutations in both groups of genes can caused uncontrolled cell division
- Most cancers (90%) are caused by mutations in somatic cells (sporadic cancer)
- These cancers are more likely to occur as we grow older
- The remaining (10%) are heredity cancer caused by germ-line mutations in tumour suppressor genes
- People with a germ-line mutation are more likely to get cancer at a young age
  (photo)

➔ Including but limited to: diseases caused by autoimmunity
  - Our immune system normally recognises and attacks infections but in some people it recognises cells of the body as 'non-self', attacking them and causing damage
  - When this occurs, it is called an autoimmune disease
    - ★ Type 1 diabetes
      - Type 1 diabetes is a chronic autoimmune disorder that normally develops in childhood and is caused by the bodys inability to produce insulin to regulate glucose levels in the blood stream
      - When we digest food, carbs are broken down into glucose which enters the bloodstream. This triggers the release of insulin from beta cellss in the islets of Langerhans in the pancreas. Insulin binds to a receptor on the surface of cells, to activate a protein called GLUT4, which movies to the plasma membrane and shuttles glucose into the cell
      - The amount of insulin secreted by beta cells is regulated by the level of glucose in the bloodstream; insulin increases after a meal when glucose levels areh igh and declines after glucose levels leaves the blood stream
      - Type 1 diabetes is caused by the immune system attacking and destroying beta cells, which means the pancreas is unable to produce insulin. The exact reason for this autoimmune response is unknown, but it may have a genetic basis or be caused by exposure to viruses or other enivornmental factors

(phot)

- If left untreated, the effects of type 1 diabetes include kidney damage (diabetic nephropathy) and loss of vision (diabetic retinopathy) which occur because of damage to and narrowing of the capillaries in the kidneys and ears
- Other symptoms include poor blood circulation and nerve damage in the legs and feet (diabetic neuropathy) which can recquire amputation
- Can be treated by insulin injections

collect and represent data to show the incidence, prevalence and mortality rates of non-infectious diseases, for example:

➔ nutritional diseases

**Prevalence:** The total number of cases of disease in a population at a specific time
**Incidence:** The number of new cases of a disease occurring in a population at a specific time
**Mortality:** Refers to the number of deaths in a given population from a particular cause over a period of time

- Helps determine what caused a disease

- Being overweight or obese is a risk factor for developing type 2 diabetes
- Population-based data can be represented on maps as a way to observe correlations between a disease and geography such as environmental conditions, wealth of a nation and standards of healthcare
- It is a global problem and more common in some areas of the world than others

➔ diseases caused by environmental exposure
  - Smoking tobacco is an environmental exposure that causes lung cancer

**Epidemiology**
**Inquiry question: Why are epidemiological studies used?**
analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to:

➔ nutritional diseases
➔ diseases caused by environmental exposure

**Epidemiology:** the scientific study of patterns of occurrence of disease in human populations and the factoes that affect these patterns.

- Can be used to study both infectious and non-infectious disease as well as events that involve injuries such as suicides
- Results are used by public health authorities to develop strategies to control disease and improve public health

Epidemiology considers the patterns of disease in terms of:

| | |
|---|---|
| Prevalence | The total number of cases of disease in a population at a specific time |
| Incidence | The number of new cases of a disease occurring in a population at a specific time |
| Geographical distribution | The extent to which cases are occurring |
| Patterns in the occurrence of disease | This can lead to its cause, revealing why people might be getting a particular disease by showing who is being affected |

- Identifies patterns in the occurrence of disease
- Possible cause of disease
- Determines the strategies that would be most effective in controlling disease

investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above

**Type 2 diabetes**

- Early diagnosis of type 2 diabetes is essential to prevent long-term health complications
- Blood glucose levels and the disease can be managed with simple lifestyle changes such as a healthy diet low in processed food, regular exercise and fewer sedentary periods
- When unsuccessful, treatment and management plans can include medications, insulin therapy or weight-loss surgery
- People with type 2 diabetes need to regularly monitor their blood glucose levels during the day, especially before  and after a meal and exercise
- This can be done with a blood glucose meter, which measures the concentration of glucose in a small drop of blood, or electronic devices that continuously mointor the blood glucose level
- A doctor may prescribe drugs or recommend insulin therapy if blood glucose levels cannot be managed through diet and exercise
- Each drug works by contrlooing either the level of blood glucose or insulin in the body

| **Drug** | **How the drug works** | **Side effects** |
|---|---|---|
| Metformin | Lowers glucose production in the liver and makes the body more | Nauseas, diarrhoeas, abdominal pain |

| | | |
|---|---|---|
| | sensitive to insulin, so it is used more efficiently | |
| Sulfonylureas | Help the body to secrete more insulin | Weight gain and low blood sugar levels |
| Glinides | Stimulate the pancreas to secrete more insulin | Weight gain and low blood sugar levels |
| Insulin therapy | Increases the amount of insulin in the body | Risk of low blood sugar and high cholestral |

- Weight loss surgery may also be part of a management plan for diabetes
- This type of surgery changes the shape of the digestive system to limit the amount of food that can be eaten
- Side effects: reduced uptake of nutrients and osteoporosis
- Best way to manage type 2 diabetes contrinuous to be through lifestyle
- This benefits individuals as it reduces the likelihood of medical intervention. Also benefits society because it reduces the cost of treating medical complications caused by the disease
- However, **technology** and **genetic engineering** developed to treat type 1 diabetes may be used to treat type 2 diabetes in the future

| **Future direction** | **Description** |
|---|---|
| Artificial pancreas | A digital device that behaves in the same way as a pancreas by releasing insulin into the body in response to changes in blood glucose levels |
| Islet cell transplantation | Healthy islet cells, which contain beta cells that produce insulin, may be transplanted into the pancreas of people who no longer produce insulin |
| New drug development | There may be more drugs available in the future. Personalised medicine may mean indiviudals are treated with different combinations of drugs |
| Genetic testing | Whole genome sequencing and GWAS studies may identify SNPs or alleles that increase the risk of type 2 diabetes. Prevention strategies could start earlier in poeple carrying these genetic variants |

evaluate the method used in an example of an epidemiological study

1. **Descriptive/cross-sectional studies** are usually the first type of study completed when investigating the cause of a disease

- It investigates ‘who, what, when, and where’
- It involves collecting large amounts of data to establish:
    - ➢ The prevalence of disease
    - ➢ The population sectors being affected, including age, gender, occupation, socioeconomic groups and shared habits such as smoking or drinking
    - ➢ The geographical location/s affected
    - ➢ The time period over which the disease occured or is occuring

| Advantages | Disadvantages |
|---|---|
| Relatively inexpensive<br>Assessing exposures/outcomes<br>Assess health needs | Causality |

Descriptive studies reveal patterns in the distribution of particular diseases within populations, for example:

- ➢ The occupations and other categories of people who develop mesothelioma
- ➢ Lifestyle habits shared by people who develop CVD
- ➢ The incidence of breast cancer in women of different age groups

2. **Longitudal or analytical study**

- Tests hypotheses to determine ‘why and how’ a disease might occur in so many people, collecting information from individuals rather than looking at a population. Groups of relevant subjects are identified, and information is gathered to establish which have been exposed to a suspected risk factor and which developed the disease. Groups are compared to assess the extent to which exposure to the suspected risk factor is linked with developing the disease.

There are two main types of analytical studies:
**Case control studies:** compare two or more similar groups of people without it (the controls) looking for differences in their exposure to suspected risk factors

| Advantages | Disadvantages |
|---|---|

| | |
|---|---|
| Quick and cheap<br>Good for uncommon disease | Not rare exposures<br>Control selection<br>Recall may be a problem |

**Cohort studies** compare two or more similar groups of people without the disease at the start of the study, one made up of people who are exposed to a suspected 'risk factor' ad the other who are not. The two groups are followed over a long time to see how many go onto develop the disease being investigated. By comparing numbers in each group that develops the disease, researchers can establish whether exposrue to the risk factor increases its incidence and assess the strength of any link between them.

| Advantages | Disadvantages |
|---|---|
| Time sequence can be determined<br>Collection of outcome/risk factors<br>Sub analysis<br>Causeality | Cost<br>Not good for rare diseases<br>Ensuring people who started study |

Both types of studies investigate the associations between a disease and any risk factor/s suspected of being the cause. Their main difference is that case control studies are inherently backward looking, attempting to identify the cause of disease by tracking backwards from outcome to exposure to a suspected risk factor. Cohort studies approach the task from the opposite direction, identifying causes by tracking forwards, from exposure to the outcome of developing the disease

3. **Intervention study**
    - This investigates the effectiveness of interventions for managing a disease. For example it can evaluate:
        - ➢ The performance of a diagnostic tool, such as a blood test or scan
        - ➢ The effectiveness of a prevention or control strategy, such as a vaccination program to reduce prevalence, or a public health campaign to change poeples behaviour
        - ➢ The effectiveness and/or safety of a treatment

| Advantages | Disadvantages |
|---|---|
| Good evidence of causality<br>Ranomisation<br>Equal chance<br>Groups are similar | Expensive<br>Many participants<br>Not always possible |

evaluate, using examples, the benefits of engaging in an epidemiological study

| **Study type** | **Benefits/Strengths** | **Limitation** |
|---|---|---|
| Case-control | > individual satisfaction from contributing to knowledge about health for future generations<br>> society benefits from the addition of knowledge about exposures and risk factors and their contribution to disease | > individuals are unlikely to benefit directly from participation<br>> retrospective nature can make data more subject to bias from recall of exposures/ lifestyle choices and unmeasured differences with control population |
| Cohort | > individual satisfaction from contributing to knowledge about health for future generations<br>> society benefits from prospective measurement of different exposures and correlating these with disease | > individuals are unlikely to benefit directly from participation<br>> large cohort studies can be expensive, with many years of follow-up required |

**Prevention**
**Inquiry question: How can non-infectious diseases be prevented?**

use secondary sources to evaluate the effectiveness of current disease-prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to:

- ➔ educational programs and campaigns

Disease prevention refers to methods or strategies that are used to stop a disease from developing

**Educational programs and campaigns**

- ★ Sun smart program
    - Skin cancer is the most common cancer in Australia but changes in behaviour that reduce our exposure to the sun also make it one of the most preventable
    - Over past year, the SunSmart program improved peoples awareness of the link between skin cancer and sun exposure, using strategies such as television advertisements (e.g. Slip, Slop, Slap, Seek, Slide) and a free app that provides daily information about UV light levels
    - One of Aus most successful health education programs
    - Studies have shown it has prevented more than 43, 000 skin cancers and 1400 skin cancer deaths and saved the public healthcare system $92 million over its first two decades

- → genetic engineering
    - Gene therapy can be used to treat or prevent non-infectious diseases
    - It replaces a defective gene with a functioning copy of the gene in the body
    - Can be done with viral vectors or CRISPR-Cas9
    - 

**Technologies and Disorders**
**Inquiry question: How can technologies be used to assist people who experience disorders?**

explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example:

- → hearing loss
    - Sound is produced when an object vibrates, creating a longitudinal wave
    - The wave is propagated by vibrating particles that move back and forth in the same direction as the movement of the wave
    - The vibration is detected by specialised structures in the ear, converted into an electrical signal and interpreted as a sound by the brain

**Characteristics of longitudinal waves**

- Wavelength is the distance between two consecutive compressions or rarefactions, and amplitude is the distance between the particles; the closer the particles, the higher the amplitude
- Frequency is the number of compressions or rarefactions that pass a point in a fixed amount of time
- Amplitude is measured in decibels (dB) and determines the volume of sound
- Frequency determines the pitch of sound; a unpleasant squeal has a high frequency → measured as the number of vibrations per second called hertz (Hz)
    - ➢ compressions are regions of high pressure due to particles being close together
    - ➢ rarefactions are regions of low pressure due to particles being spread further apart

(photo)

**Structure and function of the ear**

- Ears are our organs for hearing, but they also help us keep our balance
- It is divided into the outer ear, middle ear and inner ear
- Outer ear → captures sound waves
- Middle ear → transfers the vibrations to the inner ear
- Inner ear → converts the vibrations into nerve impulses to be interpreted by the brain

(photo)

| Outer ear | Structure | Function |
|---|---|---|
| Pinna | Fleshy, external tissue which consists of a flap of cartilage and skin | Collect and funnel sound waves into the ear canal |
| Ear canal | A passage comprised of bone and skin | Leads sound waves to the eardrum |
| Ear drum | Thin membrane between the outer and middle ear | Vibrates in response to sound to transfer these sound vibrations to the ossicles |

| Middle ear | Structure | Function |
|---|---|---|
| Ossicles | Three bones: malleus, incus, stapes | Amplify and transmit vibrations from the eardrum to the oval window |
| Oval window | Thin, flexible membrane | Transmits vibrations from the stapes to the fluid in the cochlea |
| Round window | Thin membrane between middle and inner ear | Bulge outwards to allow for equalisation of pressure in the cochlea when the oval window vibrates |
| Eustachian tube | Connects the middle ear to the pharynx at the back of the throat | Equalises air pressure on both sides of the eardrum so that it is not overly stretched |

| Outer ear | Structure | Function |
|---|---|---|
| Cochlea | Snail-shaped, spiral tube filled with fluid | When the oval window waves travel through the cochlea causing the basilar membrane which consists of hairs cells to flex. The hair bundles vibrate and press against the tectorial membrane resulting in the release of neurotransmitters. This auditory signal travels via |

| | | |
|---|---|---|
| | | the vestibulocohlear nerve to the brain to be processed and interpreted |

**Causing of hearing loss**

- Hearing loss can be congenital, which means hearing is absent at birth, or caused by event durings our lives, such as exposure to loud sounds, disease and simply getting old
- Can be mild or profound and is categoriesd based on which region of the ear is affected
    1. Conductive hearing loss occurs when sound waves cannot reach the inner ear; it is caused by defects in the outer or middle ear
        - Can be caused by blockage from ear wax or benign tumours, infections in the middle and outer ear, or a perforated eardrum
        - Results in reduced or muffled sound
        - Usually temporary and can be treated

    2. Sensorineural hearing loss occurs because of defects or damage to the cochlea in the inner ear or the auditory nerve and is often permanent
        - Can be caused by malformed inner ear at birth, exposure to damaging noises that causes hair cells in the cochlea to become fatigued/kill them, head injuries, age related hearing loss which is the gradual loss of hearing
        - Results in faint or muffled sounds
        - Often cannot be correct medically or surgically

- ➔ visual disorders
    - Eye receive and convert light into nerve impulses that are interpreted by the brain as images of objects in the environment
    - The human eye perceives the colour or visible portion of the spectrum, while other animals such as invertebrates can detect UV and infrared radiation

**Characteristics**

- The amplitude determines the brightness of light
- The wavelength determines the colour (e.g. red light has a longer wave length than blue light)
- Light waves can have different amounts of energy
- We perceive these differences in energy as different colours
- Reflection is when light 'bounces off' a surface
- Refraction is when light changes direction as it passes from one medium to another
- The angle of refraction is also dependent on the shape of an objects surface (e.g. a convex lens refract light at different angles)

**Structure and function of the eye**

- Light enters the eye through the pupil, the black opening at the centre of the iris

- Contraction and retraction of muscles in the irsis control the amount of light that enters the eye through the pupil
- Light then passes through the cornea, lens and fluid filled eyeball, which refract light to a focal point on the retina at the back of the eye
- The retina consist of photoreceptors, that detect colour and light and are localised in an area called the macula
- There are type types of photorecepotors
    - ➢ Rod cells
        - Don't detect colour, and work best at low light levels (at night)
    - ➢ Cone cells
        - Detect colour and work best in bright light (during the day)
- The retina generates a nerve impulse that is transmitted via the optic nerve to the brain
- The electrochemical message is then interpreted as images

| Eye | Structure | Function |
|---|---|---|
| Iris | Ring of pigmented muscle tissue | Works to control pupil size and hence the amount of light entering the eye → helps us deal with light as different intensities |
| Lens | Transparent, biconvex protein disc. Has a bulging shape and is made out of protein | Adjusts its thickness to bend light to that it focusses directly onto the retina<br><br>Distant objects<br>• Nearly parallel light rays<br>• Require little bending to be focussed<br><br>Close objects<br>• Light rays diverge<br>• Require more bending to be focused |
| Retina | • Thin layer of photoreceptor cells, which lines the inner back surface of the eye<br>• Specialised neurons that contain light-sensitive pigments | Responsible for detecting light, and converting the information into nerve impulses<br><br>After light has passed through the pupil, lens and jelly-like fluid, it hits the photoreceptor cells on |

| | | |
|---|---|---|
| | (photopigments) | the retina<br><br>When photoreceptor cells are exposed to light, their photopigments change shape<br><br>The photoreceptor cells generate an electrochemical signal that is sent to the brain to be interpreted |

**Causes of visual disorders**

- Refractive errors are the most common type of vision disorder. They occur when light does not focus properly on the retina
- The main types of refractive errors are myopia (short-sightedness), hyperopia (far-sightedness) and astigmatism
- They tend to run in families and are diagnosed in childhood or adolescence
- Presbyopia is another refractive error that occurs when our eyes gradually lose the ability to focus on close up objects, usually beginning in the early to mid-forties. It is caused by a hardening of the lens, which beocmes less flexible and therefore less able to accommodate light entering the eye

(phtoo)

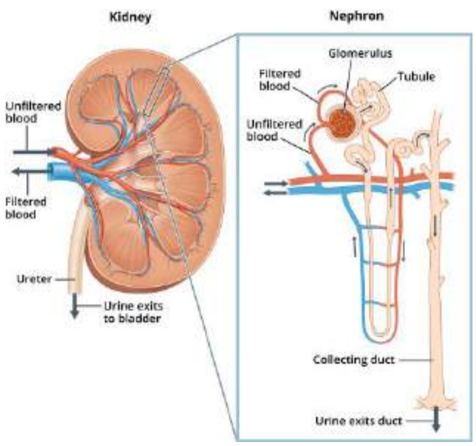
- ➔ loss of kidney function

**Structure and function of the kidney**

- Kidneys are two bean-shaped organs located below the rib cage on either side of the spine
- They are part of the urinary system, along with the bladder and ureters
- Responsible for filtering the blood
- The kidneys excrete excess wastes, salt and water in order to maintain homeostasis
- Blood is filtered out at the glomerulus (due to high blood pressure causing substances to be squeezed out), collected by the bowmans capsule and then reabsorped into capillaries in the loop of Henle. Waste products (e.g. urea) are not reabsorped

1. Blood enters the kidney via the renal artery
2. Once in the kidney the blood flows through a series of small and smaller arteries until it reaches the glomerulus
3. Filtration: Blood pressure forces plasma (water, salts, dissolved solutes) out of the blood, through the glomerulus and into the nephron at the Bowmans capsule. Larger blood components (e.g. red blood cells) stay in the blood
4. Reabsorption & secretion: useful substances (e.g. sugars, nutrients, water) are passively reabsorbed back into the blood
5. Storage: Urine is removed via the ureter and it is stored in the bladder

6. Urination: urine is released via the uretha




**Causes of kidney disorders**

- Can be divided into:
  1. acute
     - When the kidneys suddenly lose their ability to filter waste, which accumulates and changes the chemistry of the blood
     - This can be caused by:
       - ➢ Reduced blood flow to the kidneys - results from blood clots and high cholesterol (which block blood vessels leading to the kidney), severe dehydration, liver failure, infections, toxins and blood pressure medication
       - ➢ Urine blockages - the ureters can get blocked with kidney stones or tumours, which stop the flow of urine out of the body. Kidney stones are hard mineral deposits that build up in the kidney
  2. chronic failure
     - Gradual loss of function over many years. It results in the accumulation of excess fluid, electrolytes and wastes in the blood

Hypertension (high blood pressure) - the walls of thin blood vessels (arterioles) in the kidney thicken and reduce blood flow to the kidney
Diabetic nephropathy - a symptom of type 1 and 2 diabetes. Abnormal filtering from the glomerulus into the Bowmans capsule results in blood cells and large proteins leaking into the urine

investigate technologies that are used to assist with the effects of a disorder, including but not limited to:

- ➔ hearing loss: cochlear implants, bone conduction implants, hearing aids

**Cochlear implants**

- Can be used by people with profound sensorineural loss when hearing aids are not beneficial
- Cochlear implants bypass the outer and middle ear and deliver the sound or vibrations directly to the auditory nerve
- Normal hearing is not restored with a cochlear implant hence a person needs therapy and practice to learn how to hear
- The implant provides an increased awareness of sounds in the environment and improves the ability to understand through lip reading

(photo)

**Bone conduction implants**

- Are used to treat conductive or mixed hearing loss or single-sided deafness
- Similar to the cochlear implant, the system consists of an external sound processor and an implant (photo)

**Hearing aids**

- Are used to amplify sounds to improve hearing in people with sensorineural hearing loss
- They are small, removable electronic devices that ocnsist of a microphone, an amplifier and a speaker
- The microphone detects sound waves and converts them to electrical signals, which are sent to an amplifier
- The amplifier increases the power of the signals and then sends them to the ear through a speaker
- Hearing aids only work if there are some surviving hair cells in the cochlea to detect sound vibrations
- Hearing aids improve hearing and speech comprehention

- ➔ visual disorders: spectacles, laser surgery

**Glasses, contact lenses and laser surgery**

- Refractive errors can be treated with glasses or contact lenses
- People who do not want to wear glasses or contact lenses can have the shape of the cornea permanently changed with corrective laser surgery
- Each of these technologies corrects the angle of the refraction of light so that it focuses correctly on the retina

(photo)

➔ loss of kidney function: dialysis

- Each kidney performs half the kidney function in a body but a person can survive with only one kidney
- If a persons kidney function decreases to about 10-15% of a normal kidney, the only treatments are dialysis or a kidney transplant

**Dialysis**

- Dialysis is a medical techinque that replaces the normal filtering function of the kidneys
- There are two types
  1. Haemodialysis
     - The filtering of blood occurs outside the body
     - Most common type of dialysis and is ideal for people with reduced kidney function
     - Done in hospital, a dialysis centre or at home
     - Session takes about 4-5 hours and needs to be done three days per week
  2. Peritoneal dialysis
     - The filtering of blood occurs inside the body
     - It allows continuous filtration
     - Done at home or at work and is not as disruptive to a persons daily activities
     - However it may not be suitable for people with obesity or abdominal scarring

(photo)

**Kidney transplant**

- Kidney transplant can be done using a donated kidney from a living or deceased person
- If a kidney comes from a living donar, the transplant can be done when the recipients kidney is close to failing but before dialysis (a pre-emptive transplant)
- People can register as organ donors before dying
- The recipient must take anti-rejection drugs for the rest of their life, to prevent the kidney being attacked by their immune system

**Kidney stones**

- Can be treated by breaking them up with shock waves, and smaller stones are then passed naturally in the urine
- They can also be surgically removed through a small incision in the back

evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder

Technologies can vary in their efficacy in assisting with the effects of a disorder. For example, glasses are a relatively inexpensive technology that can restore normal vision to a person with myopia or hyperopia. Cochlear implants, on the other hand, are expensive and do not restore normal hearing to a person with profound hearing loss. They do, however, allow a person to perceive sound, which can enhance their ability to communicate and respond to audible stimuli in the environment.

Dialysis to treat kidney failure is provided as a more detailed example of an evaluation of a technology.

> **Note**
> This is an 'evaluate' dot point. Constructing a table is an effective way of presenting the benefits and limitations of a technology. Remember to provide a value judgement in the form of a concluding sentence.

**TABLE 4.15 Benefits and limitations of dialysis**

| Benefits | Limitations |
|---|---|
| • Extends the person's lifetime<br>• Filters out waste, toxins and excess water<br>• Increases energy levels<br>• Improves appetite | • It is not a cure – kidney function does not return to normal<br>• Need to plan life around dialysis sessions<br>• Can cause cramps and changes in blood pressure<br>• Supplements and hormone therapy may be needed to correct deficiencies (e.g. vitamin D and the hormone erythropoietin) |

In summary, haemodialysis improves the quality of life of a person with kidney failure. It does not, however, restore the normal function of the kidneys, and it is disruptive and requires daily planning around treatment times.